

Tool Learning and Function Calling for LLM Agents: A Comprehensive Survey on Architectures, Mechanisms, Applications, and Frontiers

1 Introduction and Conceptual Foundations

1.1 From Static Models to Dynamic Agents: Paradigm Shift and Limitations

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The rapid evolution of artificial intelligence has marked a definitive transition from static, knowledge-bound large language models (LLMs) to dynamic, autonomous agents capable of interacting with complex external environments. Initially, LLMs were celebrated for their ability to generate coherent text and perform zero-shot reasoning based on vast pre-training corpora. However, these foundational models operate fundamentally as static repositories of probabilistic knowledge, frozen at the moment of their training cutoff. This static nature imposes severe constraints when applied to real-world scenarios that demand up-to-date information, precise factual accuracy, and the execution of multi-step actions. Consequently, the field is undergoing a paradigm shift towards agentic architectures, where LLMs serve not merely as text generators but as the cognitive “brains” orchestrating tools, APIs, and environmental interactions to achieve specific goals. This architectural evolution lays the necessary groundwork for the mechanisms of function calling and tool learning, which will be rigorously distinguished in the subsequent section.

A primary catalyst for this shift is the pervasive issue of hallucination, an innate limitation of standalone LLMs. Despite their impressive linguistic fluency, these models frequently generate plausible-sounding yet factually incorrect information. Recent theoretical work has formalized this challenge, demonstrating that hallucination is not merely a bug to be patched but an inevitable consequence of the learning dynamics within computable models attempting to approximate complex ground truth functions [1]. This inevitability stems from the model’s reliance on statistical correlations rather than causal understanding or access to verified external data. In critical domains such as finance, healthcare, and law, where decision-making relies on absolute precision, the propensity of standalone models to fabricate facts renders them unreliable without external grounding mechanisms [2]. For instance, in medical consultations, hallucinations can lead to overly confident but incorrect diagnoses, posing significant risks to patient safety [3]. To mitigate this, agents must transcend internal parametric memory and leverage external verification, a capability realized through structured tool interaction.

Furthermore, static models suffer from knowledge staleness. Once training concludes, the model’s internal knowledge base remains fixed, unable to incorporate real-time events or emerging scientific discoveries unless retrained—a process that is computationally prohibitive and resource-intensive. This limitation is particularly acute in fast-moving fields like climate science, where outdated information can lead to detrimental policy recommendations. Research has shown that integrating external, scientifically robust sources, such as long-term memory systems linked to live databases, is essential to prevent the propagation of inaccurate or obsolete information [4]. Without such dynamic augmentation, standalone LLMs are ill-equipped to answer queries requiring current data, effectively limiting their utility to historical or general knowledge tasks. The ability to dynamically fetch and integrate fresh data represents the first critical step in transforming a static model into a responsive agent.

Beyond factual inaccuracies and outdated knowledge, standalone LLMs struggle significantly with complex sequential decision-making. Traditional models often adopt a greedy approach, generating the next token based on immediate context without a holistic plan for long-horizon tasks. This lack of strategic foresight leads to performance degradation as task complexity and planning horizons increase. In embodied environments or software development scenarios, agents must navigate a series of interdependent steps where early errors compound, leading to task failure. Existing methods that rely on static plans fail to adapt to environmental feedback, resulting in rigid execution paths that cannot recover from unexpected obstacles [5]. The inability to perform iterative reflection and plan refinement means that standalone models often hallucinate action trajectories or fail to recognize when a chosen path is untenable [6]. Overcoming these limitations requires not just better prompting, but a structural capacity to execute, observe, and adapt—core tenets of the agentic paradigm.

The transition to dynamic agents addresses these limitations by embedding the LLM within a closed-loop system capable of perception, planning, action, and reflection. Unlike static models, dynamic agents leverage tool learning and function calling to extend their capabilities beyond their parametric memory. By interacting with external tools, agents can retrieve real-time information, execute code, manipulate physical objects, and verify their own outputs. This shift transforms the LLM from a passive knowledge retriever into an active problem solver. However, this transition introduces new vectors for error, including the misalignment between the model’s pre-trained knowledge and the actual rules of the environment [7]. Moreover, as agents engage in multi-turn interactions, they face difficulties in managing long-context windows and maintaining consistent belief states about the environment, often leading to systematic failures in planning optimization [8]. These challenges highlight the gap between simply having access to tools and effectively mastering their use.

Addressing these challenges requires a fundamental rethinking of agent architecture. It is insufficient to simply prompt a static model to “think harder”; instead, systems must be designed to quantify uncertainty, detect hallucinations in real-time, and refuse to answer when knowledge boundaries are exceeded. Approaches such as Reinforcement Learning from Knowledge Feedback (RLKF) have been proposed to train models to recognize their own limitations and refuse unknown questions, thereby enhancing reliability [9]. Additionally, the development of causal world models allows agents to simulate counterfactual scenarios and reason about “what-if” questions, moving beyond spatio-temporal regularities to true causal understanding [10]. These advanced capabilities underscore the necessity of distinguishing between the mechanical act of invoking a tool and the cognitive process of learning to use it effectively.

In summary, the move from static LLMs to dynamic agents represents a necessary evolution to overcome the inherent limitations of hallucination, knowledge staleness, and poor sequential reasoning. While standalone models provide a powerful foundation for language understanding, their deployment in autonomous, high-stakes environments necessitates the integration of external tools, feedback loops, and robust verification mechanisms. The subsequent sections of this survey will explore the architectural innovations, cognitive mechanisms, and training paradigms that enable this transformation. Specifically, we will first delineate the conceptual boundaries between the operational mechanism of function calling and the higher-order cognitive process of tool learning, detailing how modern agents are being engineered to navigate the complexities of the real world with greater agency and reliability.

1.2 Distinguishing Tool Learning and Function Calling Mechanisms

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Building upon the paradigm shift from static models to dynamic agents outlined previously, it is imperative to rigorously delineate the conceptual and technical boundaries between “function calling” and “tool learning.” While these terms are often used interchangeably in casual discourse, they represent fundamentally distinct layers of capability within an autonomous system. Function calling serves as the syntactic and operational interface—the mechanical act of invoking an external capability—whereas tool learning encompasses the broader cognitive process by which an agent acquires, adapts, and masters the strategic utilization of these capabilities through interaction, feedback, and iterative refinement. This distinction is critical for understanding how agents transcend the limitations of hallucination and knowledge staleness discussed in Section 1.1, moving from mere execution to genuine agency.

Function calling, in its most basic definition, refers to the standardized mechanism that allows a language model to output a structured request to an external Application Programming Interface (API). This process transforms the LLM from a static text generator into a dynamic controller capable of executing actions in the real world. The primary challenge here lies in format alignment and intent comprehension; the model must correctly identify the appropriate tool and generate arguments that adhere to strict schema constraints. Recent research highlights that even powerful models struggle with the precision required for reliable API invocation, often achieving correctness rates far below practical thresholds without specific optimization [11]. Furthermore, the sheer volume of available APIs in real-world systems presents a significant bottleneck, as it is impossible to feed descriptions of all potential tools into the limited context window of an LLM, necessitating sophisticated retrieval mechanisms prior to the calling phase [12]. Consequently, function calling is often viewed as a discrete, atomic operation: a bridge between internal reasoning and external execution, addressing the immediate need for action but not necessarily implying strategic depth.

In contrast, tool learning represents a higher-order cognitive capability that transcends mere syntax. It is the process by which an agent learns not only *how* to call a function but *when* to call it, *which* tools to combine, and *how* to adapt its strategy based on the outcomes of previous executions. A critical distinction is that function calling can be hardcoded or prompted via few-shot examples, whereas tool learning implies a capacity for improvement and generalization over time. For instance, naive approaches often induce models to utilize tools indiscriminately, leading to error propagation in tasks that the model could otherwise solve independently. True tool learning addresses this by teaching models to selectively engage tools only when necessary, thereby mitigating excessive reliance and enhancing overall accuracy [13]. This selective utilization requires the agent to develop an internal metric of confidence and utility, a trait that goes beyond the mechanical execution of a function call and directly addresses the sequential decision-making challenges highlighted in the previous section.

The divergence between these concepts becomes even more pronounced when considering the role of feedback and adaptation. Function calling is typically a one-way street in basic implementations: the model generates a call, and the system returns a result. Tool learning, however, closes the loop by incorporating execution feedback into the agent’s decision-making policy. Frameworks that leverage simulated trial and error demonstrate that agents can significantly improve their tool usage accuracy by “imagining” scenarios and learning from the consequences of their actions, effectively orchestrating trial, error, and memory mechanisms similar to biological systems [11]. Moreover,

advanced tool learning frameworks decompose the workflow into specialized agents for grounding, execution, and observing, allowing for iterative calibration based on environmental feedback—a level of architectural complexity that simple function calling does not inherently possess [14]. This adaptive capacity is the precursor to the sophisticated planning and memory systems that will be detailed in the subsequent taxonomy.

Another key differentiator is the scope of generalization. Function calling is often tied to a specific set of known APIs, where the model is fine-tuned or prompted to recognize specific signatures. Tool learning aims for generalized proficiency, enabling agents to master unseen tools or adapt to changing API specifications without exhaustive retraining. Approaches like ToolAlpaca illustrate that compact models can achieve generalized tool-use abilities comparable to massive models by learning from diverse, simulated corpora, suggesting that the core of tool learning lies in the ability to abstract functional patterns rather than memorize specific API endpoints [15]. Similarly, methods that represent tools as embeddable tokens allow for the dynamic expansion of an agent’s toolkit, facilitating the integration of massive numbers of tools without the context limitations inherent in traditional prompt-based function calling [16].

Furthermore, the evaluation of these mechanisms reveals distinct failure modes. Evaluations focused solely on function calling often check for format alignment and successful execution, whereas comprehensive tool learning assessments, such as ToolEyes, analyze deeper dimensions including behavior planning, intent comprehension, and answer organization across complex, noisy real-world scenarios [17]. The robustness of an agent in tool learning is tested against environmental noise and ambiguity, where the ability to recover from failed function calls becomes a measure of cognitive resilience rather than just syntactic correctness [18].

Ultimately, while function calling provides the necessary infrastructure for action, tool learning constitutes the intelligence that governs those actions. Function calling is the “hand” that reaches for the tool, but tool learning is the “mind” that understands the task, selects the right instrument, adjusts the grip based on tactile feedback, and learns from mistakes to perform better next time. As the field progresses towards Artificial General Intelligence, the focus must shift from merely optimizing the syntax of function calls to cultivating the adaptive, self-improving cognitive processes that define true tool learning [19]. Recognizing this distinction is crucial for developing architectures that move beyond rigid, pre-defined workflows to flexible, autonomous agents capable of navigating the open world. Having established these conceptual boundaries, we now proceed to situate these capabilities within a unified taxonomy of agent components, dissecting how planning, memory, and tool interaction protocols interlock to form cohesive intelligent systems.

1.3 A Unified Taxonomy of Agent Components

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Having distinguished the syntactic mechanism of function calling from the cognitive breadth of tool learning, it is imperative to establish a unified taxonomy that situates these capabilities within the broader architecture of Large Language Model (LLM) agents. While early definitions of intelligent agents often relied on rigid, pre-programmed logic, modern LLM-based agents are characterized by their ability to self-evolve and engage in complex, long-term interactions with dynamic environments [20]. To systematically analyze this evolving landscape, we decompose agent architectures into three core, interdependent modules: planning strategies, memory systems, and tool interaction protocols. This tripartite structure not only provides the necessary granularity to evaluate how agents perceive, reason, and act but also serves as the structural bridge between the conceptual distinctions outlined

previously and the adaptive feedback loops discussed subsequently.

The first pillar of this taxonomy, **planning strategies**, governs the agent’s cognitive process for decomposing high-level goals into executable sequences. Planning transcends the mere generation of action lists; it involves sophisticated reasoning capabilities that allow agents to navigate uncertainty and adapt to changing contexts. Recent literature distinguishes between static planning, where a trajectory is fixed at the onset, and dynamic planning, which incorporates iterative reflection and re-evaluation. For instance, the integration of retrieval-augmented mechanisms allows agents to leverage past experiences corresponding to current situations, thereby enhancing the robustness of their decision-making processes [21]. Furthermore, the complexity of planning varies significantly across architectures; some systems employ hierarchical decomposition to manage sub-goals, while others utilize graph-based structures to model non-linear thought processes [22]. The efficacy of these strategies is often benchmarked against their ability to handle “planning hallucinations,” where models generate invalid action paths due to a lack of explicit action knowledge [6]. Therefore, within our taxonomy, planning strategies are categorized by their degree of adaptability, the presence of reflective loops, and their reliance on external knowledge bases to ground abstract reasoning.

The second critical component is the **memory system**, which serves as the repository for the agent’s context, experiences, and learned skills. Unlike traditional software systems with deterministic state storage, LLM agent memory must handle unstructured, multimodal data and support efficient retrieval under varying contextual constraints. A comprehensive taxonomy of memory must differentiate between short-term working memory, which maintains immediate conversational context, and long-term memory, which stores episodic experiences and semantic knowledge for future retrieval. The design of these systems is pivotal for enabling agents to solve real-world problems that require sustained interaction over extended periods [20]. Advanced architectures propose OS-inspired memory management systems that facilitate context switching and concurrent execution, effectively treating the LLM as the “brain” of an operating system [23]. Moreover, the role of summarization has emerged as a fundamental capacity within memory modules, allowing agents to compress vast amounts of interaction history into concise, actionable insights [24]. In embodied settings, memory systems further evolve to mediate between semantic representations and sensorimotor data streams, enabling robots to predict action effects and reason about physical environments [25]. Thus, our taxonomy classifies memory systems based on their temporal scope, modality support, and the mechanisms employed for information compression and retrieval.

The third pillar, **tool interaction protocols**, defines the interface through which agents exert influence on the external world, directly operationalizing the concepts of function calling and tool learning discussed earlier. This module encompasses the detection, selection, and execution of external APIs, software functions, or physical actuators. The distinction between simple function calling and genuine tool learning lies in the agent’s ability to dynamically discover and adapt to new capabilities without explicit retraining. Effective tool interaction requires a semantic understanding of both the agent’s internal capabilities and the affordances of external entities [26]. In multi-agent scenarios, this extends to the coordination of heterogeneous capabilities, where agents must negotiate role assignments and share resources to accomplish collective goals [27]. Recent frameworks emphasize the importance of configurable interaction layers that allow for the seamless integration of diverse tools, ranging from web search APIs to complex robotic controllers [28]. Furthermore, the reliability of tool usage is contingent upon the agent’s ability to verify preconditions and monitor execution outcomes, a challenge addressed by approaches that combine semantic capability models with formal verification techniques [29]. Consequently, tool interaction protocols in our taxonomy

are differentiated by their flexibility in handling heterogeneous tools, the sophistication of their selection mechanisms, and their robustness in error recovery.

By synthesizing these three dimensions—planning, memory, and tool interaction—this unified taxonomy offers a structured framework for dissecting the capabilities of contemporary LLM agents. It allows researchers to pinpoint specific architectural strengths and weaknesses, such as an agent with superior planning but limited memory retention, or one with extensive tool access but fragile interaction protocols. This categorization is essential for advancing the field beyond monolithic descriptions toward a nuanced understanding of how distinct components contribute to overall agent autonomy and intelligence [30]. Crucially, this static decomposition sets the stage for understanding dynamic evolution: while the taxonomy defines the *components*, the following section will explore how feedback loops and adaptation mechanisms animate these parts, transforming a structured architecture into a resilient, self-improving system [31].

1.4 The Role of Feedback and Adaptation in Agent Evolution

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While the unified taxonomy in the preceding section decomposes agents into static components—planning, memory, and tool interaction—it is the dynamic interplay of feedback and adaptation that animates these parts, transforming a structured architecture into a resilient, self-improving system. The transition from static language models to dynamic, autonomous agents is fundamentally defined by the shift from open-loop, greedy action selection to closed-loop systems capable of adaptive refinement. In early iterations, actions were often generated based on a single pass of reasoning, rendering the system brittle in the face of environmental unpredictability or execution errors. Modern agent evolution, however, is characterized by the integration of robust feedback mechanisms that allow agents to perceive the consequences of their actions, reflect on discrepancies between expected and actual outcomes, and iteratively correct their trajectories. This capacity for self-correction transforms agents from mere pattern matchers into resilient problem solvers that can navigate complex, non-stationary environments.

A critical component of this evolution is the ability to refine plans adaptively in response to environmental feedback, directly enhancing the **planning strategies** outlined previously. Traditional approaches often rely on static plans that fail to account for dynamic changes or unexpected obstacles, leading to performance degradation as task complexity increases. To address this, frameworks such as AdaPlanner have introduced closed-loop approaches where the agent does not merely execute a pre-determined sequence but actively refines its self-generated plan based on real-time feedback. By employing both in-plan and out-of-plan refinement strategies, these agents can mitigate hallucinations and adjust their strategies when the environment deviates from initial assumptions. This adaptive planning is further enhanced by code-style prompt structures that facilitate flexible plan generation across diverse tasks, allowing the agent to leverage successful past plans as few-shot exemplars for future decision-making [5].

Beyond external environmental signals, the internal mechanism of self-reflection serves as a pivotal driver of agent improvement, leveraging the **memory system** to store and retrieve corrective insights. Early self-reflection methods often struggled with knowledge-rich problems, where agents would get stuck in repetitive loops of incorrect reasoning despite receiving negative feedback. Advanced methodologies like Mirror have addressed this by introducing multiple-perspective self-reflection, utilizing a heuristic interaction between a “Navigator” and a “Reasoner” to generate diverse reasoning trajectories. This approach encourages the agent to explore alternative logical

paths and seek agreement among strategically induced perturbations, thereby avoiding local minima in the reasoning space and significantly improving reliability without access to ground truth [32]. Similarly, the concept of an “inner monologue” has been proposed to enable embodied agents to process rich sources of feedback, such as success detection and scene descriptions, allowing them to reason over their own choices and refine high-level instructions in robotic control scenarios [33].

The evolution of agents also involves learning from failures through retrospective analysis and policy optimization, effectively turning the **tool interaction protocols** into a source of continuous learning. Rather than simply discarding failed attempts, sophisticated agents now employ retrospective models that summarize the root causes of prior failures and propose corrected action plans. The Retroformer framework exemplifies this by using policy gradient optimization to fine-tune language agent prompts based on environment-specific rewards. This method allows agents to learn from rewards across multiple tasks, effectively turning negative feedback into a constructive signal for prompt refinement and strategic improvement over time [34]. Furthermore, the autonomous annotation of agent trajectories enables systems to generate their own training data by explaining the rationale behind actions, facilitating contrastive self-training loops that continuously enhance performance without heavy reliance on human annotation [35].

Adaptation in agents is not limited to correcting immediate errors but extends to the long-term alignment of agent behavior with human intentions, environmental dynamics, and self-imposed constraints. The principles of Unified Alignment for Agents (**UA**²) advocate for a holistic approach where agents simultaneously align with user profiles, personalized environmental dynamics, and runtime cost statistics. This multi-dimensional alignment ensures that agents remain effective and safe when operating in intricate, realistic environments, moving beyond simple task completion to optimized, context-aware execution [36]. Moreover, the ability to detect novelties and adapt domain models is crucial for operating in open worlds. Agents equipped with mechanisms to measure the divergence between expected and observed action executions can infer the presence of environmental novelties and revise their internal models accordingly, ensuring robustness against Distributional shift [37].

However, the integration of feedback and adaptation introduces new challenges, particularly regarding the stability of self-modifying agents. Research indicates that while perfectly rational agents might safely self-modify, agents with bounded rationality face risks of exponential performance deterioration and gradual misalignment if their self-modification processes are not carefully constrained. Imperfections in action selection or inaccurate environmental models can lead to compounding errors during self-modification, highlighting the need for rigorous safety frameworks that anticipate the consequences of an agent altering its own utility functions or policies [38] [39]. Additionally, the coupling between agent strategies and environmental feedback can drive emergent behaviors, such as group cooperation, where individuals detect inflection points in environmental states through rewards and adjust their strategies to maintain collective stability [40].

Finally, the efficiency of adaptation is being revolutionized by techniques that allow agents to learn from sparse feedback and simulate past trajectories. By combining learned feature encoders with inverse models, agents can simulate human actions backward in time to infer preferences from observed states, effectively learning “what to do” without explicit reward signals. This capability, paired with language-guided world models that allow humans to modify agent internal models via natural language, creates a powerful channel for rapid behavioral adaptation and control [41] [42]. As agents continue to evolve, the synthesis of environmental feedback, self-reflection, and adaptive planning establishes the dynamic foundation necessary for the next step in architectural design: determining whether these adaptive capabilities are best realized within a solitary cognitive loop

or distributed across a collaborative society of agents.

1.5 Single-Agent versus Multi-Agent Architectures: A High-Level Overview

1.5 Architectural Paradigms: From Monolithic Loops to Distributed Cognition

Building upon the adaptive mechanisms and feedback loops discussed previously, the architectural realization of these capabilities presents a fundamental design choice: the dichotomy between single-agent iterative loops and modular multi-agent frameworks. This distinction transcends the mere scaling of model instances; it reflects divergent philosophies regarding cognitive modularity, reasoning robustness, and the management of complex, dynamic environments. While the preceding section highlighted how agents refine trajectories through self-correction and environmental alignment, this section examines how the structural organization of these agents—whether as a solitary entity or a collaborative society—dictates the efficacy of such adaptations. At a high level, single-agent architectures rely on the internal capacity of a monolithic model to decompose tasks, reflect on errors, and execute sequential actions within a unified context window. In contrast, multi-agent frameworks distribute these cognitive responsibilities across specialized entities, leveraging interaction protocols to achieve emergent problem-solving capabilities that often surpass the limitations of individual models.

Single-agent architectures are characterized by their reliance on iterative thought-action loops, where a solitary model performs planning, tool selection, and execution in a sequential manner. This approach draws heavily from frameworks like ReAct, where the agent alternates between reasoning traces and environmental interactions. While effective for linear tasks, this monolithic structure faces significant challenges when confronted with highly complex or compositional problems. A primary limitation is the tendency toward “Degeneration-of-Thought” (DoT), where an agent, once confident in an initial solution path, struggles to generate novel perspectives or correct fundamental errors through self-reflection alone [43]. The single agent’s reasoning is inherently bounded by its own training distribution and the static nature of its internal knowledge, making it susceptible to hallucinations and logical loops when task complexity exceeds its cognitive horizon. Furthermore, as tasks require the integration of diverse skills—such as coding, mathematical reasoning, and physical planning—a single model may struggle to maintain high proficiency across all domains simultaneously without extensive and often conflicting fine-tuning.

Conversely, multi-agent architectures introduce a paradigm of distributed cognition, where the system is decomposed into heterogeneous agents, each potentially specializing in specific sub-tasks or possessing distinct roles. This modularity allows for a more flexible and robust approach to problem-solving, directly addressing the bottlenecks inherent in solitary workflows. By assigning specific responsibilities—such as grounding, execution, and observation—to different agents, systems can mitigate the fragility observed in single-agent loops. For instance, the ConAgents framework demonstrates that modularizing the tool-learning workflow into cooperative agents significantly improves performance on complex tasks by enabling adaptive calibration based on environmental feedback, a capability often lacking in solitary agents [14]. This separation of concerns mirrors biological and organizational systems, where specialized units cooperate to achieve goals that would be intractable for a generalist entity operating in isolation.

The transition from single-agent to multi-agent systems also fundamentally alters the landscape of divergent thinking and consensus building. In single-agent settings, self-correction mechanisms often fail to break local optima because the agent is essentially debating with itself, a limitation that persists even with advanced reflection strategies. Multi-agent debate frameworks, however,

instantiate multiple personas or agents that engage in argumentation, forcing the system to explore alternative solution spaces and critique initial assumptions rigorously [43]. Although recent studies suggest that the efficacy of such debates depends heavily on hyperparameter tuning and protocol design, properly configured multi-agent discussions can outperform single-agent self-consistency methods by fostering a competitive yet collaborative environment that surfaces errors and refines reasoning [44].

Beyond reasoning quality, multi-agent architectures offer superior scalability and adaptability in dynamic environments, complementing the long-term alignment goals discussed in the previous section. The concept of self-organized agents illustrates how systems can dynamically scale the number of active agents based on problem complexity, allowing for the generation and optimization of large-scale codebases that would overwhelm the context limits of a single agent [45]. This dynamic scalability is complemented by hierarchical structures, where high-level planners coordinate low-level executors, effectively managing long-horizon tasks through decomposition and parallel execution [46]. Such hierarchical and distributed arrangements align with broader systems engineering principles, where moving from monolithic to distributed architectures enhances robustness against uncertainty and facilitates the management of complex, evolving ecosystems [47].

However, the shift to multi-agent systems introduces new complexities, particularly regarding coordination overhead, communication protocols, and credit assignment. Unlike single-agent loops where the state is implicitly maintained within one context, multi-agent systems require explicit communication mechanisms to share information and synchronize actions. Research into decentralized abstractions and agent topology highlights the necessity of structured interaction patterns to ensure that cooperation emerges without leading to chaos or redundant computation [48][49]. Moreover, the trade-off between centralized control and decentralized autonomy remains a critical design consideration; while centralized planning can optimize global objectives, decentralized execution offers greater reactivity and fault tolerance in unpredictable environments [50].

Ultimately, the choice between single-agent and multi-agent architectures depends on the specific requirements of the application domain. Single-agent loops offer simplicity and lower computational costs for straightforward tasks, whereas multi-agent frameworks provide the necessary structural inductive biases for solving composite, multi-step problems that demand diverse expertise and robust error correction. As the field progresses, the integration of these paradigms—perhaps through hybrid models that combine the efficiency of single-agent reasoning with the collaborative power of multi-agent societies—will likely define the next generation of autonomous AI systems capable of operating in open-world scenarios. Having established the architectural foundations, the subsequent sections will delve deeper into the specific mechanisms, training paradigms, and evaluation metrics that govern these choices, providing a comprehensive roadmap for navigating the trade-offs inherent in designing intelligent agents.

2 Core Architectures and Orchestration Frameworks

2.1 Single-Agent Iterative Loops and Reflective Architectures

2.1 Single-Agent Iterative Loops and Reflective Architectures

The evolution of Large Language Model (LLM) agents from static knowledge repositories to dynamic problem solvers is fundamentally rooted in the development of single-agent iterative loops and reflective architectures. These foundational frameworks establish the cognitive baseline for autonomous agents, enabling them to decompose complex objectives into manageable sub-tasks

through a continuous cycle of reasoning, acting, and observing. Unlike traditional models that generate a single output sequence, iterative agents operate within a closed loop where the output of one step serves as the input for the next, allowing for real-time adaptation to environmental feedback. This paradigm shift began with frameworks that explicitly interleave reasoning traces with actionable commands, most notably the ReAct (Reasoning and Acting) paradigm. By generating verbal reasoning traces alongside actions, this approach ensures that an agent’s internal logic aligns with external tool execution, thereby mitigating hallucinations and improving task success rates in interactive environments. While these linear loops provide the essential mechanism for tool interaction, they often lack the structural complexity required for long-horizon planning, a gap addressed by subsequent architectural innovations.

Building upon these initial iterative concepts, recent research has introduced sophisticated mechanisms to enhance the autonomy and efficiency of single-agent loops, particularly addressing the scarcity of high-quality, multi-step trajectory data required for supervised fine-tuning. To overcome this bottleneck, the [35] paper proposes the A³T framework, which enables the autonomous annotation of agent trajectories. By utilizing an “ActRe” prompting agent to explain the rationale behind arbitrary actions, this framework allows a ReAct-style agent to synthesize novel trajectories by prepending posterior reasoning to sampled actions. This self-training loop facilitates contrastive learning where the agent iteratively refines its policy based on successful versus failed trajectories, achieving performance levels comparable to human experts without extensive human annotation. This demonstrates how iterative loops can be turned inward to generate the very data necessary for their own improvement, creating a self-sustaining cycle of capability enhancement that precedes more complex multi-agent coordination.

Beyond data generation, the structural organization of the agent’s internal process has evolved from simple linear sequences to state-driven workflows that impose greater control and interpretability on the decision-making process. While standard ReAct operates on a rigid thought-action-observation sequence, the [51] framework conceptualizes task solving as a state machine, distinguishing between “process grounding” via state transitions and “sub-task solving” via actions within a state. In this architecture, transitions are governed by heuristic rules or LLM decisions, allowing for dynamic progression that adapts to the current status of the running process. Empirical evaluations indicate that such state-driven approaches can significantly outperform standard ReAct baselines in terms of both success rate and computational cost, highlighting the importance of structured control flow. This move toward explicit state management serves as a critical precursor to the hierarchical decomposition strategies discussed in the following section, where high-level planners manage abstract states rather than raw actions.

Reflective capabilities represent another cornerstone of modern single-agent architectures, enabling agents to learn from their own mistakes without weight updates, thus simulating a form of meta-cognition. The [52] framework exemplifies this by introducing verbal reinforcement learning, where agents maintain an episodic memory buffer of reflective text. When an agent encounters failure or receives negative feedback, it verbally reflects on the error, summarizes the lesson learned, and stores this reflection for future trials. This mechanism allows the agent to avoid repeating identical errors in subsequent attempts, effectively simulating a reinforcement learning signal through natural language. Complementing this, the [53] paper argues that intrinsic reflection can be unstable due to model overconfidence. It proposes a method where the agent adaptively explores diverse solving perspectives, contrasts the discrepancies between them, and synthesizes a checklist to re-examine its reasoning. This approach leverages inconsistency as a signal for potential errors, leading to more robust and stable self-correction compared to standard self-evaluation methods.

Furthermore, the integration of predictive capabilities and dual-process thinking into the iterative loop has been shown to enhance strategic planning and balance efficiency with accuracy. The [54] framework integrates prediction with reasoning and action, allowing the agent to anticipate future states before committing to an action. By leveraging historical predictions, the agent can engage in more diversified and strategically oriented reasoning, guiding it toward more effective actions in complex tasks. This forward-looking capability transforms the iterative loop from a reactive mechanism into a proactive planning engine. Additionally, the concept of dual-process thinking has been adapted for agents in the [55] framework. Inspired by human cognition, SwiftSage combines a “Swift” module for fast, intuitive action selection via behavior cloning, with a “Sage” module for deliberate, slow thinking using LLMs for subgoal planning. This hybrid architecture optimizes the trade-off between efficiency and accuracy, ensuring that the agent can handle both routine operations and complex reasoning challenges within a single iterative framework.

Finally, the explicit management of work state within these loops is crucial for maintaining coherence over long-horizon tasks. The [56] paper introduces a model that employs “work notes” to record and reflect on the state throughout task execution. By maintaining a comprehensive work journal that incorporates plans, outcomes, and reflections, this architecture improves task execution efficiency and provides a solid foundation for auditing and subsequent analysis. Together, these advancements in single-agent iterative loops and reflective architectures illustrate a trajectory toward increasingly autonomous, self-improving, and cognitively sophisticated agents. However, as tasks grow in complexity and scale, the limitations of a single agent—even one equipped with reflection and state management—become apparent. This necessitates a shift from internal cognitive loops to external structural decomposition, leading naturally to the hierarchical planning and dynamic role allocation strategies explored in the next section, where complexity is managed not just by deeper reasoning, but by distributing roles across layered architectures.

2.2 Hierarchical Planning and Dynamic Role Allocation

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Building upon the reflective and state-driven capabilities of single agents discussed previously, the next evolutionary step in agent architecture involves managing complexity through structural decomposition. While single-agent iterative loops excel at local reasoning and self-correction, they often struggle with the combinatorial explosion of action spaces and sparse rewards inherent in long-horizon tasks. Hierarchical architectures address this limitation by introducing vertical abstraction, decomposing complex objectives into manageable sub-goals and primitive actions. This paradigm organizes decision-making into distinct layers: a high-level planner that operates over extended timeframes to define strategic intent, and a low-level controller that manages immediate, reactive execution [57]. By separating these temporal scales, agents can reason about abstract concepts such as “sub-goals” or “roles” rather than raw motor commands, thereby enhancing sample efficiency and generalization in uncertain domains.

A primary mechanism within this hierarchical framework is the master-slave or manager-worker coordination strategy, which effectively bridges the gap between centralized planning and decentralized execution. In these semi-centralized systems, a “master” agent or module observes the global state and assigns specific sub-tasks or latent goals to “slave” or “worker” agents. This decoupling allows the system to scale to multi-agent scenarios without requiring every agent to possess full global awareness. For instance, the Feudal Multi-Agent Hierarchies (FMH) framework exemplifies this approach, where a manager agent learns to communicate subgoals to multiple worker agents

that operate concurrently to maximize environmental rewards [58]. Similarly, recent advancements have revisited the master-slave architecture by integrating composed action representations and learnable communication protocols, demonstrating that such hierarchical structures can leverage the benefits of both centralized coordination and decentralized reasoning to outperform purely decentralized methods in complex micromanagement tasks [59].

Complementing vertical decomposition, dynamic role allocation introduces horizontal specialization, enabling agents to adapt their behaviors based on real-time environmental feedback and team composition. Unlike static hierarchies where roles are predefined, dynamic allocation allows agents to fluidly switch functions—such as transitioning from exploration to coverage or from offense to defense—based on the current task state. This adaptability is crucial in heterogeneous teams with diverse capabilities or in homogeneous teams facing evolving challenges. Research has shown that integrating role concepts into upper-level task planning enables robots to learn optimal role selection strategies through deep reinforcement learning, utilizing intelligent switching mechanisms to promote interchangeability between exploration and coverage objectives [60]. Furthermore, role-based learning can be optimized by decomposing joint action spaces into restricted role action spaces, where a bi-level learning hierarchy allows a role selector to operate at a lower temporal resolution while role policies execute specialized behaviors, significantly boosting learning efficiency and policy generalization [61].

The efficacy of these hierarchical systems often hinges on the quality of sub-goal representations and the mechanisms used to discover them. Unsupervised learning schemes, such as those based on asymmetric self-play, have been proposed to automatically learn meaningful representations of sub-goals, allowing high-level policies to direct low-level policies via continuous sub-goal vectors [62]. In more complex, pixel-based environments, methods like Director plan inside the latent space of a learned world model, selecting latent goals that are interpretable because the world model can decode them back into images, thus bridging the gap between abstract planning and visual perception [57]. Additionally, human feedback can be incorporated to guide the discovery of suitable subgoals, ensuring that the hierarchical decomposition aligns with human-intuitive task structures and stabilizes the learning process in sparse-reward settings [63].

Adaptive task assignment strategies further refine these architectures by optimizing the distribution of work among agents or sub-modules through advanced attention and optimization techniques. In multi-human multi-robot teams, attention-enhanced hierarchical reinforcement learning approaches have been developed to decompose complex initial task assignment problems into structured sub-problems, leveraging cross-attribute attention mechanisms to discern nuances in the state space crucial for decision-making [64]. Moreover, architectural frameworks combining Behavior Trees for reactive planning with Mixed-Integer Linear Programs for dynamic role allocation have demonstrated the ability to handle large heterogeneous teams in industrial settings, balancing metrics such as makespan and ergonomics through negotiation phases [65]. These systems illustrate how hierarchical planning can be augmented with formal optimization techniques to ensure robust and efficient task execution in dynamic, real-world scenarios.

Ultimately, the synergy between hierarchical planning and dynamic role allocation provides a powerful framework for building scalable and adaptable agents capable of navigating complex, open-ended environments with human-like flexibility. Whether through the autonomous discovery of intrinsic options [66] or the structured decomposition of tasks via language instructions [67], these methods represent a critical intermediate step between individual cognition and collective intelligence. However, while these structured approaches rely on explicit high-level controllers or predefined role assignments, the next frontier lies in architectures where stable patterns and dynamic coalitions

emerge spontaneously. This necessity drives the transition toward self-organizing and decentralized multi-agent systems, where global order arises from local interactions without the need for a central orchestrator, as explored in the following section.

2.3 Self-Organizing and Decentralized Multi-Agent Systems

2.3 Self-Organizing and Decentralized Multi-Agent Systems

While the hierarchical planning and dynamic role allocation mechanisms discussed in the previous section provide a structured approach to decomposing complex tasks, they often rely on explicit high-level controllers or predefined role assignments. The transition to self-organizing and decentralized multi-agent systems represents a fundamental paradigm shift, moving from centralized orchestration to architectures where stable patterns and dynamic coalitions emerge spontaneously through local interactions. This subsection analyzes the mechanisms underpinning such decentralization, focusing on the emergence of organic hierarchical structures, the formation of adaptive coalitions, and the critical trade-offs between efficiency and robustness, thereby setting the stage for the sophisticated communication protocols required to sustain these distributed societies.

Self-organization in multi-agent systems is defined as a process where global order arises from the cooperative behavior of initially disordered components via local rules, eliminating the need for a predefined global controller. A primary challenge in this domain is predicting global system behavior based on local agent interactions, given the autonomy of individual agents. To address this, researchers have proposed logic-based frameworks that reason about the dependence relations between coalitions regarding their contributions to global behavior. These frameworks often reveal that verifying such self-organizing systems is computationally complex, typically falling within exponential time, necessitating decomposition strategies such as graph theory to analyze agents' full contributions across different layers [68]. Consequently, the structural integrity of these systems relies heavily on the communication topology, which plays a pivotal role in learning capabilities. For instance, structural self-adaptation allows agents to reposition themselves within a communication structure, such as a tree, to expand learning capacity and optimize decision-making in complex combinatorial problems [69].

In the context of LLM agents operating in open-ended environments, rigid, task-oriented workflows are increasingly viewed as insufficient compared to fluid, agent-centric organizations. Recent innovations introduce self-organizing agent systems, such as S-Agents, which utilize a "tree of agents" structure to enable dynamic workflow adjustments. This architecture, combined with an "hourglass agent architecture" for balancing information priorities, allows agents to autonomously coordinate and execute tasks asynchronously without human intervention. Such systems have demonstrated proficiency in collaborative building and resource collection tasks in dynamic simulations like Minecraft, validating the effectiveness of agent-centric organizational structures over fixed workflows [70]. The formation of these structures often mirrors natural phenomena, where rational agents self-organize to connect to common targets. In game-theoretic models, agents activating outgoing edges to form trees demonstrate that self-organization can yield equilibrium structures with a beneficial trade-off between low height and low maximum degree, ensuring fair cost distribution even if the total cost is slightly higher than a centralized optimum [71].

A critical mechanism in these decentralized systems is the dynamic formation of coalitions, allowing agents to group together and operate as a single entity when necessary. This approach balances the performance degradation of purely decentralized control with the practical costs of

full centralization. Coalitional control schemes employ robust distributed model predictive control, where controllers reach a consensus on coalition sets based on consensus theory and potential games. This approach guarantees recursive feasibility and closed-loop stability, provided there is a sufficient dwell time for coalition reconfiguration [72]. Furthermore, in scenarios involving heterogeneous agents with rank-deficient coupling, the system can be transformed into a singularly perturbed form where the emergent “blended dynamics” predict the collective behavior. This allows for plug-and-play operations where the stability of the synthesized system is an outcome of trading off stability among individual agents, rendering the system initialization-free [73].

The trade-off between centralized efficiency and decentralized robustness remains a recurring theme in the literature. While centralized approaches may be more efficient for systems with low interdependence, decentralized approaches significantly outperform them in highly dependent systems where adaptability is crucial. Experimental evaluations suggest that a large number of small-size cooperative groups often improve system performance more effectively than a small number of large-size groups, highlighting the importance of optimal group size identification [74]. Moreover, decentralized systems exhibit inherent resilience against disruptions. For example, algorithms inspired by social dynamics, such as the Deffuant Model, enable efficient consensus formation even in the presence of unknown disruptors by filtering out extremist opinions, a capability often lacking in centralized architectures [75].

However, self-organization is not without its challenges, particularly regarding stability and coordination. The stability of hierarchical networks emerging from self-organization is complex; while stability generally increases with hierarchy and efficiency, these networks can be paradoxically vulnerable to targeted attacks on specific hierarchical nodes [76]. Additionally, the coordination of autonomic functionalities running in parallel requires sophisticated mechanisms to ensure stability, often modeled through control loops and Lyapunov stability analysis to prevent instability in non-linear systems [77]. To mitigate the computational hardness of coordination in decentralized settings, novel methods like Self-Organized Polynomial-time Coordination Graphs (SOP-CG) have been developed. These methods use structured graph classes to guarantee computational efficiency in action selection, bypassing the NP-hard nature of decentralized constraint optimization problems [78].

Ultimately, the evolution of decentralized multi-agent systems points toward architectures that can dynamically reconfigure their interaction networks to match the frequency of environmental changes. Research indicates that optimal collective response often requires a dynamic rewiring of the interaction network, where the number of interacting agents adjusts monotonically with the frequency of the driving signal [79]. As LLM agents increasingly operate in physical and digital swarms, the integration of self-organizing nervous systems allows for the autonomous establishment of multi-level hierarchies, combining the scalability and fault tolerance of swarms with the coordinated decision-making of localized centralization [80]. This synthesis of local autonomy and emergent global order defines the frontier of decentralized agent architectures. However, the efficacy of such self-organizing behaviors is inextricably linked to the quality of information exchange, leading naturally to the exploration of advanced communication protocols and graph-based coordination mechanisms in the following section.

2.4 Communication Protocols and Graph-Based Coordination

2.4 Communication Protocols and Graph-Based Coordination

Building upon the decentralized architectures and self-organizing principles established in the previous section, the practical realization of emergent global order hinges critically on the sophistication of inter-agent communication protocols. While Section 2.3 highlighted how local interactions give rise to stable patterns and dynamic coalitions without centralized control, this subsection details the underlying mechanisms that sustain these distributed societies: the graph-based coordination frameworks that govern information exchange. As agent collectives scale in size and complexity, simple broadcast or pairwise messaging schemes often fail to capture the nuanced dependencies required for synchronized action and coherent planning. Consequently, recent advancements have increasingly turned to treating agents as nodes within a structured topology, where edges represent communication channels that can be dynamically weighted, pruned, or reconfigured based on task requirements and environmental states. By leveraging Graph Neural Networks (GNNs) and heterogeneous attention models, these approaches facilitate the dynamic, context-aware information flow necessary to maintain the structural integrity of self-organizing systems.

A fundamental challenge in graph-based coordination is the design of message aggregation mechanisms that allow agents to prioritize salient information while filtering out noise from less relevant neighbors, a necessity for the stability of the organic hierarchies discussed previously. Vanilla GNNs often rely on simplistic aggregation functions that treat all incoming messages equally, which can lead to performance degradation in large-scale or heterogeneous settings. To address this, advanced attention mechanisms have been integrated into graph architectures. For instance, the Message-Aware Graph Attention Network (MAGAT) introduces a key-query-like mechanism that enables agents to determine the relative importance of features in messages received from neighboring robots [81]. This approach allows for decentralized path planning that rivals centralized expert algorithms, demonstrating that selective attention is vital for maintaining stability and efficiency as system scales increase. Furthermore, theoretical analyses suggest that standard attention mechanisms may fail to distinguish certain graph structures due to the ignorance of cardinality information; thus, incorporating cardinality-preserved attention models can significantly enhance the discriminative capacity of agent communications [82].

In environments where agents possess diverse capabilities, observation spaces, or roles—common in the sophisticated LLM agent societies emerging from self-organization—homogeneous communication protocols are insufficient. Heterogeneous graph architectures have emerged as a robust solution for modeling these complex interactions, bridging the gap between the abstract topologies of Section 2.3 and the concrete capability matching required in Section 2.5. By representing multi-agent systems as directed labeled heterogeneous graphs, where node and edge labels denote agent classes and communication types respectively, systems can learn specialized transformations for messages exchanged between different agent types [83]. Frameworks such as HetNet utilize heterogeneous graph attention networks to learn diverse communication models, ensuring that information sharing is customized according to sender and receiver types, thereby preventing the deterioration of cooperative performance often seen in mismatched teams [84]. This capability is further extended by models like HetGPPO, which leverage GNNs to enable fully decentralized training of heterogeneous policies, allowing agents with distinct physical and behavioral traits to coordinate effectively in partially observable environments [85].

Beyond static heterogeneity, dynamic coordination requires mechanisms that can adapt to changing connectivity and information flow patterns, mirroring the adaptive coalition formation described

earlier. Connectivity-Driven Communication (CDC) exemplifies this by modeling agents as nodes in a weighted graph where state-dependent edges encode pair-wise messages. This framework employs a graph-dependent attention mechanism that accounts for the current system state and utilizes a diffusion process to capture information flow, learning the graph topology concurrently with the policy in an end-to-end fashion [86]. Similarly, adaptive graph structures have been applied to wireless network optimization, where graphs serve as communication-inducing structures to mitigate convergence challenges in multi-agent reinforcement learning, introducing a relational inductive bias that enhances collective decision-making [87].

The integration of temporal dynamics into graph-based coordination is another critical frontier, enabling agents to reason over historical interactions and predict future states, thus reinforcing the resilience of decentralized systems against disruptions. Architectures like the Spectral Temporal Graph Neural Network (SpecTGNN) capture inter-agent correlations and temporal dependencies in both time and frequency domains, utilizing multi-head spatio-temporal attention to mitigate error propagation over long horizons [88]. In navigation tasks involving dynamic crowds, spatial-temporal graphs combined with recurrent neural networks allow robots to infer the future intentions of surrounding agents by leveraging past trajectory information, resulting in more proactive and safe navigation behaviors [89]. Moreover, hierarchical approaches such as Soft Hierarchical Graph Recurrent Networks (HGRN) combine hierarchical graph attention with recurrent units to enable agents to remember historical information while communicating with neighbors, proving effective in large-scale partially observable environments [90].

To further enhance the expressiveness and efficiency of these protocols, researchers are exploring hybrid architectures that combine the global receptive fields of Transformers with the structural inductive biases of GNNs. The TransGNN model, for example, integrates Transformer and GNN layers alternately to broaden the receptive field and disentangle information aggregation, allowing agents to attend to globally relevant nodes while preserving local structural knowledge [91]. Additionally, the concept of a shared global workspace has been proposed to coordinate neural modules through a bandwidth-limited channel, forcing specialist agents to compete for access and thereby encouraging specialization and synchronization [92]. Finally, to prevent the homogenization of strategies where a single “core” agent dominates the collective behavior, regularization techniques such as Normalized Tensor Nuclear Norm Regularization (NTNNR) have been introduced to actively enrich the diversity of message aggregation during training, ensuring robust coordination in both cooperative and competitive scenarios [93]. These advancements collectively establish a foundation for resilient, scalable, and intelligent communication protocols. By defining *how* agents exchange information through structured, adaptive, and heterogeneous topologies, this section sets the stage for the subsequent exploration of *who* participates in these exchanges and how specific combinations of diverse capabilities are orchestrated in heterogeneous teaming.

2.5 Heterogeneous Teaming and Capability Integration

2.5 Heterogeneous Teaming and Capability Integration

Building upon the graph-based coordination mechanisms and dynamic communication protocols established in the previous section, the evolution from homogeneous swarms to heterogeneous multi-agent systems represents a critical architectural shift. While Section 2.4 detailed *how* agents exchange information through structured topologies, this section addresses *who* participates in these exchanges and *why* specific combinations of agents are formed. Moving beyond simple redundancy, heterogeneous teaming leverages synergistic collaboration where diverse capabilities—such as per-

ception, manipulation, navigation, and reasoning—are orchestrated to solve intractable problems that no single agent type can address in totality. Consequently, the core challenge expands from designing efficient message passing to the sophisticated orchestration of heterogeneous teams, necessitating frameworks that effectively match capabilities to task requirements, facilitate dynamic group formation, and integrate specialized agents into a cohesive unit capable of adhering to rigorous constraints.

A foundational aspect of heterogeneous teaming is the accurate modeling and matching of agent capabilities to specific task demands, a prerequisite for the effective graph structures discussed previously. Traditional approaches often rely on manually specified constraints, which fail to capture the nuanced and implicit preferences inherent in complex problem-solving. To address this, recent research has introduced learning-based frameworks that infer trait preferences directly from expert demonstrations. For instance, algorithms have been developed to analyze the consistency with which experts allocate specific traits to tasks, thereby deriving a computationally efficient model of implicit trait importance that outperforms baseline methods treating all capabilities equally [94]. Building on this, unified frameworks like STRATA (Stochastic TRAI-based Task Assignment) model agents and tasks within a continuous trait space, allowing for the reasoning of both species-level and agent-level variability. By categorizing agents into species based on their traits and defining measures of diversity, such systems can compute optimal assignments that satisfy complex, multi-dimensional task requirements in large-scale teams [95]. Furthermore, hierarchical approaches such as ALMA decompose composite tasks into isolated subtasks, simultaneously learning high-level allocation policies and low-level agent policies to encourage the reuse of subtask-specific skills across different team compositions [96].

Beyond static allocation, effective heterogeneous teaming requires dynamic group formation mechanisms that allow agents to self-organize based on emerging needs, directly influencing the topology of the communication graphs described earlier. In open-ended environments, fixed workflows are often insufficient; instead, agents must employ handshake-based or negotiation protocols to form transient coalitions. The concept of “handshake-based group communication” has been successfully applied in embodied settings, where heterogeneous agents with different sensing and actuation capabilities must coordinate to detect and tidy misplaced objects, leveraging their distinct roles through explicit communication signals [97]. Similarly, in the realm of Large Language Model (LLM) agents, frameworks like AutoAgents adaptively generate specialized agents and coordinate them based on task content, effectively creating a dynamic team structure where an observer role reflects on and refines the collaboration strategy [98]. This dynamism extends to reinforcement learning settings where methods like LDSA (Learning Dynamic Subtask Assignment) utilize ability-based selection strategies to dynamically group agents with similar competencies for specific subtasks, balancing training complexity with behavioral diversity [99]. Moreover, self-organizing systems like S-Agents employ a “tree of agents” structure and non-obstructive collaboration methods to autonomously coordinate groups in open-ended environments without human intervention, demonstrating the viability of decentralized team formation [70].

The integration of specialized agents also demands robust conflict resolution mechanisms, particularly when agents possess overlapping or complementary capabilities that may lead to runtime conflicts within the coordinated graph. Advanced controller synthesis frameworks utilizing temporal logic have been proposed to guarantee safety and task completion by redirecting agents with specific capabilities to resolve obstacles encountered by teammates, effectively turning environmental conflicts into opportunities for collaborative assistance [100]. This intersection of capability integration and logical constraint_handling serves as a natural bridge to the formal verification

techniques discussed in the following section. In scenarios involving uncertainty, stochastic programming frameworks optimize task decomposition and scheduling by representing agent capabilities as random distributions, ensuring robust plan generation even when task requirements and agent availabilities are uncertain [101]. Additionally, communication efficiency remains paramount; heterogeneous graph attention networks (HetNet) have been introduced to learn diverse communication models tailored to sender and receiver types, significantly improving coordination performance in composite robot teams by addressing the heterogeneity in action and observation spaces [84].

The human element further complicates heterogeneous teaming, necessitating architectures that can integrate human operators as unique agents within the team, adding another layer of heterogeneity to the system. Frameworks for mixed-initiative teaming employ online Bayesian adaptation to infer human willingness to comply and adjust robot intervention styles accordingly, enhancing both objective performance and subjective trust in suboptimal human-agent pairs [102]. Augmented Reality (AR) interfaces have also been deployed to facilitate command and control of large heterogeneous teams, allowing human supervisors to localize and communicate with dozens of autonomous agents seamlessly, thereby bridging the gap between human strategic oversight and robotic tactical execution [103]. Finally, the optimization of delegation between human and AI agents involves training manager models that learn to authorize actions based on observed performance deficiencies, effectively balancing the load between human intuition and automated precision [104]. Together, these advancements illustrate a maturing field where heterogeneous teaming is not merely about aggregating different agents, but about crafting intelligent, adaptive, and resilient architectures. As these teams grow in complexity and autonomy, the need for the rigorous safety guarantees and temporal consistency provided by formal methods becomes indispensable, setting the stage for the orchestration strategies detailed next.

2.6 Formal Verification and Temporal Logic Orchestration

2.6 Formal Verification and Temporal Logic Orchestration

Building upon the sophisticated heterogeneous teaming architectures and capability integration strategies detailed in Section 2.5, the deployment of Large Language Model (LLM) agents in high-stakes, real-world environments introduces a critical challenge: ensuring that the flexible, probabilistic outputs of generative models adhere to rigorous safety and temporal constraints. While heterogeneous teams effectively match diverse capabilities to complex tasks, the orchestration of these agents requires more than just dynamic group formation; it demands mathematical guarantees against hallucinations, logical inconsistencies, and temporal violations. To bridge the gap between the adaptive synergy of heterogeneous swarms and the deterministic reliability required for safe execution, recent architectures increasingly leverage formal methods, specifically Linear Temporal Logic (LTL) and Behavior Trees (BT). These frameworks provide the essential logical scaffolding that transforms the collaborative potential discussed previously into verifiable, safe operational realities, serving as a necessary precursor to the resilient execution mechanisms explored in the subsequent section.

Linear Temporal Logic (LTL) has emerged as a standard specification language for defining reactive system behaviors, allowing for the precise encoding of complex temporal requirements such as sequencing, liveness, and safety constraints within these multi-agent teams. In the context of LLM agents, LTL serves as a high-level constraint layer that guides or verifies the plans generated by the language model, ensuring that the synergistic actions of heterogeneous agents do not violate global safety policies. A critical challenge in this domain is the computational complexity associated

with synthesizing planners that guarantee goal satisfaction from LTL specifications. To address this, researchers have developed methods to translate goal-oriented LTL formulas directly into executable Behavior Trees. For instance, specific approaches demonstrate how to convert a subset of finite trace LTL into Behavior Trees that guarantee any successful execution trace satisfies the original logical goal [105]. This synthesis allows for a relaxed behavior problem where various planners can implement the action nodes, yet the overall structure ensures formal compliance with the temporal logic specification, enabling agents to handle complex sequential dependencies that pure LLM prompting might miss even in highly coordinated teams.

Beyond simple translation, the integration of LTL into agent architectures facilitates robust planning in dynamic and partially observable environments where the static assumptions of traditional teaming may falter. Traditional planning methods often struggle when environmental changes render a strict LTL specification infeasible, potentially causing the heterogeneous team to stall. Advanced orchestration frameworks address this by incorporating mechanisms for incremental replanning and minimal violation. Algorithms such as LTL-D* have been designed to handle dynamically changing environments by finding optimal replanning solutions that minimally violate desired task specifications when strict satisfaction is impossible [106]. By leveraging distance metrics within synthesized automata, these systems can quantify the degree of task violation and adjust the agent’s trajectory accordingly, ensuring resilience without abandoning the core logical objectives. Furthermore, to handle the inherent uncertainty in real-world event detection common in open-ended domains, novel reinforcement learning frameworks like LAQBL (Learning Action and Query over Belief LTL) have been proposed. These frameworks enable agents to learn policies that account for branching possibilities in LTL instructions caused by uncertain event detectors, simultaneously optimizing action and query policies to avoid task failure [107].

The scalability of LTL-based orchestration is another focal point of current research, particularly for the large-scale multi-agent systems and high-dimensional robotic control scenarios envisioned in heterogeneous teaming. Constructing discrete abstractions for complex systems often leads to state-space explosion, making traditional model checking intractable for large teams. To overcome this, abstraction-free methods utilizing sampling-based algorithms have been introduced. These approaches incrementally build trees to explore the product state-space without requiring explicit discrete abstractions, significantly enhancing scalability while maintaining probabilistic completeness and asymptotic optimality [108]. In multi-robot scenarios, hierarchical temporal logic specifications allow for the decomposition of lengthy, complex formulas into manageable sub-tasks. This hierarchical structure facilitates simultaneous task allocation and planning, where search spaces are approximated by loosely interconnected sub-spaces corresponding to individual LTL specifications, thereby improving execution times and solution quality [109].

Moreover, the fusion of neural networks with formal logic creates neurosymbolic architectures that combine the learning capacity of deep learning with the verifiable guarantees of formal methods, effectively merging the adaptability of LLMs with the rigor of logic. Neurosymbolic frameworks train neural network-based planners that inherit correctness guarantees from symbolic models, effectively using the symbolic component as a discrete memory necessary for satisfying temporal logic formulas [110]. This hybrid approach is crucial for generalizing to unseen tasks where pure meta-reinforcement learning techniques might fail. Additionally, to ensure that LLM-driven agents adhere to safety constraints in collaborative environments, queryable safety constraint modules based on LTL have been developed. These modules enable the encoding of natural language instructions into temporal constraints, perform safety violation reasoning, and prune unsafe actions, thus providing a “safety chip” that enforces compliance with international standards [111].

Finally, the verification of agent behavior extends to runtime monitoring and explainability, which are vital for maintaining trust in autonomous heterogeneous teams. Efficient decentralized monitoring frameworks using tableau techniques allow processes to propagate observations optimally, ensuring that distributed agents adhere to global LTL specifications without excessive communication overhead [112]. Furthermore, the integration of explanations into the learning process helps bridge the gap between human intent and formal specifications. By combining LLMs with optimization-based methods, systems can translate human explanations and demonstrations into robust LTL specifications, enhancing the reliability of agents in safety-critical domains [113]. Through these diverse architectural innovations, formal verification and temporal logic orchestration provide the essential scaffolding for deploying LLM agents that are not only capable but also trustworthy, safe, and logically consistent. However, while formal methods establish the theoretical boundaries of correct behavior, they must be complemented by operational strategies that handle the stochastic disruptions of the physical world, setting the stage for the resilient execution and adaptive recovery mechanisms discussed next.

2.7 Resilient Execution and Adaptive Recovery Mechanisms

2.7 Resilient Execution and Adaptive Recovery Mechanisms

While Section 2.6 established the formal guarantees necessary for safety and temporal consistency through Linear Temporal Logic and Behavior Trees, the practical deployment of Large Language Model (LLM) agents in dynamic, real-world environments demands a complementary layer of operational resilience. Formal verification ensures that a plan is logically sound prior to execution, but it cannot fully anticipate the stochastic disruptions, resource constraints, and unforeseen environmental shifts that occur during runtime. Consequently, resilient execution architectures are designed to maintain system stability when these formal specifications are threatened by reality. These architectures employ adaptive mechanisms such as interleaved task allocation, targeted repair strategies, and stochastic programming frameworks to handle unexpected failures without catastrophic collapse. Drawing from established literature in complex engineering systems, robotics, and distributed control, these patterns provide the essential blueprint for next-generation agentic orchestration that bridges the gap between theoretical correctness and practical robustness.

A foundational element of resilient agent architecture is the capability for **interleaved execution**, which allows an agent to simultaneously address multiple layers of a problem—such as coalition formation, scheduling, and motion planning—while avoiding incompatibilities between them. In static architectures, these layers are often solved sequentially, leading to suboptimal solutions when dynamic changes occur; however, resilience requires a tightly coupled loop where high-level planning and low-level tool invocation inform one another continuously. The Dynamic Iterated Task Allocation Graph Search (D-ITAGS) framework exemplifies this approach by enabling effective search solutions at each layer while maintaining consistency across the system [114]. For LLM agents, this translates to an architecture where, upon encountering a failure in tool execution, interleaved mechanisms allow the agent to instantly re-evaluate the planning layer without discarding the entire trajectory, thereby preserving computational effort and maintaining mission continuity even when strict temporal sequences defined in Section 2.6 become temporarily infeasible.

Closely related to interleaved execution is the concept of **targeted repair**, a strategy designed to identify and rectify only the specific components of a solution impacted by a disruption, rather than recomputing the entire plan from scratch. This is critical for agents operating in latency-sensitive environments where full re-planning is computationally prohibitive and where restarting a formally

verified sequence would be inefficient. Research into resilient task allocation demonstrates that targeted repair can significantly accelerate recovery times with negligible loss in solution quality compared to full recomputation [114]. In the context of LLM agents, targeted repair manifests as localized context updating, where the agent isolates the failed step in its reasoning chain and invokes specific corrective sub-agents or tools, leaving the valid portions of the plan intact. This mirrors approaches in software self-healing, where pattern-based adaptation rules resolve issues incrementally to maintain optimal utility without expensive global optimization [115].

To manage the inherent uncertainty of external environments that often defies deterministic modeling, resilient agents increasingly rely on **stochastic programming frameworks**. Unlike deterministic models that assume a fixed world state, stochastic approaches explicitly model the probability distributions of potential disruptions, such as sensor failures, communication loss, or fluctuating resource availability. For instance, stochastic optimization techniques have been successfully applied to restoration frameworks in complex engineering systems to handle the temporal sparsity of decision-making during extreme events [116]. By integrating similar stochastic models, LLM agents can generate policies that are robust to a range of possible futures rather than optimizing for a single expected outcome. This is particularly relevant for agents managing distributed resources, where stochastic coded offloading schemes have been shown to minimize costs and energy consumption amidst weather and demand uncertainties [117].

Furthermore, the integration of **Markov Decision Processes (MDPs)** and **Partially Observable MDPs (POMDPs)** provides a rigorous mathematical foundation for agents to make sequential decisions under the very uncertainties that formal logic struggles to quantify dynamically. These models allow agents to account for the long-term consequences of immediate actions, incorporating “lookahead” capabilities that myopic approaches lack [118]. In scenarios where the agent has incomplete information about the environment—a common challenge when interacting with unpredictable human partners or noisy sensors—POMDP-based frameworks enable the derivation of optimal policies by maintaining a belief state over possible world configurations [119]. Advanced implementations, such as those utilizing deep decentralized multi-agent actor-critic methods, allow systems to learn policies that intrinsically consider system-level effects and probabilistic dependencies among components [120].

Resilience also requires mechanisms for **graceful degradation and recovery**, ensuring that an agent can maintain a safe, albeit reduced, level of functionality when faced with severe constraints or attacks that render full task satisfaction impossible. A requirement-driven adaptation framework treats degradation as the temporary weakening of system requirements and recovery as their subsequent strengthening, allowing for automated coordination between these dual operations [121]. For LLM agents, this implies the ability to dynamically switch to simpler, more reliable tools or heuristic strategies when complex reasoning paths become unreliable due to context window limits or API failures, effectively acting as a fail-safe when the formal constraints of Section 2.6 cannot be strictly met. Additionally, the use of **risk-averse optimization** ensures that agents prioritize avoiding catastrophic failures over maximizing average rewards, a crucial trait for safety-critical applications [116].

Finally, the scalability of resilient mechanisms in large-scale multi-agent systems often relies on **distributed and approximate dynamic programming (ADP)** techniques. These methods address the curse of dimensionality inherent in large state spaces by decomposing problems into manageable sub-problems and utilizing heuristic approximations to identify near-optimal restoration decisions [122]. Such approaches enable communities of agents to coordinate recovery efforts across interdependent infrastructure networks, balancing local autonomy with global stability [123]. By

synthesizing these diverse architectural patterns—interleaved execution, targeted repair, stochastic modeling, and distributed optimization—the field is moving toward LLM agent systems that are not only intelligent and formally verified but also fundamentally resilient to the unpredictability of the real world. This operational robustness serves as a critical prerequisite for the subsequent challenge of integrating these agents into mixed teams with humans, where adaptability and trust become paramount, as discussed in the following section.

2.8 Human-Agent Collaboration and Mixed-Team Architectures

2.8 Human-Agent Collaboration and Mixed-Team Architectures

Building upon the operational resilience mechanisms detailed in Section 2.7, which equip agents to handle stochastic disruptions and environmental uncertainties autonomously, the next critical frontier lies in integrating these robust agents into cohesive mixed teams with humans. While resilient execution ensures system stability against technical failures, mixed-team architectures must address the even more complex variability of human behavior, including fluctuating cognitive states, emotional factors, and the paramount need for trust preservation. Unlike homogeneous multi-agent systems or purely autonomous setups, these architectures require a shift from static, pre-defined role assignments to dynamic frameworks capable of real-time negotiation and adaptive role allocation. These systems operate on a spectrum of autonomy, ranging from shared control paradigms to decentralized cooperative models where humans and agents negotiate task ownership based on evolving contextual constraints, effectively extending the concept of “graceful degradation” to include human-centric adaptability.

A foundational element of effective mixed-team architectures is the mechanism for dynamic role allocation and task planning, which mirrors the interleaved execution strategies discussed previously but incorporates human performance metrics. Traditional approaches relying on offline optimization often fail to accommodate the unpredictable variability of human operators. To address this, recent architectures have adopted reactive, modular planning methods grounded in Behavior Trees (BTs) combined with online optimization solvers. For instance, novel frameworks utilize centralized reactive planning based on Behavior Trees to manage action scheduling, while simultaneously solving role allocation problems through Mixed-Integer Linear Programming (MILP) [65]. This hybrid approach allows the system to dynamically assign individual roles or collaborative tasks to team members based on fluctuating metrics such as makespan, ergonomics, and explicit human preferences. Furthermore, to handle the complexity of large-scale industrial applications, some systems decompose the allocation problem into simplified online sub-problems, enabling rapid re-planning without the computational burden of cross-schedule dependencies [124], thereby applying the principle of targeted repair to human-robot interaction flows.

Beyond mere efficiency, successful mixed-team architectures must incorporate human preferences and psychological states into the decision-making loop, recognizing that ignoring factors such as fatigue, stress, or fairness can lead to team dysfunction and rejection of the automated system. Architectures designed for “Factories of the Future” increasingly employ bi-level optimization strategies that consider not only task execution time but also job quality and human satisfaction [125]. In these systems, the robot actively monitors human activities and dynamically reschedules tasks to mitigate ergonomic risks and accommodate learning curves [126]. The concept of fairness has also been formalized within these architectures; for example, deep reinforcement learning approaches have been developed to learn allocation policies that maximize pick efficiency while ensuring workload fairness among human pickers in warehouse environments [127]. Additionally, research indicates

that humans are more willing to join and collaborate with robot teams that demonstrate “legible” and fair subtask allocations, even if such allocations result in slightly lower overall efficiency [128], highlighting that optimal team performance is a function of both utility and social acceptance.

Negotiation and mutual adaptation serve as critical protocols in advanced mixed-team architectures, moving beyond unilateral command structures to bidirectional influence, akin to the decentralized coordination seen in resilient multi-agent systems. In scenarios where humans and agents possess asymmetric capabilities or incomplete knowledge, architectures employing online Bayesian adaptation have proven effective. These systems allow robots to infer a human partner’s willingness to comply with assistance and adjust their intervention styles accordingly, significantly enhancing both objective team performance and subjective measures of trust [102]. Similarly, game-theoretic models have been integrated to capture the dynamics of human partial adaptation, where the robot optimally balances revealing its capabilities against taking immediate optimal actions based on the human’s current mental model [129]. This mutual adaptation extends to shared autonomy frameworks, where the agent guides the human toward effective strategies while retaining the flexibility to comply with human preferences to maintain trust when the human is unadaptable [130].

Communication interfaces and semantic interaction layers are also pivotal in these architectures, serving as the bridge between high-level planning and human intent, much like the semantic layers required for tool learning in LLMs. Augmented Reality (AR) interfaces have been successfully deployed to facilitate bilateral communication, allowing human agents to accept or refuse assigned tasks and providing intuitive coordination cues during complex assembly phases [65]. In more uncertain environments, frameworks like HARPS (Human Assisted Robotic Planning and Sensing) combine online sampling-based POMDP policies with multimodal semantic interaction, enabling humans to opportunistically impose model structure on the environment through sketching and labeling, which the robot then uses to refine its belief states and planning strategies [131]. Furthermore, the integration of Large Language Models (LLMs) into preference-based reinforcement learning has emerged as a powerful tool for reducing the sample complexity of aligning robot behaviors with human values, allowing agents to reason from text prompts to infer complex social preferences [132].

Finally, the architectural robustness of mixed teams often relies on sophisticated modeling of human state and intent, leveraging the same probabilistic frameworks used for environmental uncertainty in Section 2.7. Architectures utilizing heterogeneous graph neural networks and recurrent neural schedule propagators have demonstrated superior performance in scheduling stochastic human-robot teams by effectively modeling team dynamics and temporal constraints [133]. Other approaches leverage dynamic Bayesian networks to construct trust models that integrate robot performance, safety coefficients, and human cognitive workload, using these trust values to guide symbolic motion planning and task allocation paths [134]. By synthesizing dynamic role allocation, preference-aware optimization, negotiation protocols, and advanced human-state modeling, modern mixed-team architectures are evolving into resilient, adaptive systems capable of thriving in complex, unstructured environments alongside human partners, setting the stage for the broader societal and ethical implications of deploying such autonomous collaborators.

3 Cognitive Mechanisms: Reasoning, Planning, and Memory

3.1 Advanced Chain-of-Thought Variations and Iterative Reasoning

3.1 Advanced Chain-of-Thought Variations and Iterative Reasoning

The evolution of reasoning strategies in Large Language Models (LLMs) has progressed significantly beyond the foundational Chain-of-Thought (CoT) paradigm. While standard CoT prompting enables models to decompose complex problems into linear sequences of intermediate steps, it often struggles with long-horizon tasks, hallucinations, and the rigidity of sequential logic. To address these limitations, recent research has introduced advanced variations that integrate external knowledge retrieval, algorithmic structures, and non-linear graph-based thinking. These enhanced strategies aim to mimic human cognitive flexibility, allowing agents to verify facts, explore multiple solution paths, and correct errors dynamically during the inference process. This section explores how these iterative and structured reasoning mechanisms serve as a critical precursor to more formalized search algorithms, bridging the gap between simple prompting and complex planning architectures.

A critical advancement in this domain is the integration of retrieval mechanisms directly into the reasoning loop to mitigate hallucinations and ground arguments in factual evidence. Traditional Retrieval-Augmented Generation (RAG) typically retrieves information once before generation begins, which can be insufficient for multi-step reasoning where new questions arise at each step. The Retrieval-Augmented Thoughts (RAT) framework addresses this by iteratively revising each thought step with information retrieved relevant to the task query, the current thought, and past reasoning steps. [135] demonstrates that applying this iterative revision process substantially improves performance on long-horizon generation tasks, including code generation, mathematical reasoning, and embodied task planning, while hugely mitigating hallucination. Similarly, the Retrieval-Augmented Thought Process (RATP) formulates thought generation as a multiple-step decision process, leveraging Monte-Carlo Tree Search to optimize the retrieval and reasoning trajectory. [136] highlights that this approach is particularly effective in scenarios involving private data where ethical constraints limit model training, achieving significant improvements over existing in-context methods. Furthermore, the concept of adaptive retrieval has gained traction, where models learn to retrieve information only when necessary. The Rowen approach, for instance, employs a multilingual semantic-aware detection module to evaluate response consistency; upon detecting inconsistencies indicative of hallucinations, it activates external retrieval to rectify outputs. [137] This selective mechanism ensures a balanced integration of parametric knowledge and external evidence, reducing the risk of introducing irrelevant information that could lead to external hallucinations.

Beyond retrieval integration, researchers have developed frameworks that structure reasoning as algorithmic pathways rather than simple linear chains. The Algorithm of Thoughts (AoT) represents a novel strategy that propels LLMs through algorithmic reasoning pathways by employing algorithmic examples within the prompt. [138] This method exploits the innate recurrence dynamics of LLMs, enabling them to expand their idea exploration with minimal query overhead. Intriguingly, results suggest that instructing an LLM using an algorithm can lead to performance surpassing that of the algorithm itself, hinting at the model’s ability to weave intuition into optimized searches. This contrasts with methods that require extensive external search procedures, offering a more computationally efficient alternative for complex problem solving. Another variation, Logical Thoughts (LoT), leverages principles rooted in symbolic logic, specifically *Reductio ad Absurdum*, to systematically verify and rectify reasoning processes step by step. [139] By constraining reasoning with logical principles, LoT effectively reduces the unconstrained hallucinations

often observed in standard CoT approaches across diverse domains such as arithmetic and causal inference.

The limitations of linear reasoning have also spurred the development of graph-based and non-linear thinking paradigms, which naturally extend the capabilities of linear CoT into more flexible topologies. Human thought processes are rarely strictly sequential; they involve branching, merging, and revisiting previous ideas. The Graph-of-Thought (GoT) framework models these processes by representing thought units as nodes and their dependencies as edges in an arbitrary graph. [140] This structure allows for the combination of arbitrary thoughts into synergistic outcomes and the distillation of entire networks of information, offering advantages over linear chains in tasks requiring elaborate problem solving. Extending this concept to multimodal contexts, the Aggregation-Graph-of-Thought (AGoT) mechanism models each reasoning step as an aggregation graph to cope with multiple aspects of thinking simultaneously. [141] This approach has shown strong domain generalization performance in tasks like visual question answering. Furthermore, the Hierarchical Graph of Thoughts (HGOT) utilizes a divide-and-conquer strategy to break down complex queries into manageable sub-queries, refining self-consistency majority voting by incorporating citation quality metrics. [142] This hierarchical structure significantly enhances the factuality of LLMs in retrieval-augmented settings.

Iterative refinement and self-correction are also central to these advanced variations, providing the mechanism for agents to recover from errors without external intervention. The Boosting of Thoughts (BoT) framework iteratively explores and self-evaluates many trees of thoughts to acquire an ensemble of trial-and-error reasoning experiences. [143] By using error analysis to explicitly revise prompting, BoT enhances reasoning step generation until a final answer is attained. Similarly, the THOUGHTSCULPT method introduces revision actions within a search tree, allowing the model to revise parts of its previous output rather than merely continuing forward. [144] This capability to backtrack and edit intermediate states is crucial for solving tasks where early errors can propagate catastrophically in linear models. Additionally, the State Machine of Thoughts (SMoT) leverages past reasoning trajectories by recording successful and failed states in a state machine, enabling the agent to select optimal sub-solutions for analogous future problems. [145]

These advanced variations collectively represent a shift from static, one-pass reasoning to dynamic, iterative, and grounded cognitive processes. By integrating external knowledge sources like knowledge graphs [146] and employing sophisticated search algorithms, these methods address the core weaknesses of standalone LLMs. Whether through the algorithmic precision of AoT, the factual grounding of RAT, or the structural flexibility of GoT, these innovations are essential for building robust agents capable of handling the complexity and uncertainty of real-world applications. While these techniques introduce structured thinking and iterative refinement within the generation process, they often rely on heuristic guidance or limited search depths. Consequently, to fully navigate complex, multi-step decision spaces with rigorous optimality guarantees, these reasoning frameworks must be coupled with dedicated tree-search algorithms, a transition explored in the following section.

3.2 Tree-Search Algorithms for Planning and Decision Making

3.2 Tree-Search Algorithms for Planning and Decision Making

Building upon the iterative and structured reasoning mechanisms discussed in Section 3.1, which introduce dynamic cognitive processes but often rely on heuristic guidance or limited search depths, this section explores the transition to formalized **Tree-Search Algorithms**. While advanced

Chain-of-Thought variations enable models to branch and revise thoughts, they lack the rigorous optimality guarantees and systematic exploration capabilities required for navigating complex, multi-step decision spaces. The integration of Monte Carlo Tree Search (MCTS) with Large Language Models (LLMs) addresses this gap, representing a pivotal advancement in enabling agents to balance exploration and exploitation through recursive rollouts. Although LLMs possess vast parametric knowledge, they frequently suffer from hallucinations and struggles with deep sequential reasoning when acting as standalone policy generators. MCTS, a heuristic search algorithm renowned for its structured planning, offers a robust framework to mitigate these limitations. By coupling the generative capabilities of LLMs with MCTS, researchers have developed systems where the LLM serves dual roles: acting as a policy guide to prune the action space and functioning as a value function to evaluate the quality of simulated trajectories without necessitating additional model training [147].

In this synergistic architecture, the LLM effectively replaces traditional hand-crafted heuristics. The model provides a commonsense prior belief about the world, which informs the MCTS expansion phase, while its policy output acts as a heuristic to direct the search toward promising branches, vastly improving computational efficiency [148]. This approach is particularly effective in domains requiring long-horizon planning, such as software engineering and web navigation. For instance, the Language Agent Tree Search (LATS) framework unifies reasoning, acting, and planning by employing LLMs as agents, value functions, and optimizers within an MCTS loop. Crucially, LATS leverages external environmental feedback to refine the search tree, allowing the agent to adaptively correct errors and achieve superior performance in programming and browsing tasks compared to standard chain-of-thought prompting [149]. Furthermore, recent innovations have demonstrated that MCTS can be guided by lightweight energy functions derived from the LLM itself to rank decision steps, significantly boosting mathematical reasoning capabilities on benchmarks like GSM8k without requiring further fine-tuning or reinforcement learning from human feedback [150].

The theoretical underpinnings of integrating MCTS with neural models have also seen significant refinement to address convergence and optimality concerns inherent in these large search spaces. Traditional MCTS relies on the Upper Confidence Bound for Trees (UCT) selection policy, which balances exploration and exploitation based on visit counts and value estimates. However, in the context of LLMs, where the action space is vast and the cost of simulation is high, standard UCT may be suboptimal. Research has shown that the suboptimality of estimated values in LLM-guided MCTS scales with the cardinality of the pruned action space and the errors incurred by the LLM acting as a pruner and critic [147]. To address this, various modifications to the backup and selection mechanisms have been proposed. For example, the use of power mean operators in backup strategies, known as Power-UCT, has been shown to yield more accurate node value estimates than simple averaging, accelerating convergence in complex Markov Decision Processes [151]. Additionally, novel backup operators utilizing Wasserstein barycenters have been introduced to propagate uncertainty more effectively in stochastic and partially observable environments, ensuring that the search remains robust under uncertainty [152].

Beyond standard tree structures, the evolution of MCTS for LLM agents extends to continuous domains and graph-based representations to handle more complex environmental dynamics. Continuous Monte Carlo Graph Search (CMCGS) addresses the challenge of high branching factors in continuous state spaces by clustering similar states into stochastic action bandit nodes, forming a layered directed graph rather than a strict tree. This approach allows for the sharing of action policies across similar states, enhancing sample efficiency in continuous control tasks [153].

Similarly, in scenarios involving decentralized cooperation, such as autonomous driving, decoupled UCT variants enable agents to evaluate action values in a distributed manner while respecting the interdependence of actions among multiple participants [154]. These adaptations are critical for embodied agents that must operate in physical worlds with continuous dynamics, bridging the gap between abstract planning and concrete execution.

The efficiency of the search process is further ameliorated by dynamic resource allocation and adaptive sampling strategies, ensuring that computational budgets are utilized optimally. Methods such as AmEx-MCTS decouple value updates from visit count updates to prevent the algorithm from wasting computational resources on re-evaluating already explored regions, thereby facilitating a broader search coverage [155]. Moreover, the integration of temporal difference learning into MCTS, as seen in MCTS-TD, allows for online adaptivity without the need for extensive pre-training, enabling agents to adjust their strategies rapidly in non-stationary environments [156]. The concept of “value of computation” has also been revisited to explicitly quantify the impact of specific simulations on the final decision quality, moving beyond myopic exploration strategies to optimize the entire planning budget [157].

Finally, the application of MCTS extends to self-improvement loops where LLMs generate their own training data, creating a virtuous cycle of capability enhancement. Frameworks like AlphaLLM utilize MCTS to create a self-improving cycle where the model imagines solutions, searches for optimal paths, and critiques its own outputs to generate high-quality data for subsequent refinement. This approach addresses data scarcity and the subjective nature of feedback in language tasks, demonstrating that search algorithms can drive autonomous capability enhancement [158]. Similarly, in text generation, value-guided decoding algorithms like PPO-MCTS integrate value networks from reinforcement learning to guide the decoding process, reducing the mismatch between training objectives and inference-time generation to produce more preferable text [159]. Collectively, these advancements illustrate that tree-search algorithms are foundational components for transforming static LLMs into dynamic, reasoning-capable agents. However, while these algorithms excel at exploring decision spaces, their efficacy in long-horizon tasks is often contingent upon the granularity of the action space. This necessity leads directly to the topic of **Hierarchical Decomposition and Subtask Management**, discussed in the following section, which bridges high-level strategic planning enabled by tree search with low-level tool execution.

3.3 Hierarchical Decomposition and Subtask Management

3.3 Hierarchical Decomposition and Subtask Management

While tree-search algorithms discussed in the previous subsection provide a robust mechanism for exploring decision spaces and evaluating trajectories, their efficacy in long-horizon tasks is often contingent upon the granularity of the action space. This necessitates **Hierarchical Decomposition and Subtask Management**, a cognitive process that bridges high-level strategic planning with low-level tool execution. Unlike the static organizational structures of agent architectures, hierarchical decomposition dynamically structures task plans to ensure “subtask-tool atomicity alignment.” This alignment is critical: without it, agents risk generating subtasks that are either too abstract to be executable by specific tools or too granular to leverage the LLM’s high-level reasoning capabilities, leading to planning failures despite sophisticated search mechanisms.

Drawing inspiration from human cognition, which naturally employs hierarchical structures to trade off utility and computational cost, recent research aims to replicate this efficiency in LLM agents. Humans utilize heuristics such as betweenness centrality to identify optimal subgoals that bridge

initial states and final objectives [160]. Translating this normative framework to artificial agents involves developing mechanisms that infer these latent structures without explicit supervision. For instance, frameworks like ProTIP address the limitations of rigid, single-step decomposition by implicitly performing task breakdown through contrastive learning. This approach not only maintains subtask-tool atomicity as the toolbox evolves but also significantly improves recall and plan generation accuracy compared to standard decomposition methods [161]. This implicit decomposition serves as a vital precursor to the dynamic retrieval strategies discussed in the following section, ensuring that the agent understands the sequential logic required before attempting to fetch specific tools.

A central challenge in this domain is discovering subtask boundaries within unstructured demonstrations or open-ended queries. Traditional top-down approaches often enforce decomposition based on predefined rules, limiting flexibility in novel environments. In contrast, bottom-up frameworks leverage diverse world models to automatically decompose complex solutions into modular subpolicies. The Model Primitive Hierarchical Lifelong Reinforcement Learning framework exemplifies this by concurrently learning modular subpolicies and a coordinator controller, enabling effective task decomposition from a single source task [162]. Similarly, the Ordered Memory Policy Network (OMPN) demonstrates the ability to discover subtask hierarchies by learning from demonstrations, effectively recovering subtask boundaries in unstructured data and achieving superior performance in both unsupervised and weakly supervised settings [163]. These methods highlight a paradigm shift towards data-driven discovery of hierarchical structures, moving away from reliance on human-engineered prompts.

The robustness of hierarchical decomposition is further enhanced when agents can learn parameterized task structures that generalize to unseen entities. Real-world tasks are often compositional, involving subtasks defined by entities that can be recombined. The Parameterized Subtask Graph Inference (PSGI) method models these dependencies using first-order logic, allowing agents to infer subtask orderings and generalize them to new entities and unseen tasks [164]. This capability is essential for managing subtasks in dynamic environments where specific tool parameters or object attributes may change. Furthermore, the integration of natural language guidance facilitates the automatic construction of task-specific planning representations. The Ada (Action Domain Acquisition) framework interacts with language models to learn libraries of high-level action abstractions, aligning low-level controllers with high-level goals to produce more accurate and generalizable plans [165].

In multi-agent contexts, hierarchical decomposition extends to the allocation of subtasks among specialized agents, mirroring the distributed nature of complex problem-solving. The ALMA framework illustrates how sophisticated coordination can be achieved by decomposing decision-making into high-level subtask allocation and low-level agent policies. This structural inductive bias reduces observation spaces and encourages the reuse of subtask-specific policies, leading to advanced coordination behaviors in composite tasks [96]. Similarly, the OKR-Agent system utilizes a hierarchical structure of Objects and Key Results to drive self-collaboration, where high-level objectives are recursively decomposed into sub-objects assigned to specific agents, ensuring that strategic planning is tightly coupled with detailed execution [166].

However, the efficacy of these hierarchical approaches depends heavily on the quality of the decomposition and the management of inter-subtask dependencies. Benchmarks such as STEPS reveal that many current models struggle with order reasoning in sequential tasks, often failing to determine the rationality of action sequences without extensive tuning [167]. To mitigate this, researchers have explored script generation tasks that explicitly incorporate subgoals, finding that

while discovering subgoals is challenging, their inclusion significantly improves the quality of generated procedures [168]. Moreover, the concept of “procedures as programs” offers a formalism for representing hierarchical procedural knowledge, where a planner converts natural language intents into executable programs composed of nested sub-tasks, outperforming reactive baselines in data efficiency and iterative development [169].

Ultimately, successful subtask management requires a synergy between high-level planning and low-level execution feedback, creating a closed loop that informs both the search strategies of Section 3.2 and the retrieval mechanisms of Section 3.4. Hierarchical Reinforcement Learning (HRL) frameworks, such as Hierarchical Actor-Critic (HAC), address the instability of joint policy learning by training hierarchy levels independently, assuming lower-level policies are optimal during higher-level training. This method has proven effective in learning multi-level hierarchies in parallel, accelerating the solution of sequential decision-making tasks [170]. Additionally, the emergence of hierarchical modularity in neural networks, uncovered through pruning and analysis, suggests that deep networks naturally organize sub-functions into reusable hierarchies, providing a biological inspiration for artificial agent architectures [171]. By aligning these cognitive mechanisms with tool capabilities, LLM agents achieve the robustness and flexibility required for complex, real-world problem solving, setting the stage for precise tool retrieval and context grounding.

3.4 Dynamic Tool Retrieval and Context Grounding

3.4 Dynamic Tool Retrieval and Context Grounding

Building upon the hierarchical subtask structures established in Section 3.3, where complex objectives are decomposed into atomic units aligned with tool capabilities, the subsequent critical challenge lies in dynamically retrieving and grounding the specific tools required to execute these subtasks. While early paradigms often relied on static, one-time retrieval strategies—attempting to identify all necessary tools based solely on an initial user query—such approaches frequently falter when faced with ambiguous intents or evolving task requirements inherent in long-horizon planning. The transition toward **Dynamic Tool Retrieval and Context Grounding** represents a pivotal evolution in agent architecture, shifting from passive lookup mechanisms to active, iterative processes that refine tool selection through continuous planning and contextual analysis. This subsection explores advanced mechanisms that optimize tool selection, focusing on progressive retrieval techniques and planning-based paradigms that dynamically shortlist relevant tools, thereby serving as the essential bridge between the abstract subgoals defined previously and the concrete memory management systems discussed in the following section.

A primary limitation of traditional retrieval methods is their inability to handle inter-tool dependencies or decompose complex queries into actionable subtasks without explicit supervision, a gap that hierarchical decomposition alone cannot fully close without precise tool matching. To address this, recent research has introduced progressive retrieval frameworks that implicitly perform task decomposition while maintaining atomicity between subtasks and tools. For instance, the ProTIP framework demonstrates that progressive tool retrieval can significantly outperform single-step retrieval and explicit task decomposition methods. By employing a lightweight, contrastive learning-based approach, ProTIP manages to shortlist relevant tools without requiring explicit subtask labels, thereby achieving substantial improvements in recall and tool accuracy for plan generation [161]. This suggests that dynamic retrieval is not merely about filtering a list but involves an implicit understanding of the sequential logic required to execute a plan, effectively operationalizing the hierarchical structures discussed in the preceding section.

Furthermore, the quality of tool retrieval is heavily dependent on the richness of the context provided to the retrieval engine. Standard semantic search often fails when user queries are incomplete or lack the specific contextual cues necessary to distinguish between similar tools, particularly when the agent must operate within the constraints of limited working memory. To mitigate this, context tuning mechanisms have been proposed to augment the retrieval process with additional signals. The concept of Context Tuning for RAG introduces a smart context retrieval system that fetches relevant information—such as numerical, categorical, and habitual usage signals—to enhance both tool retrieval and subsequent plan generation. Empirical results indicate that such context augmentation can lead to significant improvements in recall metrics and reduce hallucination during the planning phase, proving that grounding tools in a richer contextual window is essential for robust agent performance [172]. This enrichment of context acts as a precursor to the virtual context management strategies detailed in Section 3.5, ensuring that the information fed into retrieval engines is both pertinent and structured.

Beyond simple context expansion, advanced architectures now incorporate explicit planning modules that interact with retrieval systems in a closed loop. The PLUTO approach exemplifies this by integrating a “Plan-and-Retrieve” paradigm, where a neural retrieval module shortlists tools while an LLM-based query planner simultaneously decomposes complex queries. This dual-process ensures that the retrieval step is guided by a structural understanding of the task rather than just keyword matching. Additionally, PLUTO introduces an “Edit-and-Ground” paradigm, utilizing LLMs to dynamically enrich tool descriptions based on specific user scenarios. This bridging of the gap between abstract user queries and concrete tool functionalities allows for more precise grounding, particularly in scenarios where tool documentation may be generic or misaligned with the user’s specific intent [173]. Such dynamic editing mechanisms are crucial for maintaining coherence as agents navigate the vast information spaces that will be addressed by memory augmentation techniques in the next section.

The challenge of grounding is further compounded when dealing with large-scale tool libraries containing both seen and unseen tools, a scenario where static ranking mechanisms often fail to adapt to the hierarchical nature of tool libraries or the novelty of specific tools. ToolRerank addresses these issues by proposing an adaptive, hierarchy-aware reranking method. This approach utilizes adaptive truncation to handle seen and unseen tools differently and employs hierarchy-aware reranking to ensure that retrieval results are concentrated for single-tool queries while maintaining diversity for multi-tool scenarios. By dynamically adjusting the retrieval strategy based on the query type and tool visibility, ToolRerank enhances the quality of the shortlisted tools, leading to better execution outcomes [174].

Efficiency in dynamic retrieval is also paramount, especially under strict budget constraints where the number of tool calls must be minimized to prevent cognitive overload and excessive latency. Recent work on budget-constrained tool learning highlights the necessity of formulating a feasible plan before tool utilization. This involves estimating the usefulness of candidate tools based on past experience and employing dynamic programming to allocate the budget effectively. Such planning-based retrieval ensures that the agent does not exhaust its resources on low-utility tools, thereby optimizing the overall decision-making process [175]. This focus on resource optimization foreshadows the computational efficiency goals central to the memory management systems discussed subsequently.

Moreover, the integration of retrieval with generative capabilities allows for more sophisticated grounding. In non-knowledge-intensive tasks, where the relevance of tools may not be immediately obvious from the query alone, prompt-guided retrieval augmentation has shown promise. By using

a task-agnostic retriever to build a shared index followed by a prompt-guided reranker, systems can dynamically adjust relevance scores based on task-specific contexts. This two-stage framework effectively balances the trade-off between training costs and task performance, ensuring that the retrieved tools are not only semantically similar but functionally appropriate for the specific generation task at hand [176].

In summary, dynamic tool retrieval and context grounding have evolved from simple semantic matching to complex, multi-stage processes involving planning, context tuning, and adaptive reranking. By leveraging progressive retrieval techniques that implicitly decompose tasks and grounding mechanisms that enrich tool descriptions with situational context, modern LLM agents can achieve higher precision and robustness. These advancements collectively enable agents to navigate large toolspaces efficiently, adapting their retrieval strategies to the nuances of user intent and environmental constraints. Crucially, this dynamic retrieval layer provides the structured, context-rich input necessary for the advanced memory augmentation and virtual context management systems explored in Section 3.5, ensuring that agents can sustain coherent operation over extended horizons without succumbing to information loss or context fragmentation.

3.5 Memory Augmentation and Virtual Context Management

3.5 Memory Augmentation and Virtual Context Management

While the preceding section detailed mechanisms for dynamically retrieving and grounding external tools, the efficacy of these processes ultimately relies on the agent’s capacity to maintain and manipulate vast amounts of information over time. The transition from static language models to dynamic agents has exposed a fundamental bottleneck: the fixed context window of Large Language Models (LLMs). Even with optimal tool selection, agents struggle with long-horizon tasks, extended conversations, and comprehensive document analysis if they cannot retain information beyond a specific token limit. To overcome these architectural constraints, recent research has pivoted toward memory augmentation and virtual context management, drawing heavy inspiration from operating system (OS) principles to create the illusion of infinite memory through intelligent data movement and hierarchical structuring. This evolution is critical for supporting the complex, multi-step planning strategies discussed in the subsequent section, as robust memory systems provide the necessary state persistence for efficient search and decision-making.

A primary paradigm in this domain is the adaptation of OS-inspired virtual memory management for LLMs. Just as traditional operating systems swap data between fast RAM and slower disk storage to handle processes larger than physical memory, new frameworks propose managing LLM context through tiered memory systems. The seminal work [177] introduces a system that intelligently manages different memory tiers, treating the LLM’s limited context window as a register or cache while utilizing external storage for long-term memory. This approach allows the agent to dynamically load relevant historical context, retrieve documents, and maintain state across multi-session interactions, effectively decoupling the agent’s cognitive scope from the model’s native token limit. By implementing interrupts and system calls, such architectures enable the agent to control its own memory flow, deciding when to evict irrelevant information to slow storage and when to recall critical details into the active context window, thereby ensuring that the context provided to retrieval engines remains pertinent and uncluttered.

Complementary to hierarchical tiering are techniques that modify the attention mechanism itself to process inputs exceeding standard limits without requiring full model retraining. Methods such as Parallel Context Windows (PCW) attempt to alleviate context restrictions by carving

long inputs into chunks and restricting attention to apply only within each window while reusing positional embeddings. This allows off-the-shelf LLMs to process sequences significantly longer than their training configuration [178]. However, the efficacy of such parallel approaches is not universal; subsequent analysis has highlighted limitations, noting that while PCW may assist in retrieval tasks, it can lead to deterioration in complex Chain-of-Thought reasoning due to question miscomprehension and false inference when context is fragmented [179]. This underscores the necessity for memory mechanisms that preserve global coherence rather than simply concatenating isolated windows, a requirement that becomes increasingly stringent as agents engage in the deep deliberative planning explored later.

To address the scalability issues of quadratic attention complexity in long contexts, researchers have explored recurrent memory augmentation. Approaches like Recurrent Memory Transformer (RMT) integrate recurrent units into pre-trained transformers, enabling linear scaling of compute while storing information in memory for sequences reaching up to two million tokens [180]. Similarly, innovations in positional encoding, such as LongRoPE, have extended context windows to over two million tokens by exploiting non-uniformities in positional interpolation and employing progressive extension strategies, allowing models to maintain performance on short contexts while mastering extreme lengths [181]. Furthermore, contrastive training techniques like the Focused Transformer (FoT) have been developed to mitigate the “distraction issue,” where an abundance of irrelevant keys in a large memory pool dilutes attention; FoT enhances the structure of the key-value space to ensure the model focuses on semantically relevant information even as the context scales [182]. These architectural advancements directly support the computational efficiency goals of modern agents by reducing the overhead associated with processing extensive historical trajectories.

Beyond architectural modifications, effective memory management also requires sophisticated strategies for handling knowledge conflicts between internal parameters and external context. When an agent retrieves external information that contradicts its pre-trained knowledge, naive integration can lead to hallucinations or ignored context, undermining the reliability of tool usage. Research indicates that specific attention heads are responsible for recalling internal memory versus retrieving external context, and conflicts arise at the integration point of these flows [183]. Mechanisms like Pruning Head via Path Patching (PH3) offer a way to mitigate these conflicts by selectively pruning conflicting attention heads, thereby allowing the agent to flexibly prioritize either internal knowledge or external context without parameter updates. Additionally, training frameworks such as Knowledge Aware FineTuning (KAFT) have been proposed to improve an LLM’s controllability and robustness, teaching models to give precedence to task-relevant context while falling back on internal knowledge when the context is irrelevant [184].

Interactive reading methods and episodic memory further enhance an agent’s ability to navigate extensive knowledge bases and reuse past experiences. In computer control tasks, agents like Synapse utilize exemplar memory to store embeddings of past trajectories, retrieving them via similarity search to generalize to novel tasks despite limited context windows [185]. Similarly, for summarization tasks where GPU memory is constrained, frameworks like AWESOME employ external memory mechanisms to track previously encoded segments and identify global salient content, ensuring coherence across long documents without sacrificing efficiency [186]. Lighter alternatives, such as MemoryPrompt, complement LLMs with small auxiliary recurrent networks that pass information via soft prompts, improving context tracking in long-distance dialogues without the catastrophic forgetting associated with full fine-tuning [187]. Notably, the concept of caching and reusing action sequences here foreshadows the optimization strategies discussed in the next section, where such memory-based reuse becomes a cornerstone for reducing computational redundancy in

search algorithms.

Finally, the challenge of benchmarking these advanced memory systems has led to the creation of datasets like XL²Bench, which evaluates models on extremely long contexts with long-range dependencies, revealing that current LLMs still lag significantly behind human levels in tasks requiring deep comprehension of hundreds of thousands of tokens [188]. As the field evolves, the integration of OS-like memory management, efficient attention mechanisms, and conflict resolution strategies will be pivotal in transforming LLMs from static text generators into persistent, context-aware agents. These memory capabilities form the essential substrate for the high-efficiency planning and search optimization techniques that follow, enabling agents to operate in open-ended environments without succumbing to context limitations or computational exhaustion.

3.6 Optimization of Search Efficiency and Computation

3.6 Search Efficiency Optimization: Stopping, Pruning, and Reuse

Building upon the robust memory infrastructures and context management strategies established in the previous section, the deployment of Large Language Model (LLM) agents in complex, real-world environments necessitates a rigorous approach to optimizing search efficiency. While the preceding discussion highlighted how OS-inspired memory systems and extended context windows provide the *capacity* for long-horizon reasoning, the practical application of deliberative planning algorithms like Monte Carlo Tree Search (MCTS) often encounters a new bottleneck: prohibitive computational costs. The exponential growth of the search space, compounded by the high latency of LLM inference, renders naive search applications infeasible for real-time interaction. Consequently, recent research has pivoted toward strategies that enhance computational efficiency through dynamic simulation stopping criteria, intelligent pruning of search trees, and the strategic reuse of action sequences. These mechanisms are critical for transforming the theoretical planning capabilities enabled by advanced memory systems into practical, scalable agent architectures.

A primary avenue for optimization involves the implementation of dynamic simulation stopping criteria, which allows agents to terminate search processes early when confidence in the optimal action is sufficiently high. Traditional MCTS implementations often rely on fixed computational budgets or iteration counts, leading to wasted resources on states where the optimal policy is already evident—a inefficiency that persists regardless of context window size. Addressing this, the concept of “Learning to Stop” introduces mechanisms to predict the uncertainty of the current search status and decide dynamically whether to continue searching. For instance, [189] demonstrates that for a significant proportion of states, the best action predicted by deep neural networks remains unchanged even after extended search durations. By training a model to predict this stability, agents can halt simulations prematurely, achieving speedups of up to 2.5 times while maintaining competitive performance levels. Furthermore, the theoretical underpinnings of such decisions are explored in [190], which formalizes the “value of computation” within a Markov decision process. This work distinguishes between static and dynamic computation values, providing a principled method for agents to determine not only when to act but also what specific computations are worth the resource expenditure, thereby optimizing the trade-off between planning depth and execution latency.

Parallel to dynamic stopping, the pruning of search trees represents a fundamental strategy for managing the branching factor inherent in agentic planning. In domains with expansive action spaces, exhaustively evaluating all potential branches is computationally infeasible, even with efficient memory retrieval. To address this, researchers have developed methods that integrate heuristic guidance

and learned models to prune suboptimal paths effectively. The [191] algorithm exemplifies this by formulating the action space as a decision tree and employing A* search with task-specific cost functions. This approach efficiently prunes high-cost branches that likely involve incorrect actions, significantly reducing the time required to identify valid solution paths compared to unidirectional exploration. Similarly, [192] proposes a Primal-Dual MCTS technique that utilizes sampled information relaxation upper bounds. This method creates the possibility of “ignoring” parts of the tree stemming from highly suboptimal choices, allowing the algorithm to converge to optimal actions even while expanding only a partial decision tree. Additionally, [193] introduces convex regularizers to drive exploration more efficiently, offering theoretical guarantees of exponential convergence rates and reducing the sample inefficiency often plaguing standard UCT algorithms. By integrating these pruning strategies, agents can focus their computational budget on the most promising regions of the state space, drastically improving planning throughput.

Beyond stopping and pruning, the reuse of action sequences offers a powerful mechanism for reducing redundancy in long-horizon planning tasks, directly leveraging the memory capabilities discussed previously. Agents often encounter recurring sub-goals or structural patterns within complex tasks, yet standard search algorithms frequently recompute solutions for these familiar scenarios from scratch. The concept of “Habits of Mind,” as detailed in [194], posits that exercised action sequences can be cached and reused to accelerate future planning. By employing a flexible Bayesian action chunking mechanism, agents can identify statistically reliable structures at different scales and embed these routines directly into the search tree. This transforms deep planning problems into shallower ones via multi-step jumps, significantly enhancing both the speed and accuracy of decision-making. This principle is further extended in [195], where the Trajectory Autoencoding Planner (TAP) learns low-dimensional latent action codes. By searching over these discrete latent actions rather than raw high-dimensional controls, the agent effectively reuses compressed representations of successful trajectories, achieving low decision latency regardless of the underlying action space dimensionality. Moreover, [196] investigates the viability of reusing sub-trees as action recommenders without immediate re-planning. By analyzing the statistics of sub-trees, agents can determine when it is safe to execute a sequence of actions derived from a previous search step, thereby avoiding the computational overhead of rebuilding the tree at every time step.

The integration of these optimization strategies is essential for the viability of agentic systems in resource-constrained environments. As highlighted in [197], while planning is crucial for learning and data distribution, the specific choices regarding tree depth and complexity must be tailored to the task difficulty; shallow trees with efficient rollouts often suffice for many applications, whereas complex reasoning tasks demand more sophisticated search adaptations. Furthermore, in multi-agent settings, the scalability of these methods is paramount. [198] addresses the exponential growth of joint action spaces by combining MCTS with factored representations and coordination graphs, enabling dynamic coordination with significantly lower computation costs. Similarly, [153] adapts MCTS for continuous domains by clustering similar states into stochastic action bandit nodes, effectively sharing policy information across states to prevent the explosion of the search graph. Collectively, these advancements in dynamic stopping, tree pruning, and sequence reuse form the backbone of efficient cognitive mechanisms, enabling LLM agents to perform robust, real-time planning. However, while these techniques optimize *how* agents search, they do not inherently enable agents to learn *from* the search. The transition from static optimization to adaptive improvement leads naturally to the next critical evolution: self-reflection and experience-based learning, where agents refine their strategies by internalizing the lessons from past successes and failures.

3.7 Self-Reflection and Experience-Based Learning in Search

3.7 Self-Reflection and Experience-Based Learning in Search

While the optimization strategies discussed previously—dynamic stopping, tree pruning, and sequence reuse—significantly enhance the computational efficiency of search algorithms, they primarily address *how* to search faster rather than *how* to search smarter over time. To transcend the limitations of static planning paradigms, modern agents increasingly integrate self-reflection and experience-based learning, evolving from isolated decision-makers into adaptive systems that refine their strategies through interaction. In traditional search frameworks like Monte Carlo Tree Search (MCTS) or heuristic forward search, agents typically operate with fixed policies that do not inherently learn from the specific failures or successes encountered during a trajectory. Closing the loop between execution and learning is therefore critical, enabling agents to summarize guidelines from past experiences to refine future planning policies without the prohibitive computational burden of extensive retraining. This capability is particularly vital in complex, non-stationary environments where pre-trained models may quickly become obsolete or where the cost of failure is high.

A fundamental aspect of this evolution is leveraging historical data to accelerate policy optimization. Reusing historical trajectories obtained from previous policies is essential for expediting this process, though it introduces challenges related to the interdependence of trajectories across different iterations. Recent theoretical work has demonstrated that reusing past trajectories via importance sampling can significantly improve convergence rates in natural policy gradient methods, provided that the bias of the gradient estimator is properly managed [199]. By mathematically characterizing how past experiences influence current policy updates, agents can effectively “reflect” on their historical performance to guide future search directions, ensuring that valuable data is not discarded after a single planning episode. This approach transforms the search process from an isolated instance of decision-making into a cumulative learning experience.

Central to robust self-reflection is the concept of “learning from failure.” Agents often develop policies effective only under specific conditions, leading to policy confounding, where agents pick up spurious correlations that do not generalize beyond their specific training trajectories [200]. To mitigate this, advanced search mechanisms incorporate strategies to explicitly explore parts of the decision tree that the current policy deems unlikely or suboptimal. For instance, agents can be designed to randomly deviate from an improved policy during training to experience rewards associated with unexplored states, thereby correcting flawed value expectations and improving the underlying model used for planning [201]. This form of active exploration acts as a reflective mechanism, forcing the agent to question its own assumptions and update its internal model based on the divergence between predicted and actual outcomes.

The integration of hindsight learning further enhances the agent’s ability to extract value from search trajectories, particularly in sparse-reward environments where successful outcomes are rare. Hindsight experience replay allows agents to reinterpret failed trajectories by relabeling goals to match what was actually achieved, effectively turning failures into successful training examples for alternative goals [202; 203]. When applied to search, this means that even if an agent fails to reach its intended target during a planning rollout, the trajectory generated can still be utilized to learn how to reach the state that was actually visited. This dramatically increases sample efficiency and allows the agent to build a richer library of “skills” or sub-goals derived from its own search history. Moreover, frameworks like MapGo extend this concept by using learned dynamics models to infer foresight goals, relabeling past experiences with future-looking objectives to enhance the diversity

of training data and improve sample efficiency in goal-oriented tasks [204].

In the context of hierarchical planning, self-reflection often manifests as the discovery and refinement of temporal abstractions or “options.” Instead of learning a single monolithic policy, agents can engage in “option iteration,” where they learn a set of locally strong policies (options) that match search results over specific horizons [205]. This creates a virtuous cycle: better options lead to more effective search results, which in turn provide higher-quality data for training better options. Similarly, agents can learn reusable options from trajectories generated by solving a small number of problems, allowing them to quickly adapt to novel but related tasks by composing these learned skills [206]. This modular approach to experience-based learning enables agents to decompose complex search spaces into manageable components, refining their high-level planning strategies based on the success rates of lower-level skill execution.

For language-based agents, the paradigm of self-reflection has recently been formalized through retrospective models that utilize policy gradient optimization to tune prompts based on environmental feedback. The Retroformer framework, for example, introduces a mechanism where the agent automatically summarizes the root causes of prior failed attempts and proposes refined action plans, effectively learning a retrospective model that improves performance over time without requiring full model retraining [34]. This aligns with the broader trend of enabling agents to critique their own reasoning chains and adjust their planning policies dynamically. Additionally, in scenarios where agents must adapt to open worlds with novel situations, the ability to detect divergences between expected and observed action executions allows for the rapid revision of domain models, ensuring that planning remains robust in the face of unpredictability [37].

Finally, the efficiency of search can be further enhanced by learning the search strategies themselves. Rather than relying on fixed heuristics, agents can employ policy gradient methods to learn when to switch between different search approaches based on the current state of the search process [207]. This meta-learning perspective treats the selection of the search algorithm as a policy optimization problem, where the agent reflects on the efficacy of different strategies across a distribution of problems to discover domain-specific search behaviors. By combining these diverse mechanisms—historical trajectory reuse, hindsight relabeling, option discovery, and retrospective prompt tuning—modern agents are becoming increasingly capable of self-improvement, transforming every search trajectory into an opportunity for learning and refinement. However, as agents accumulate more experience and rely on external contexts for decision-making, they face a new set of challenges regarding the consistency and reliability of the information they process. The transition from optimizing search efficiency to ensuring the faithfulness of knowledge integration leads naturally to the critical issue of handling knowledge conflicts and maintaining coherence in long-context interactions.

3.8 Handling Knowledge Conflicts and Long-Context Reliability

3.8 Handling Knowledge Conflicts and Long-Context Reliability

While the self-reflection mechanisms discussed in the previous section enable agents to refine their search strategies and learn from historical trajectories, these adaptive capabilities rely heavily on the consistency and reliability of the information processed during decision-making. As agents accumulate experience and increasingly depend on external contexts for real-time guidance, they face a critical vulnerability: the potential for knowledge conflicts. The integration of Large Language Models (LLMs) with external tools and retrieval mechanisms has fundamentally expanded their operational boundaries, allowing access to up-to-date information and specialized knowledge beyond

parametric memory. However, this augmentation introduces a significant challenge when the internal parametric knowledge of an LLM, acquired during pre-training, clashes with non-parametric information provided by external contexts or tools. Such discrepancies can severely undermine the reliability, faithfulness, and coherence of agent outputs, particularly in dynamic environments where facts evolve rapidly. Addressing these conflicts requires a nuanced understanding of how models process competing information streams and the development of robust mechanisms to ensure that external context is utilized appropriately without sacrificing the model’s inherent reasoning capabilities.

Knowledge conflicts manifest in various forms, ranging from direct contradictions between stored facts and retrieved evidence to more subtle inconsistencies involving outdated information or misleading noise. Recent surveys have categorized these challenges into context-memory, inter-context, and intra-memory conflicts, highlighting their detrimental impact on trustworthiness in real-world applications [208]. When an agent encounters conflicting information, it faces a dilemma: should it rely on its internal “world knowledge” or defer to the provided external context? Empirical studies reveal that many state-of-the-art models exhibit a strong bias toward their parametric memory, a phenomenon often described as the “Dunning-Kruger effect” in retrieval-augmented models, where the model persistently favors its own faulty internal knowledge even when correct evidence is explicitly provided [209]. Furthermore, models may display confirmation bias, preferentially selecting external evidence that aligns with their pre-existing beliefs while ignoring contradictory but accurate data [209].

To mitigate these issues, researchers have proposed various adaptive decoding and intervention strategies. One promising approach involves discerning whether a conflict actually exists before attempting resolution. The Contextual Information-Entropy Constraint Decoding (COIECD) method, for instance, adaptively adjusts the decoding process based on the entropy of contextual information, allowing the model to maintain high performance in non-conflicting scenarios while improving faithfulness to conflicting contexts when necessary [210]. Another line of research focuses on the mechanistic interpretation of information flow within the transformer architecture. Studies have identified specific attention heads in later layers that act as “memory heads” retrieving internal knowledge and “context heads” retrieving external information. By precisely pruning or patching these conflicting attention heads via methods like Pruning Head via PatH Patching (PH3), it is possible to mitigate conflicts without updating model parameters, effectively controlling the balance between internal memory and external context [183].

Beyond decoding strategies, training methodologies play a pivotal role in enhancing a model’s ability to handle conflicts. Knowledge Aware Fine-Tuning (KAFT) incorporates counterfactual and irrelevant contexts into standard supervised datasets to strengthen both controllability and robustness, ensuring that models learn to prioritize relevant context while ignoring noise [184]. Similarly, instruction-based approaches have been developed to explicitly train models to identify conflicts, pinpoint conflicting segments, and provide distinct answers or viewpoints, although achieving robust responses in complex scenarios remains an open research question [211]. The construction of specialized benchmarks, such as KNOT, which divides reasoning with conflicting knowledge into levels of direct extraction, explicit reasoning, and implicit reasoning, provides a structured framework for evaluating and improving these capabilities [212].

In the realm of long-context reliability, maintaining semantic coherence across extended interactions poses additional hurdles that compound the challenge of knowledge conflicts. As context windows expand to accommodate richer tool outputs and longer interaction histories, models struggle with “in-context interference,” where the continuous flow of new information causes the forgetting of

previously learned knowledge within the session, degrading performance over time [213]. This issue is compounded by the difficulty models face in binding entities to their attributes in long contexts, a fundamental requirement for accurate reasoning [214]. To address coherence in long-form generation and dialogue, advanced modeling techniques are being explored. For example, unified neural coherence models that incorporate sentence grammar, inter-sentence relations, and global patterns have shown superior performance in discriminating coherent from incoherent discourse compared to traditional entity-grid approaches [215]. Furthermore, frameworks like SituationSupervision train models to construct explicit representations of entity states, significantly improving coherence by ensuring generated text remains compatible with the described world state [216].

Evaluating true context utilization and coherence also requires sophisticated metrics that go beyond simple accuracy. Traditional evaluation often fails to distinguish whether a model’s answer stems from genuine context comprehension or merely from its parametric knowledge, especially when the context aligns with internal facts. To overcome this, researchers advocate for using imaginary data based on fictitious facts to evaluate linguistic abilities independent of world knowledge, revealing that models often struggle with hypothetical and modal contexts where knowledge conflicts are triggered [217]. Additionally, novel metrics such as the Brownian Bridge Based Score (BBScore) offer reference-free methods to assess both local and global text coherence, providing a more holistic view of an agent’s ability to maintain thematic consistency over long sequences [218].

Ultimately, ensuring reliable tool learning and function calling necessitates a hybrid approach combining architectural interventions, adaptive decoding, specialized training, and rigorous evaluation. As agents increasingly operate in open-world environments filled with noisy and contradictory information, the ability to dynamically resolve knowledge conflicts and maintain long-context coherence becomes a defining factor in their success. These mechanisms serve as the foundational guardrails that allow the self-reflective and experience-based learning processes described earlier to function effectively; without reliable information integration, an agent’s capacity to learn from its own history is compromised. Future research must continue to bridge the gap between statistical mimicry and true causal reasoning, enabling agents to not only detect conflicts but to reason through them with human-like discernment and reliability, thereby paving the way for the next generation of autonomous systems capable of operating safely in complex, real-world scenarios.

4 Training Paradigms and Capability Enhancement

4.1 Supervised Fine-Tuning on Trajectory Datasets

4.1 Supervised Fine-Tuning on Trajectory Datasets

The transition from static language models to dynamic agents capable of tool use and function calling fundamentally relies on the quality and structure of the training data. Supervised Fine-Tuning (SFT) on interaction trajectories serves as the foundational paradigm for equipping Large Language Models (LLMs) with the ability to decompose complex tasks, select appropriate tools, and execute sequential actions. However, the efficacy of SFT is not merely a function of model scale but is critically dependent on the curation strategies employed to construct high-quality trajectory datasets. A primary challenge in this domain is the scarcity of expert-level, human-annotated trajectories that cover the vast space of potential tool interactions and failure modes. Consequently, recent research has pivoted toward sophisticated methodologies for leveraging negative examples, synthesizing data through self-training loops, and optimizing sample selection to reduce reliance on costly human annotation. These data-centric innovations establish the necessary behavioral priors that subsequent parameter-efficient adaptation techniques, discussed in the following section, will

refine and specialize.

A crucial aspect of curating robust trajectory datasets involves the strategic inclusion of negative examples. Standard SFT approaches often treat all training data as positive demonstrations of correct behavior, yet growing evidence suggests that models significantly benefit from explicitly learning what *not* to do. In the context of tool learning, negative examples can manifest as incorrect API calls, hallucinated parameters, or logically flawed planning sequences. The concept of learning from negative data is well-supported in broader machine learning contexts, where procedures like the CRINGE loss (ContRastive Iterative Negative GEneration) have demonstrated that training with examples of what a model should avoid can alleviate issues in safe generation and contradiction avoidance [219]. By integrating such negative trajectories into the fine-tuning process, agents can develop a more nuanced understanding of tool constraints and safety boundaries, thereby reducing the frequency of catastrophic failures during deployment. Furthermore, in contrastive learning frameworks, the selection of “hard” negative samples—those that are semantically similar to correct actions but ultimately incorrect—has been shown to be both necessary and sufficient for learning discriminative representations [220]. Applying this principle to trajectory datasets implies that curating difficult failure cases, rather than random errors, yields superior generalization in agent behavior.

To address the bottleneck of human annotation, self-training loops and data generation strategies have emerged as powerful alternatives for expanding trajectory corpora. These methods often employ a teacher-student framework where a pre-trained model generates synthetic trajectories that are subsequently filtered and used to train a student model. For instance, in low-resource information extraction tasks, methods like STAR leverage Large Language Models to synthesize data instances from limited seed demonstrations, utilizing self-reflection mechanisms to identify and refine errors iteratively [221]. This approach of generating target structures followed by content generation, refined through self-correction, can be directly adapted for creating tool-use trajectories where the “structure” corresponds to the sequence of function calls and the “content” represents the natural language reasoning steps. Similarly, self-training pipelines that restrict the number of positive or negative self-labeled examples have shown comprehensive performance improvements in causality classification, suggesting that careful balancing of synthetic data is vital for maintaining model stability [222].

Beyond simple generation, the quality of self-taught data can be enhanced by identifying and utilizing ambiguous or uncertain instances that traditional filtering methods might discard. The STAD approach, designed for low-resource relation extraction, proposes classifying auto-annotated instances into confident and uncertain groups, then applying set-negative training on the ambiguous instances to extract useful supervisory signals [223]. In the realm of agent training, this suggests that trajectories where the agent is uncertain about the next tool to call should not be discarded but rather used to teach the model how to resolve ambiguity, perhaps by triggering retrieval mechanisms or asking clarifying questions. Moreover, the synthesis of counterfactual negatives provides another avenue for data augmentation. The SCENE method demonstrates that synthesizing negative examples zero-shot from positive ones by perturbing input data can significantly close performance gaps in detecting unanswerable questions or false claims [224]. For tool-learning agents, generating counterfactual trajectories where a slight modification in the user query leads to a completely different valid tool path can help the model learn robust decision boundaries.

The issue of negative transfer, where irrelevant source data degrades target performance, also necessitates rigorous sample selection mechanisms during the curation of trajectory datasets. Autoencoder-based sample selection strategies have been proposed to measure the relevance

between source samples and target domains, ensuring that only sufficiently related data is used for fine-tuning [225]. In multi-task agent settings, this translates to filtering synthetic trajectories that, while syntactically correct, may introduce domain-specific biases or irrelevant reasoning patterns that hinder performance on the target tool suite. Additionally, the dynamics of user interests and model updates require that negative sampling strategies remain adaptive. In sequential recommendation systems, generating negative samples based on the model’s current learned preferences ensures that the training signal remains informative as the model evolves [226]. Analogously, agent training loops should dynamically generate negative tool-use trajectories that challenge the agent’s current policy, preventing stagnation and encouraging continuous adaptation.

Finally, the integration of human feedback into these loops remains paramount for aligning synthetic trajectories with human intent. While automated methods can scale data production, human-in-the-loop mechanisms are essential for correcting subtle misalignments in reasoning or tool selection. Research into modeling human annotation errors suggests that the ordering and presentation of instances can significantly impact the quality of feedback received, which in turn affects the reliability of the resulting training data [227]. Therefore, a comprehensive SFT strategy for tool learning must not only automate the generation of positive and negative trajectories but also incorporate robust mechanisms for validating these trajectories against human expectations, ensuring that the resulting agents are not only capable but also safe and aligned. By synthesizing these diverse strategies—leveraging hard negatives, employing iterative self-reflection, and carefully managing sample relevance—the community can construct the high-fidelity trajectory datasets necessary to propel LLM agents from simple prompt followers to robust, autonomous tool users. Once these high-quality datasets are established, the focus shifts to efficiently adapting massive models to learn from them, leading to the development of advanced Parameter-Efficient Fine-Tuning architectures.

4.2 Advanced Parameter-Efficient Fine-Tuning Architectures

4.2 Advanced Parameter-Efficient Fine-Tuning Architectures

Building upon the high-fidelity trajectory datasets curated through the strategies discussed in Section 4.1, the challenge shifts to efficiently adapting massive pre-trained models to learn from this data. The rapid escalation in the scale of Large Language Models (LLMs) has rendered full parameter fine-tuning increasingly prohibitive due to excessive memory consumption and computational costs. Consequently, the field has pivoted toward Parameter-Efficient Fine-Tuning (PEFT) methodologies, with Low-Rank Adaptation (LoRA) emerging as a foundational paradigm. The original LoRA framework operates on the hypothesis that the intrinsic dimensionality of model adaptation is low, freezing pre-trained weights while injecting trainable rank decomposition matrices into Transformer layers [228]. While this approach drastically reduces trainable parameters and GPU memory requirements compared to full fine-tuning, the initial implementation relies on fixed, uniform rank assignments across all layers, which may not be optimal for diverse downstream tasks or the complex reasoning required for tool learning in agents. To address these limitations, a rich ecosystem of advanced architectures has evolved, focusing on dynamic rank allocation, structural optimization, and enhanced training dynamics, thereby bridging the gap between data availability and model specialization.

A primary avenue of innovation involves overcoming the rigidity of fixed-rank constraints. Standard LoRA requires an exhaustive search to determine the optimal rank, a process that is computationally expensive and often yields suboptimal results if the rank is uniform across layers. To mitigate this, researchers have developed dynamic adaptation mechanisms. For instance, DyLoRA introduces

a dynamic search-free technique that trains LoRA blocks across a range of ranks simultaneously, allowing the model to perform consistently well over a larger spectrum of ranks without retraining [229]. Similarly, AutoLoRA employs meta-learning to automatically identify the optimal rank for each layer by associating rank-1 matrices with selection variables, effectively pruning unnecessary components during training [230]. Extending this concept, SoRA (Sparse Low-Rank Adaptation) incorporates gate units optimized via proximal gradient methods to dynamically adjust the intrinsic rank during the adaptation process, eliminating zeroed-out parameter blocks during inference to achieve a concise, rank-optimal configuration [231]. Furthermore, ALoRA proposes a strategy to estimate the importance score of each LoRA rank, gradually pruning abundant ranks and reallocating budgets to Transformer modules that require higher capacity [232]. These dynamic approaches are particularly critical for tool-learning agents, which must adapt to heterogeneous API schemas and varying complexity levels in function calling sequences, ensuring that model capacity is allocated where it is most needed for precise tool execution.

Beyond dynamic ranking, recent architectures have sought to enhance the expressive power and stability of low-rank updates through structural modifications. The standard LoRA formulation applies a single scaling factor that can impede learning in large-width networks; LoRA+ corrects this by assigning distinct learning rates to the adapter matrices A and B , significantly improving both performance and finetuning speed [233]. Addressing the issue of overfitting, which is prevalent in low-parameter regimes, BiLoRA introduces a bi-level optimization framework that splits the training of pseudo singular vectors and values across different data subsets, thereby enhancing generalization on test data [234]. Structural innovations also include ResLoRA, which integrates residual paths during training to shorten the calculation path and accelerate convergence, merging these paths away during inference to maintain zero latency overhead [235]. Another notable advancement is Delta-LoRA, which propagates learning not only through the low-rank matrices but also to the pre-trained weights via the delta of the product of these matrices, effectively addressing the limitation where incremental updates are insufficient for complex representations [236]. These structural refinements lay the groundwork for the deeper theoretical analysis of optimization stability and hyperparameter sensitivity explored in the subsequent section.

The evolution of PEFT also encompasses methods that decouple parameter counts from architectural constraints or leverage ensemble techniques to handle the multifaceted nature of agent tasks. NOLA (Networks as Linear Combination of Low Rank Random Basis) overcomes the rank-one lower bound of traditional LoRA by re-parameterizing updates as linear combinations of random basis matrices, allowing for extreme parameter reduction without sacrificing performance [237]. In the realm of ensemble methods, MELoRA (Mini-Ensemble Low-Rank Adapters) trains a group of mini-LoRAs with few parameters to capture diversity, achieving superior generalization with significantly fewer trainable parameters than standard LoRA [238]. Additionally, Hydra explores a multi-head architecture combining parallel and sequential adaptation branches to integrate novel and general features more effectively than single-branch methods [239]. For scenarios requiring the composition of multiple skills, such as an agent mastering various tools, LoraHub provides a framework for dynamically composing multiple LoRA modules trained on diverse tasks to achieve cross-task generalization without additional gradients [240].

Finally, efficiency in memory and serving has driven architectures designed for scalable deployment. VeRA uses a single pair of low-rank matrices shared across all layers while learning small scaling vectors, drastically reducing storage challenges for deploying numerous per-task models [241]. For serving environments, S-LoRA and CaraServe introduce system-level architectural optimizations to handle thousands of concurrent LoRA adapters with unified paging and rank-aware schedul-

ing, essential for scalable multi-tenant agent deployments [242; 243]. Collectively, these advanced architectures transform LoRA from a static fine-tuning trick into a flexible, robust, and scalable foundation for equipping LLM agents with the specialized capabilities required for sophisticated tool learning and function calling. However, the mere existence of these sophisticated structures does not guarantee success; their practical efficacy is fundamentally contingent upon the stability and efficiency of the underlying optimization processes, a critical topic addressed in the following section.

4.3 Optimization Dynamics, Stability, and Hyperparameter Tuning

4.3 Optimization Dynamics, Stability, and Hyperparameter Tuning

While Section 4.2 established advanced architectural paradigms for Parameter-Efficient Fine-Tuning (PEFT), the practical efficacy of equipping Large Language Models (LLMs) with tool-learning capabilities is fundamentally contingent upon the stability and efficiency of the underlying optimization processes. Even with sophisticated structures like dynamic rank allocation or multi-head adapters, the optimization landscape remains fraught with challenges related to instability, hyperparameter sensitivity, and computational overhead. Recent theoretical and empirical advancements have begun to unravel the complex dynamics governing these processes, offering robust solutions that range from modified scaling factors to sophisticated bi-level optimization frameworks, thereby bridging the gap between architectural potential and realized performance.

A primary source of instability in fine-tuning, particularly within Low-Rank Adaptation (LoRA) architectures discussed previously, stems from improper scaling mechanisms relative to the adapter rank. Traditional implementations often employ a scaling factor that divides the adapter output by the rank, which inadvertently suppresses learning signals as the rank increases, thereby stunting performance and limiting the practical utility of higher-rank adapters. To address this, researchers have proposed the rank-stabilized LoRA (rsLoRA) method, which theoretically proves and empirically demonstrates that adapters should be scaled by the square root of the rank rather than the rank itself [244]. This adjustment allows for the utilization of larger ranks to trade increased computational resources during training for significantly improved fine-tuning performance, effectively decoupling the learning capacity from the instability previously observed in high-rank settings. Furthermore, the inherent asymmetry in the roles of the low-rank matrices within LoRA has been identified as a critical factor in optimization dynamics; specifically, the matrix responsible for projecting features to the output space (B) plays a more pivotal role than the input extraction matrix (A), suggesting that fine-tuning strategies can be optimized by focusing computational effort on the more influential components [245].

Beyond scaling adjustments, the stability of the fine-tuning trajectory is often compromised by optimization difficulties that lead to vanishing gradients or catastrophic forgetting, where the model loses its pre-trained general knowledge while acquiring task-specific skills. Theoretical analyses indicate that fine-tuning instability is not merely a result of small dataset sizes but is deeply rooted in the optimization landscape, where models with identical training losses can exhibit vastly different generalization capabilities [246]. To mitigate these issues, advanced strategies such as the Chain of LoRA (COLA) have been introduced. Inspired by the Frank-Wolfe algorithm, COLA employs an iterative residual learning procedure that merges learned LoRA modules into the backbone parameters and re-initializes new modules, effectively bridging the performance gap between parameter-efficient and full fine-tuning without additional memory overhead [247]. Additionally, periodic unloading strategies, such as those found in PeriodicLoRA, accumulate low-rank updates

over multiple stages to break the low-rank bottleneck, thereby enhancing the learning ability of the model while maintaining memory efficiency [248].

Hyperparameter tuning represents another critical frontier in optimizing agent training, particularly given the sensitivity of PEFT methods to rank selection and learning rates. Traditional grid search approaches are computationally prohibitive for large-scale models. Consequently, bi-level optimization frameworks have gained traction as a powerful mechanism for automating this process. These frameworks formulate the hyperparameter selection as an outer-loop optimization problem nested within the inner-loop model training. For instance, AutoFT utilizes bi-level optimization to search for objective functions and hyperparameters that maximize out-of-distribution generalization, significantly outperforming hand-crafted regularization techniques [249]. Similarly, the BiLoRA framework addresses the tendency of LoRA to overfit by splitting the training of pseudo-singular vectors and values across different data subsets within a bi-level structure, thereby enhancing robustness and generalization [234]. Theoretical surveys on gradient-based bi-level optimization further elucidate how these methods can efficiently compute hypergradients to optimize regularization parameters and meta-knowledge, providing a unified approach to handling complex optimization scenarios [250].

Adaptive backpropagation strategies also play a pivotal role in enhancing both the stability and efficiency of the training process. Recognizing that standard backpropagation can be computationally expensive and environmentally taxing, novel approaches like GreenTrainer have been developed to adaptively evaluate the backpropagation costs of different tensors against their contributions to model accuracy. By selectively updating only the most impactful tensors, such methods can reduce the floating-point operations (FLOPs) required for fine-tuning by up to 64% without compromising accuracy, aligning training dynamics with “Green AI” objectives [251]. Moreover, alternative optimization dynamics that deviate from standard gradient descent, such as zeroth-order methods, offer memory-efficient pathways for fine-tuning extremely large models where backpropagation is infeasible. Techniques like MeZO-SVRG combine zeroth-order estimation with variance reduction to stabilize convergence, demonstrating that effective fine-tuning can be achieved even when gradient computation is approximated or entirely bypassed [252].

Finally, the interplay between optimization dynamics and fairness cannot be overlooked. Fine-tuning processes can inadvertently amplify biases present in pre-trained models. Recent frameworks address this by employing transfer learning strategies that neutralize the importance of weights influential to specific demographic groups, often leveraging Fisher information to guide the optimization trajectory towards fairer outcomes [253]. Collectively, these advancements in scaling factors, bi-level frameworks, and adaptive backpropagation signify a maturation of the field, moving from heuristic trial-and-error toward theoretically grounded, stable, and efficient optimization paradigms. By resolving these foundational training challenges, the community establishes a robust baseline necessary for the subsequent integration of Reinforcement Learning and feedback-driven alignment mechanisms, which require stable policies to effectively process sparse rewards and human preferences as discussed in the following section.

4.4 Reinforcement Learning and Feedback-Driven Alignment

4.4 Reinforcement Learning and Feedback-Driven Alignment

Building upon the stable optimization paradigms and hyperparameter tuning strategies established in Section 4.3, the integration of Reinforcement Learning (RL) into Large Language Model (LLM)

agents marks a critical evolution from static knowledge retrieval to dynamic, feedback-driven adaptability. While the parameter-efficient fine-tuning techniques discussed previously provide a robust foundation for tool use, they often fail to capture the nuanced, multi-step decision-making required in complex environments where rewards are sparse or delayed. Consequently, RL mechanisms, particularly actor-critic architectures, have become indispensable for aligning agent behaviors with environmental constraints and human preferences. A primary challenge in this domain remains the sparsity of reward signals, which can stall learning progress and lead to suboptimal policies. Traditional approaches often rely on binary rewards that merely indicate success or failure; however, research suggests that introducing controlled adversity or varying reward magnitudes can significantly accelerate the learning speed of model-based actor-critic methods [254]. By deviating from classical deterministic binary reward functions and incorporating external environmental pressure, agents can achieve higher performance with reduced sample complexity and variance, effectively leveraging the stable training dynamics enabled by the optimization techniques of the prior section.

To address the limitations of sparse rewards without resorting to brittle reward shaping, advanced actor-critic frameworks have begun to incorporate auxiliary critics that leverage prior information. For instance, the Potential Field Guided Actor-Critic approach extends the standard architecture by introducing a potential-field-based critic alongside the traditional reward-based critic [255]. This dual-critic system allows the agent to trust potential-field guidance in states with strong priors, such as obstacle avoidance, while relying on reward-based evaluation in uncertain regions, thereby accelerating policy improvement and reducing the reliance on meticulously engineered reward functions. Similarly, in multi-agent settings where the joint action space grows exponentially, ensuring a reliable critic is paramount. Regularizing the critic with spectral normalization has been shown to enable robust learning even in complex, sparse-reward multi-agent scenarios, highlighting the importance of stability mechanisms within the critic network [256]. Furthermore, sharing experiences among agents through frameworks like Shared Experience Actor-Critic (SEAC) can dramatically improve exploration efficiency, making the difference between solving a task and failing to learn altogether in difficult environments [257].

Beyond environmental feedback, aligning agents with human intent requires sophisticated methods for Processing Human Feedback (RLHF). The conventional pipeline of learning a reward model followed by policy optimization via algorithms like Proximal Policy Optimization (PPO) faces challenges regarding reward overoptimization and distributional shift. Recent studies indicate that Direct Preference Optimization (DPO), a reward-free method, often struggles to surpass PPO in challenging domains such as code generation, suggesting that actor-critic methods remain superior when properly tuned [258]. However, the reliance on a singular reward model in standard RLHF fails to capture the diversity of human preferences. To mitigate this, novel approaches like MaxMin-RLHF employ an expectation-maximization algorithm to learn a mixture of preference distributions, ensuring equitable alignment across diverse user groups and improving win-rates for minority preferences without compromising majority performance [259].

The fidelity of the reward model itself is another critical area of innovation. Standard assumptions that human preferences are based solely on partial returns have been challenged by evidence suggesting humans evaluate trajectories based on regret. Modeling preferences via regret rather than cumulative reward has been proven to yield better identifiability and alignment with real human judgments [260]. Moreover, the uncertainty inherent in human feedback can be leveraged to drive exploration. By measuring disagreement across an ensemble of learned reward models, agents can generate intrinsic rewards that guide exploration toward areas of high uncertainty, significantly improving both feedback and sample efficiency in complex manipulation tasks [261]. To further

combat reward overoptimization, regularization techniques such as Uncertainty-Penalized RLHF utilize diverse Low-Rank Adaptation (LoRA) ensembles to quantify reward uncertainty, penalizing the policy when uncertainty is high to prevent exploitation of flawed reward models [262]. This integration of LoRA ensembles directly capitalizes on the parameter-efficient architectures and stability solutions detailed in Section 4.3, demonstrating a seamless continuity between optimization dynamics and alignment strategies.

Architectural innovations also extend to how agents process and utilize feedback. The Planner-Actor-Critic architecture for huMAN-centered planning and learning (PACMAN) unifies symbolic knowledge-based planning with RL, allowing agents to jump-start learning using prior knowledge while fine-tuning behaviors through human feedback and environmental rewards [263]. This hybrid approach demonstrates robustness against inconsistent or misleading human feedback, a common issue in real-world deployment. Additionally, the concept of “shaping advice” has been formalized in deep multi-agent reinforcement learning, where potential functions provide additional reward signals that guide agents without altering the optimal policy of the original environment [264]. In text generation tasks, where rewards are typically sparse and provided only at the end of a sequence, leveraging the critique capabilities of LLMs to generate intermediate, token-level rewards has shown to significantly enhance sample efficiency and overall performance [265].

Finally, the dynamics of the environment itself can be encoded to improve preference learning. Methods that learn dynamics-aware state-action representations via self-supervised temporal consistency tasks can bootstrap preference-based reward functions, reducing the number of required human labels by an order of magnitude while recovering a higher percentage of ground-truth reward performance [266]. This suggests that the future of feedback-driven alignment lies not just in better reward models, but in richer representations that integrate environmental dynamics, human diversity, and architectural stability to create agents that are both capable and aligned. By establishing these robust, feedback-tuned policies, we lay the essential groundwork for the next critical challenge: ensuring that these aligned capabilities are retained over time. As agents encounter new tools and evolving tasks, the risk of catastrophic forgetting becomes paramount, necessitating the lifelong adaptation strategies and memory preservation mechanisms discussed in the following section.

4.5 Multi-Task Learning, Lifelong Adaptation, and Catastrophic Forgetting

4.5 Multi-Task Learning, Lifelong Adaptation, and Catastrophic Forgetting

While the preceding section detailed how reinforcement learning and feedback mechanisms align agent behaviors with environmental constraints and human preferences, a critical prerequisite for sustained agent efficacy is the ability to retain these aligned capabilities over time. The transition from static, single-task models to dynamic agents capable of lifelong adaptation represents a fundamental shift in artificial intelligence architecture. In the context of tool learning and function calling, agents must not only master a diverse array of APIs and external functions but also retain these competencies as they encounter new tools and evolving environments. The primary obstacle to this continuous accumulation of knowledge is catastrophic forgetting, a phenomenon where the optimization of neural network parameters for new tasks leads to a significant degradation in performance on previously learned skills. This challenge is particularly acute in agentic systems, where the loss of a specific tool-use skill can render the agent incapable of completing complex, multi-step workflows that rely on historical competencies, effectively undoing the alignment achieved through the feedback-driven processes discussed .

A comprehensive understanding of catastrophic forgetting reveals that it is not merely a failure of memory but a structural conflict within the model’s representational space. Research indicates that deeper layers of neural networks are disproportionately responsible for forgetting, as these layers encode high-level semantic features that are highly susceptible to interference when task distributions shift [267]. Furthermore, theoretical analyses suggest that the severity of forgetting is correlated with the similarity between tasks; specifically, tasks with intermediate similarity often induce the maximal degree of interference due to the overlapping yet conflicting gradient updates required for their respective solutions [268]. Consequently, effective lifelong adaptation strategies must address both the geometric properties of the loss landscape and the semantic overlaps between distinct tool-use scenarios to ensure that new learning does not overwrite the specialized knowledge acquired during earlier training phases.

To mitigate these issues, recent advancements have increasingly turned to Mixture-of-Experts (MoE) architectures and parameter isolation techniques. These approaches operate on the principle that different tasks should utilize disjoint or sparsely overlapping subsets of the model’s parameters, offering a structural solution complementary to the reward-shaping methods of the previous section. For instance, the concept of sub-network discovery allows an agent to isolate knowledge for specific tasks by identifying and activating unique sub-networks within a shared backbone, thereby preventing interference while still enabling knowledge transfer where appropriate [269]. Building on this, soft-masking mechanisms have been proposed to partially block parameter updates based on their importance to prior tasks, allowing the full network to be utilized without any single task monopolizing the capacity, thus fostering efficient knowledge transfer across both similar and dissimilar domains [270]. Similarly, the Remembering Transformer employs a mixture-of-adapters coupled with a generative routing mechanism to dynamically direct task data to relevant expert modules, achieving state-of-the-art performance in vision-based continual learning while maintaining high parameter efficiency [271]. Another innovative approach, the Information Bottleneck Masked sub-network (IBM), leverages information bottleneck principles to construct redundancy-free sub-networks, effectively eliminating unimportant weights and significantly reducing the parameter count required for each task while robustly mitigating forgetting [272].

Beyond architectural modifications, replay-based strategies remain a cornerstone of lifelong adaptation, serving as a data-centric counterpart to the model-centric isolation techniques. While traditional experience replay buffers store a subset of past data to interleave with new training samples, the quality and diversity of these stored exemplars are critical for maintaining the fidelity of learned policies. Methods such as Diverse Sample Selection (DSS) emphasize the necessity of selecting diverse samples to better preserve the decision boundaries of previous classes, outperforming random selection strategies especially in scenarios with fuzzy task boundaries [273]. Further refinements include Memory Outlier Elimination (MOE), which filters the replay buffer to remove noisy or non-representative samples, ensuring that the stored memory consists of label-homogeneous subpopulations that accurately reflect class distributions [274]. In scenarios where storing raw data is prohibited due to privacy concerns—a common constraint in real-world agent deployment—generative replay offers a viable alternative. By training generative models to synthesize pseudo-samples of past tasks, agents can rehearse old knowledge without accessing original data. Advanced implementations, such as the Class-Prototype Conditional Diffusion Model, integrate learnable class prototypes into the denoising process to ensure high-fidelity generation of past task data, significantly reducing the risk of forgetting in classifiers [275]. Additionally, generative negative replay has shown promise by using generated data not just for positive reinforcement but as negative examples to sharpen the decision boundaries for new classes [276].

Cognitive-inspired mechanisms also provide powerful frameworks for managing the stability-plasticity dilemma, drawing parallels between artificial agents and biological learning systems. Drawing inspiration from the human brain’s complementary learning systems, researchers have developed methods like Cognitive Replay (CORE), which adaptively allocates replay buffer quantities based on forgetting rates and selects high-quality data to mimic human cognitive review processes [277]. Similarly, the TriRE paradigm emulates multiple neurophysiological processes simultaneously—including neurogenesis, active forgetting, and context-dependent gating—to retain prominent neurons, revise extracted knowledge, and promote less active neurons for future tasks, thereby surpassing approaches that rely on isolated mechanisms [278]. Furthermore, the integration of meta-learning priors has proven effective; by exposing networks to sequential tasks during a meta-training phase, models can learn parameters that are inherently more resistant to future interference, slowing down the rate of catastrophic forgetting [279].

Finally, the challenge of lifelong adaptation extends to unsupervised and open-world settings where task boundaries are unknown, a scenario increasingly relevant as agents move toward the training-free paradigms discussed in the following section. In such stochastic environments, methods like Mask and Visual Prompt tuning (MVP) have been devised to handle blurry task boundaries by employing instance-wise logit masking and contrastive visual prompt tuning to discern current learning objectives without explicit task identifiers [280]. Moreover, hierarchical information-theoretic principles have been applied to create task-agnostic continual learning systems that optimize the trade-off between learning and forgetting without requiring prior knowledge of task identities [281]. As agents become more autonomous, the synthesis of these diverse strategies—ranging from sparse expert activation and sophisticated replay buffers to bio-inspired cognitive mechanisms—will be essential for developing robust, lifelong learning systems. These systems must be capable of mastering an ever-expanding suite of tools without sacrificing previously acquired competencies, thereby laying the groundwork for the efficient, inference-time adaptation techniques explored next.

4.6 Training-Free and In-Context Capability Enhancement

4.6 Training-Free and In-Context Capability Enhancement

Building upon the architectural strategies for mitigating catastrophic forgetting discussed in the previous section, this subsection explores a complementary paradigm: achieving robust tool-use capabilities without modifying model weights. While lifelong adaptation addresses the challenge of retaining knowledge during continuous updates, a critical frontier in agent development lies in training-free and in-context capability enhancement, where models acquire new skills or adapt to dynamic environments solely through inference-time mechanisms. This approach is particularly vital for deploying agents in scenarios involving black-box models, rapid task switching, or resource-constrained edge devices where full model updates are prohibitive. By shifting the focus from modifying the internal knowledge representation of the model to strategically manipulating the input context and decoding dynamics, these methods elicit desired tool-interaction behaviors while inherently sidestepping the stability-plasticity trade-offs inherent in parameter updates.

A primary mechanism within this domain is the adaptation of Decision Transformers (DT) via prompt-based strategies, which reframes reinforcement learning problems as sequence modeling tasks. Traditional DT approaches often require extensive offline training to learn policies, but recent advancements demonstrate that few-shot policy generalization can be achieved without any parameter updates. By constructing trajectory prompts that encode segments of expert demonstrations, agents can guide policy generation for unseen tasks effectively. For instance, the work

titled [282] illustrates how a Prompt-based Decision Transformer can leverage the sequential modeling inherent in Transformer architectures to adapt to novel environments using only a handful of demonstration timesteps. This method encodes task-specific information directly into the prompt, allowing the frozen model to generalize to out-of-distribution environments and varying prompt lengths, thereby bypassing the need for retraining while maintaining robust performance in control benchmarks.

Beyond static prompting, dynamic and input-dependent prompt generation represents a sophisticated evolution in training-free adaptation. Static prompts often fail to account for the variability inherent in complex tool-use scenarios, where the optimal strategy depends heavily on the specific instance or environmental state. To address this, researchers have proposed architectures capable of generating high-performing, input-dependent prompts dynamically. The [283] paper introduces a Prompt Generation Network (PGN) that samples from an end-to-end learned library of tokens to create prompts tailored to specific inputs. Although initially demonstrated in vision, the underlying principle of “prompt inversion” and latent space optimization is directly transferable to tool-learning agents, enabling them to construct context-aware tool invocation sequences without modifying the backbone model weights. This capability is crucial for agents that must interact with diverse APIs where the correct function arguments vary significantly based on user intent and current context.

Furthermore, the integration of hypernetworks offers a powerful framework for task-conditioning without full fine-tuning, serving as a lightweight alternative to the MoE and parameter isolation techniques discussed previously. By utilizing a small auxiliary network to generate hyper-prompts, agents can achieve flexible information sharing and task-specific feature mapping. The [284] study demonstrates that such hyper-prompts serve as global memories, allowing the model to attend to task-relevant features with minimal additional parameters. In the context of tool learning, this architecture enables an agent to rapidly switch between different tool-use protocols—such as switching from a database query tool to a code execution tool—by merely updating the hyper-prompt generator rather than the entire model. This approach supports efficient multi-task learning and is competitive with strong baselines while utilizing a fraction of the parameters, making it ideal for agile agent deployment.

The theoretical underpinnings of these methods suggest that Transformers possess an inherent capacity for in-context algorithm selection, functioning effectively as statisticians that can identify and execute appropriate procedures based on input sequences. The research presented in [285] provides a statistical theory showing that a single Transformer can adaptively select different algorithms or perform qualitatively different tasks without explicit prompting of the right algorithm. This implies that for tool-learning agents, the model can implicitly learn to select the correct tool or combination of tools based on the context provided in the prompt, effectively performing “in-context tool selection” akin to a human expert choosing the right instrument for a job.

Efficiency in these training-free methods is further enhanced by techniques that optimize the prompt structure itself. Dynamic prompting strategies, as explored in [286], reveal that optimizing the position, length, and representation of prompts based on specific tasks and instances can capture semantic information that fixed prompts miss. By employing lightweight learning networks to determine instance-dependent guidance, agents can improve their tool-use accuracy significantly. Additionally, methods like [287] propose guiding large base models during decoding through lightweight policy adapters trained via reinforcement learning. This allows for the optimization of arbitrary user objectives, such as reducing toxicity in tool outputs or adhering to specific formatting constraints, without the cost of fine-tuning the extreme-scale language model itself.

Finally, the robustness of these approaches is supported by findings that prompt-tuned models can exhibit improved resilience against adversarial attacks compared to fully fine-tuned counterparts, as noted in [288]. This is a critical consideration for tool-learning agents operating in open environments where malicious inputs might attempt to hijack tool execution. By relying on in-context enhancement and dynamic prompt matching, agents can maintain high performance and safety standards while avoiding the catastrophic forgetting often associated with continuous weight updates. Collectively, these training-free paradigms represent a shift towards more flexible, efficient, and adaptable AI agents capable of mastering complex tool-use landscapes through intelligent context management. These inference-time adaptation strategies lay the essential groundwork for the specialized applications in embodied and multimodal domains, where bridging the gap between abstract language and physical action requires the rapid, low-latency responsiveness that only training-free methods can provide, as detailed in the following section.

4.7 Specialized Adaptation for Embodied and Multimodal Agents

4.7 Specialized Adaptation for Embodied and Multimodal Agents

Building upon the training-free and in-context enhancement strategies discussed in the previous section, the application of Large Language Models (LLMs) to embodied and multimodal agents introduces a distinct set of challenges centered on the “reality gap.” While prior approaches successfully leveraged prompt engineering and hypernetworks to adapt text-based models to dynamic tasks without weight updates, physical embodiment requires bridging the disconnect between abstract language tokens and the continuous, high-dimensional state spaces of real-world environments. Specialized adaptation techniques are therefore critical to enable agents not only to perceive multimodal inputs but also to reason about physical constraints and execute precise motor commands. This subsection synthesizes emerging methodologies that adapt foundation models for embodied intelligence, focusing on conditional mixture models, probabilistic motion policies, and efficient sampling strategies that facilitate robust real-world control.

A primary hurdle in this domain is equipping agents with the ability to generalize across diverse tasks and objects without the prohibitive cost of retraining for every new scenario. Recent advancements have leveraged the reasoning capabilities of Multimodal Large Language Models (MLLMs) to enhance manipulation stability and generalization. For instance, frameworks like ManipLLM introduce fine-tuning paradigms that preserve the inherent common sense of MLLMs while injecting specific capabilities for object-centric pose prediction and affordance reasoning [289]. By utilizing chain-of-thought prompting combined with active impedance adaptation, these models can predict end-effector poses and adjust to real-world scene configurations through test-time adaptation strategies. Similarly, other approaches have demonstrated that LLMs can function as zero-shot trajectory generators, directly predicting dense sequences of end-effector poses for manipulation tasks when grounded with object detection models, thereby challenging the assumption that LLMs lack low-level control knowledge [290].

Beyond direct trajectory generation, a sophisticated line of research focuses on adapting LLMs to define reward functions or optimization objectives for traditional control frameworks. This hybrid approach allows the semantic flexibility of language models to guide mathematically rigorous controllers, effectively extending the “in-context” philosophy to the domain of control theory. The “Language to Rewards” paradigm illustrates this by utilizing LLMs to synthesize reward parameters that optimize low-level policies for diverse robotic tasks, effectively bridging high-level instructions with hardware-dependent actions [291]. In a similar vein, the NARRATE architecture layers an

LLM over Model Predictive Control (MPC), where the language model frames safety constraints and objective functions as mathematical expressions. This ensures that the resulting motor control is not only flexible but also adheres to strict safety guarantees and interpretability requirements [292]. Furthermore, frameworks such as REAL demonstrate how LLMs can be integrated into autonomy stacks to provide resilience against unmodeled dynamics and parameter errors, acting as a high-level supervisor that adapts control inputs based on natural language diagnostics and prior knowledge [293].

To handle the inherent multimodality and uncertainty of robot motion, conditional mixture models represent another pivotal technique for specialized adaptation. Traditional Gaussian Mixture Models (GMMs) have long been used for Learning from Demonstration (LfD), yet they often struggle with complex, cluttered environments or non-Euclidean data. To address this, researchers have developed advanced variants such as the Directionality-Aware Mixture Model (DAMM), which applies Riemannian metrics to blend directional data with Euclidean states, enabling efficient parallel sampling for Linear Parameter Varying Dynamical Systems [294]. These models allow for the incremental learning of multi-behavior policies, crucial for robots operating in dynamic settings. Additionally, the Elastic-DS approach embeds task parameters directly into GMM-based formulations, allowing motion policies to generalize to new task instances while preserving stability guarantees through Laplacian editing of the underlying probabilistic model [295].

Efficiency in adaptation is further enhanced through specialized sampling and optimization techniques, which are essential for deploying agents on resource-constrained edge devices where computational latency must be minimized without sacrificing control fidelity. In scenarios requiring rapid decision-making under uncertainty, methods like log-MPPI improve trajectory feasibility by utilizing normal log-normal mixture distributions for sampling, significantly outperforming standard Gaussian-based approaches in cluttered environments [296]. Moreover, the integration of layerwise importance sampling and biased sampling distributions has proven effective in overcoming local minima in stochastic optimal control. The Biased-MPPI method, for example, fuses ancillary controllers to inform the sampling distribution, thereby enhancing convergence and robustness against model uncertainties [297].

The adaptation of agents also extends to interactive and collaborative settings, where robots must interpret human intent and adjust their behaviors accordingly, effectively closing the perception-action loop. Frameworks like InCoRo utilize in-context learning within a classical feedback loop to enable robots to adapt to dynamic environmental changes and correct controller errors without iterative optimization [298]. Similarly, interactive perception systems such as Matcha leverage LLMs to instruct epistemic actions, allowing agents to actively query multimodal sensors (vision, sound, haptics) to resolve ambiguities before executing tasks [299]. These systems highlight the necessity of combining the generative power of LLMs with the precision of probabilistic robotics to achieve true adaptability.

Finally, the scaling of these adaptation techniques to large-scale systems necessitates novel architectural choices that build upon the modular principles seen in earlier sections. For very-large-scale robotic systems, skew-normal mixture models have been proposed to better represent agent distributions in complex environments compared to traditional GMMs, facilitating more effective path planning [300]. Furthermore, the concept of Local Mixture of Experts allows for the specialization of skill libraries, where components focus on local context regions to achieve high accuracy and versatility in skill acquisition [301]. Collectively, these specialized adaptation strategies—ranging from conditional mixture modeling and reward synthesis to efficient sampling and interactive perception—form the backbone of modern embodied agents, enabling them to operate reliably

in the unstructured and dynamic physical world.

5 Optimization, Efficiency, and Deployment Strategies

5.1 Model Compression and Quantization for Resource-Constrained Deployment

5.1 Model Compression and Quantization for Resource-Constrained Deployment

The realization of privacy-preserving, low-latency artificial intelligence on mobile and edge devices hinges on the ability to execute Large Language Models (LLMs) within strict hardware budgets. However, the immense memory footprint and computational demands of modern LLMs often exceed the capacity of individual edge nodes, creating a fundamental barrier to localized agentic AI. To bridge this gap, the research community has developed a robust toolkit of model compression techniques—primarily quantization, pruning, and knowledge distillation. These methods serve as a critical prerequisite for the distributed architectures discussed in the subsequent section, as reducing model size and complexity directly alleviates the memory and communication bottlenecks inherent in splitting models across heterogeneous clusters.

Quantization stands as the most prominent strategy for mitigating memory constraints by converting high-precision floating-point weights and activations into lower-bit integers. Recent advancements have evolved from simple post-training quantization to sophisticated, hardware-aware approaches. For instance, Activation-aware Weight Quantization (AWQ) introduces a hardware-friendly method that protects salient weights based on activation observations, allowing for effective low-bit weight-only quantization without requiring backpropagation [302]. This approach preserves generalization capabilities across diverse domains, including coding and mathematics, enabling the deployment of massive models like the 70B Llama-2 on mobile GPUs. Similarly, OmniQuant offers an omnidirectionally calibrated quantization technique that optimizes various parameters through learnable weight clipping and equivalent transformations, achieving superior performance in extremely low-bit settings such as W4A4 and W2A16 [303].

A specific and critical bottleneck in autoregressive generation is the Key-Value (KV) cache, which grows linearly with sequence length and batch size, often becoming the limiting factor for edge deployment. Addressing this, KIVI proposes a tuning-free asymmetric 2-bit quantization algorithm specifically designed for the KV cache, distinguishing between per-channel quantization for keys and per-token quantization for values [304]. This innovation significantly reduces peak memory usage, enabling larger batch sizes and higher throughput on real workloads. Furthermore, WKVQuant extends these efforts by incorporating a two-dimensional quantization strategy and cross-block reconstruction regularization to handle the complex distribution of KV caches, balancing memory savings with computational efficiency [305].

While quantization reduces precision, pruning aims to directly reduce the number of parameters. Structured pruning, which removes entire layers or blocks, has proven particularly effective for improving inference latency under memory-constrained conditions where width pruning falls short [306]. Research indicates that LLMs possess a surprising robustness to depth reduction, with minimal performance degradation even when a substantial fraction of layers is removed, provided that parameter-efficient fine-tuning is applied to “heal” the network [307]. Innovations like SpQR (Sparse-Quantized Representation) take a hybrid approach by isolating outlier weights that cause quantization errors and storing them in higher precision, while compressing the remaining weights to 3-4 bits, achieving near-lossless compression [308]. Additionally, Rapid Deployment of DNNs for Edge Computing via Structured Pruning at Initialization demonstrates that structured pruning at

the initialization phase can rapidly generate models that are significantly smaller and faster without the extensive computational cost of neural architecture search [309].

Knowledge Distillation (KD) serves as another pivotal mechanism, transferring knowledge from a large teacher model to a compact student model. However, combining KD with quantization presents unique challenges, as the regularization effect of KD can diminish the already reduced representation power of quantized models. To mitigate this, Quantization-aware Knowledge Distillation (QKD) coordinates quantization and distillation in three phases—self-studying, co-studying, and tutoring—to recover full-precision accuracy even at very low bit-widths [310]. Further refining this synergy, Oh! We Freeze analyzes signal propagation to stabilize the KD-quantization process, proposing techniques to freeze specific components during training to prevent low-bit quantization errors from destabilizing the learning process [311].

Beyond algorithmic innovations, the interplay between compression techniques and hardware architecture is crucial for practical deployment. The compressibility of quantized LLMs involves a trade-off between data movement reduction and inference latency, especially when models must be partially loaded from storage on memory-constrained devices [312]. Techniques like Agile-Quant leverage activation-guided strategies and token pruning to optimize both weights and activations for specific edge hardware, achieving significant speedups on devices with limited SIMD capabilities [313]. Moreover, Dual Grained Quantization addresses the inefficiency of fine-grained quantization in matrix multiplication by dequantizing weights to a coarse-grained representation for computation, thus ensuring fast inference speeds alongside high accuracy [314].

The deployment landscape is further enhanced by specialized frameworks that integrate these compression techniques into cohesive systems. For example, Transformer-Lite implements FP4 quantization and sub-tensor techniques to eliminate KV cache copying overhead, achieving order-of-magnitude speedups on mobile phone GPUs [315]. Similarly, EdgeMoE optimizes Mixture-of-Experts models for on-device inference by strategically partitioning expert weights and utilizing bitwidth adaptation to manage memory hierarchy efficiently [316]. Collectively, these advancements in quantization, pruning, and distillation not only dismantle the barriers to standalone edge AI but also lay the foundational groundwork for the distributed inference architectures and model partitioning strategies explored in the following section, where compressed models are strategically split across heterogeneous edge-cloud continuums to maximize system-wide throughput.

5.2 Distributed Inference Architectures and Model Partitioning

5.2 Distributed Inference Architectures and Model Partitioning

While Section 5.1 established the foundational techniques for compressing models to fit within individual device constraints, the sheer scale of modern Large Language Models (LLMs) often necessitates a paradigm shift beyond monolithic execution, even when quantized. When a single edge node cannot accommodate a model despite aggressive compression, the solution lies in distributed inference architectures that split models across heterogeneous edge clusters and cloud tiers. This section explores the architectural frameworks that leverage pipeline parallelism and optimal partitioning strategies to minimize bottleneck latency and maximize throughput, effectively bridging the gap between the compressed models discussed previously and the dynamic orchestration mechanisms detailed in the subsequent section.

A primary strategy in this domain is pipeline parallelism, which divides a neural network into sequential stages executed across different devices to overcome memory capacity limits. The [317]

paper introduces EdgePipe, a distributed framework specifically designed to handle the heterogeneity of edge systems in terms of compute power, memory, and network bandwidth. By employing an optimal partition strategy, EdgePipe enables the execution of large models, such as ViT-Large and ViT-Huge, across multiple edge devices that individually cannot fit the model in memory. Empirical evaluations demonstrate that this approach can achieve significant speedups, reaching up to $10.59\times$ and $11.88\times$ for specific vision transformer models using 16 edge devices, while maintaining accuracy. Similarly, the [318] study presents Pipe-it, a framework that employs a pipelined design to split convolutional layers across heterogeneous core clusters within ARM big.LITTLE architectures. By limiting parallelization to assigned clusters and utilizing a performance-prediction model to balance the pipeline, Pipe-it achieves a 39% higher throughput compared to antecedent methods, highlighting the efficacy of intra-device pipelining alongside inter-device distribution.

Beyond simple pipeline splitting, advanced orchestration systems aim to automate the partitioning and placement of model operators to optimize system-wide performance. The [319] framework addresses the lack of design tools for distributed CNN inference by automatically splitting trained models into sub-models and generating code for collaborative execution on heterogeneous edge devices. This automation reduces energy consumption and memory usage per device while improving overall system throughput. Complementing this, the [320] research proposes an algorithm specifically targeting the minimization of bottleneck latency to maximize inference throughput across networked edge clusters. This system demonstrates robustness and scalability, reducing bottleneck latency by 10x compared to random algorithms and showing fault tolerance across multi-node configurations. Furthermore, the [321] paper isolates the structured optimization problem inherent in device placement for modern pipelined settings, providing algorithms that solve for optimality in partitioning neural network computational graphs across heterogeneous devices.

The challenge of heterogeneity extends to the integration of diverse hardware accelerators, such as CPUs and GPUs, within a single inference pipeline, a critical consideration for LLM agents operating on mixed-device clusters. The [322] system formulates model deployment as an NP-hard scheduling problem, partitioning recurrent neural network graphs to execute operators in parallel on both CPUs and GPUs. This collaborative utilization reduces execution time by up to 19.4% and significantly lowers GPU memory footprints. In the specific context of Large Language Models (LLMs), the [323] framework introduces a principled approach to heterogeneous parallel computing using CPUs and GPUs. By employing asynchronous overlap techniques, HeteGen mitigates I/O bottlenecks common in offloading strategies, achieving inference speed improvements surpassing state-of-the-art methods by over 317%.

Optimizing these distributed architectures also requires addressing the trade-offs between communication overhead and computation, particularly when scaling from single-model execution to multi-tenant serving. The [324] system explores the use of model parallelism not just for scaling single models, but for the statistical multiplexing of multiple models across a distributed cluster. AlpaServe navigates the trade-off between parallelism overhead and burstiness handling, processing requests at up to 10x higher rates while adhering to latency constraints. Additionally, the [325] describes an industry-scale prototype that automates the splitting of Deep Neural Networks between cloud servers and edge devices. This framework ensures high model accuracy while keeping end-to-end latency low, validating the practical viability of deep model splitting in real-world applications.

Resource management and latency optimization are further refined through holistic modeling and static scheduling policies that define the initial deployment topology. The [326] approach introduces a graph-based hardware representation and a hierarchical orchestrator to map tasks across the edge-

cloud continuum, improving latency by up to 47% over baselines. For scenarios with strict Service Level Agreements (SLAs), the [327] framework redefines physical resources into logically isolated units to maximize throughput while guaranteeing prioritized services. Moreover, the [328] paper formulates a multi-objective optimization problem to find Pareto efficient solutions for splitting CNNs between edge and cloud, ensuring feasible execution under tight resource constraints.

Finally, the efficiency of these distributed systems relies on accurate performance modeling and adaptive strategies to inform the static partitioning decisions discussed above. The [329] tool aids in this by generating micro-benchmarks to estimate ideal execution latency, thereby identifying optimization opportunities in model serving pipelines. Adaptive approaches like [330] further enhance efficiency by adapting model architectures post-deployment to suit specific edge environments, achieving superior accuracy-latency trade-offs. Collectively, these frameworks illustrate that effective distributed inference requires a synergistic combination of optimal model partitioning, heterogeneous resource orchestration, and adaptive communication strategies. While this section focuses on the structural design and static placement of split models, these fixed architectures serve as the essential substrate upon which the dynamic, learning-based resource allocation and service orchestration techniques discussed in Section 5.3 operate to handle real-time variability.

5.3 Dynamic Resource Allocation and Service Orchestration

5.3 Dynamic Resource Allocation and Service Orchestration

While Section 5.2 established the architectural foundations for splitting models across heterogeneous devices through static partitioning and pipeline parallelism, the deployment of Large Language Model (LLM) agents in real-world scenarios demands a complementary layer of dynamic adaptability. Static strategies, though effective for fixed workloads, often fail to address the stochastic nature of agent interactions, time-varying network conditions, and fluctuating resource availability. Consequently, Deep Reinforcement Learning (DRL) has emerged as a pivotal methodology for enabling adaptive task offloading and multi-dimensional resource scheduling. These approaches allow systems to learn optimal policies that dynamically balance computation, communication, and energy costs without relying on predefined mathematical models of the environment, which are often intractable in the highly variable landscapes where LLM agents operate.

A primary challenge in orchestrating LLM agent workflows is the efficient distribution of computational tasks across a hierarchy of end devices, edge servers, and cloud data centers. This problem is frequently formulated as a Markov Decision Process (MDP) where the agent must decide whether to execute a task locally or offload it to a remote node based on current system states. Research has demonstrated that DRL algorithms can effectively solve these joint optimization problems. For instance, in mobile edge computing contexts, DRL-based frameworks have been proposed to jointly determine offloading decisions and resource allocation under time-varying wireless channels and stochastic computing capabilities [331]. These methods significantly outperform conventional numerical optimization techniques, particularly when dealing with the combinatorial complexity of task dependencies and large state spaces. Furthermore, the integration of DRL allows for the minimization of system energy consumption while adhering to strict latency constraints, a critical factor for sustaining long-running agent operations on battery-powered devices [332].

The complexity of resource orchestration increases exponentially in multi-user and multi-agent environments where resources are shared and competitive. In such scenarios, decentralized or multi-agent DRL approaches are essential to mitigate the curse of dimensionality and preserve user privacy. Multi-agent systems enable individual nodes to learn collaborative or competitive

strategies through localized observations. For example, combinatorial client-master multi-agent DRL algorithms have been developed to handle distinct constraints on user devices and server storage simultaneously, allowing for more nuanced resource negotiation than homogeneous agent models [333]. Similarly, decentralized strategies utilizing algorithms like Deep Deterministic Policy Gradient (DDPG) have shown superior performance in minimizing long-term computation costs by allowing devices to adaptively allocate power for local execution and offloading without requiring global system knowledge [334]. These architectures are particularly relevant for LLM agent swarms that must coordinate inference tasks across a distributed fabric without a central bottleneck.

Beyond simple offloading decisions, advanced orchestration frameworks must address the heterogeneity of hardware and the dynamic nature of workloads. Hierarchical frameworks have been introduced to decouple global resource allocation from local power management, utilizing global DRL agents for virtual machine placement and local predictors for fine-grained power control [335]. This separation of concerns enables scalable management of warehouse-scale data centers, which is increasingly important as LLM training and inference jobs consume vast amounts of energy. Moreover, the concept of “green” orchestration has gained traction, where DRL is employed to schedule workloads based on carbon intensity and energy prices. Multi-agent reinforcement learning has been successfully applied to geo-distributed data centers to maximize GPU utilization while minimizing carbon emissions, demonstrating that intelligent scheduling can align operational efficiency with sustainability goals [336].

To further enhance the adaptability of resource orchestration, recent studies have incorporated risk quantification and meta-learning into DRL frameworks. In highly dynamic environments such as UAV-assisted edge computing, standard reward functions may not adequately penalize catastrophic failures like deadline violations for critical tasks. By integrating financial risk concepts such as Conditional Value at Risk (CVaR) into the learning objective, agents can learn to avoid risky behaviors that lead to irreversible consequences, thereby ensuring higher reliability for time-sensitive agent interactions [337]. Additionally, the volatility of network conditions and workload patterns can cause traditional DRL models to degrade. Meta-reinforcement learning approaches address this by enabling agents to rapidly adapt to new environments with minimal data, leveraging parallel deep neural networks to make fine-grained offloading decisions that remain effective even as system parameters shift [338].

The orchestration of services also extends to specialized domains such as serverless computing and integrated energy systems. In serverless architectures, DRL is utilized to optimize function scheduling and automatic resource scaling, tackling challenges like cold starts and fluctuating demand [339]. Similarly, in integrated energy systems powering edge infrastructure, state-adversarial DRL methods have been developed to maintain stable demand-supply balances despite cyber-attacks and renewable energy uncertainties, ensuring the resilience of the underlying infrastructure supporting AI agents [340]. These advancements collectively illustrate that dynamic resource allocation via DRL serves as the critical bridge between the static model partitioning discussed previously and the high-throughput serving mechanisms explored next. While Section 5.4 focuses on optimizing throughput within the server through batching and early exiting, the DRL-driven orchestration detailed here ensures that requests are routed to the most appropriate resources dynamically, forming a foundational component for the reliable, efficient, and scalable deployment of autonomous LLM agents in diverse and unpredictable computational landscapes.

5.4 Multi-Tenancy, Batching, and Early Exiting for High-Throughput Serving

5.4 Multi-Tenancy, Batching, and Early Exiting for High-Throughput Serving

While Section 5.3 established how Deep Reinforcement Learning dynamically orchestrates resources across distributed nodes to handle stochastic workloads, the actual execution of Large Language Model (LLM) agents within individual serving instances requires a complementary layer of intra-server optimization. Once requests are routed to a specific compute node, the challenge shifts from *where* to execute to *how* to process concurrent demands efficiently. In production environments, this necessitates sophisticated serving strategies that balance the competing imperatives of high throughput and low latency, particularly under multi-tenant workloads where request arrivals are heterogeneous and unpredictable. Traditional inference systems often falter due to the “head-of-line” blocking problem inherent in run-to-completion processing, where long-generation requests obstruct shorter ones, severely degrading average and tail job completion times (JCT). To mitigate these bottlenecks, modern serving architectures have evolved to integrate dynamic batching, early exiting mechanisms, and intelligent caching, creating a robust foundation for the uncertainty-aware scheduling discussed in the subsequent section.

Dynamic batching serves as a cornerstone for enhancing throughput in multi-tenant settings by aggregating multiple inference requests into a single computational kernel execution. However, coarse-grained graph batching frequently fails to adapt to the fluid nature of real-world traffic, leaving significant performance gains untapped. Advanced systems like [341] propose an SLA-aware approach that performs batching at the granularity of individual graph nodes rather than the entire model graph. This fine-grained flexibility allows the system to intelligently determine which nodes can be efficiently batched together, yielding substantial improvements in response time and throughput while strictly adhering to Service Level Agreements. Furthermore, integrating batching with preemptive scheduling is critical for minimizing JCT. [342] introduces a system that exploits the autoregressive nature of LLMs to enable preemption at the token level. By utilizing a skip-join Multi-Level Feedback Queue scheduler, such systems can prioritize incoming short requests over long-running ones, effectively mitigating head-of-line blocking and significantly reducing tail latency compared to state-of-the-art solutions.

Complementing batching, early exiting (EE) mechanisms offer a powerful paradigm for adaptive computation, allowing models to terminate inference at intermediate layers once a confidence threshold is met. This approach is particularly effective in multi-tenant scenarios where input complexity varies widely; simple queries can exit early, freeing up computational resources for more demanding tasks. However, harmonizing early exiting with batching presents technical hurdles, as traditional token-level exit methods often require waiting for the slowest sample in a batch to exit, thereby negating latency benefits. [343] overcomes this limitation by establishing a singular exit point for every token in a batch at each sequence position, guaranteeing a monotonic decrease in exit points. This design enables seamless compatibility with Key-Value (KV) caching and batch inferencing, yielding inference speedups of 2x to 5x with negligible performance regression. Similarly, [344] repurposes early exits to provide continual feedback for runtime monitoring, dynamically adapting to time-varying overheads to lower median response latencies by up to 91.5% without compromising throughput.

The synergy between batching and early exiting is further refined through hardware-aware optimizations and specialized scheduling algorithms. [345] introduces a novel scheduling algorithm that accounts for the dynamicity introduced by both request arrivals and early-exiting processes. By enabling sample preemption at runtime and utilizing stackable processing elements, this approach

significantly improves Neural Processing Unit (NPU) utilization and SLO satisfaction under spiking loads. Moreover, the efficiency of these systems relies heavily on effective memory management, particularly regarding KV caches which scale linearly with sequence length. Techniques such as [346] accelerate inference by reusing attention states across different prompts that share overlapping text segments, such as system messages or templates. This modular caching strategy drastically reduces the time-to-first-token, especially for long-context applications, without requiring model parameter modifications.

Beyond algorithmic refinements, cost-effective serving in multi-tenant clouds often depends on strategic heterogeneous resource allocation. [347] demonstrates that intelligently employing a diverse set of cloud computing instances can meet QoS targets while maximizing cost savings. By using Bayesian Optimization to select the optimal mix of heterogeneous instances, such systems outperform homogeneous pools, achieving significant reductions in inference service costs. Furthermore, the complexity of managing vast model variants is addressed by model-less systems like [348], which automatically navigates the trade-off space of model variants, hardware architectures, and optimizations to meet specific application objectives. This system combines VM-level horizontal autoscaling with model-level autoscaling, allowing multiple different model variants to serve queries within a single machine, thereby achieving higher throughput and lower costs.

Finally, the optimization of shared prefixes in batched requests represents a critical frontier in high-throughput serving. [349] introduces a hardware-aware attention implementation that computes attention over shared prefixes and unique suffixes separately. This decomposition reduces redundant memory reads and enables the use of hardware-friendly matrix multiplications, improving end-to-end throughput by up to 32x against competitive baselines. Such innovations underscore that the future of high-throughput serving lies not merely in scaling hardware, but in the algorithmic co-design of batching policies, early exit strategies, and memory caching mechanisms. Collectively, these techniques tame the latency-throughput tension inherent in multi-tenant LLM deployment, providing the stable, high-efficiency substrate necessary to address the stochastic inference times and uncertainty challenges explored in the following section on real-time latency optimization.

5.5 Uncertainty-Aware Scheduling and Real-Time Latency Optimization

5.5 Uncertainty-Aware Scheduling and Real-Time Latency Optimization

While the previous section detailed dynamic batching, early exiting, and caching to maximize throughput in multi-tenant environments, these techniques often assume a degree of predictability in request patterns that does not always hold for Large Language Model (LLM) agents. The deployment of LLM agents in latency-critical applications—such as real-time conversation systems, autonomous driving decision modules, and emergency response coordination—introduces a fundamental challenge: the stochastic nature of language generation leads to highly variable inference times. Unlike traditional software services with deterministic execution paths, the computational cost of an LLM inference is directly proportional to the number of tokens generated, which is inherently uncertain at the time of request arrival. This variability can cause severe bottlenecks even in optimized serving architectures, leading to violated Service Level Agreements (SLAs) and degraded user experiences. To address this, recent research has pivoted toward uncertainty-aware scheduling frameworks that quantify input-induced uncertainty to predict output lengths and dynamically adjust resource allocation strategies, bridging the gap between static optimization and

dynamic reality.

A primary avenue for optimization involves quantifying the uncertainty intrinsic to the input prompt to forecast the complexity and length of the subsequent generation. The variance in output length is not merely random noise but is often correlated with specific characteristics of the input data. For instance, the [350] framework explicitly identifies that input uncertainties adversely affect latency, frequently resulting in increased output lengths. By developing lightweight methods to correlate input text uncertainties with output length at runtime, systems can utilize these quantifications as latency heuristics. This approach allows system-level schedulers to implement uncertainty-aware prioritization, where requests predicted to be short and low-uncertainty are processed with higher priority, while complex, high-uncertainty tasks are strategically offloaded or consolidated to prevent queue saturation. This dynamic correlation is essential for maintaining throughput under high-traffic workloads, as it transforms unpredictable latency into a manageable variable, complementing the SLA-aware batching strategies discussed previously.

Beyond simple length prediction, effective scheduling requires robust mechanisms to handle tail latencies and rare, high-cost events that can overwhelm even the most efficient batching systems. Traditional system designs often rely on average metrics, such as mean queue length or average latency, which fail to capture the risks associated with low-probability, high-impact delays. In contrast, advanced frameworks incorporate probabilistic constraints to characterize the occurrence of such extreme events. The [351] study proposes a novel network design that invokes results from extreme value theory to model the probability of queue length violations. By formulating the resource allocation problem as a minimization of power and computation subject to strict latency and reliability constraints, these systems can guarantee performance bounds even in the presence of stochastic demand. This is particularly relevant as we transition toward edge computing scenarios, where resources are finite, and the cost of a latency spike can be catastrophic for time-sensitive agent actions—a theme that becomes central in the following discussion on communication-efficient edge protocols.

Furthermore, the integration of predictive models into the scheduling loop enables proactive rather than reactive resource management. In multi-tiered computing environments, such as fog or cloud-edge continua, the decision to offload tasks depends heavily on the anticipated computational burden. The [352] approach demonstrates that exploiting unique problem structures through traffic prediction can minimize time-average power consumption while ensuring queue stability. Although originally designed for IoT workloads, the principles of PORA are directly applicable to LLM agents, where predicting the “traffic” of token generation allows for near-optimal offloading decisions. By requiring only mild predictive information, such schemes can achieve notable latency reductions even when prediction errors occur, providing a robust buffer against the inherent unpredictability of generative models and laying the groundwork for the distributed inference architectures explored next.

The challenge of uncertainty is further compounded in dynamic environments where system conditions fluctuate rapidly, akin to the variable wireless channels faced by edge devices. Static resource allocation policies are ill-suited for such scenarios, necessitating frameworks that can adapt to changing information states. The [353] framework, known as Stochastic Processing under Imperfect Information (SPII), provides a theoretical foundation for understanding how information constraints impact capacity. It highlights that the presence of noisy channels—analagous to imperfect uncertainty estimates in LLMs—always reduces system capacity, but that increasing memory for allocation policies can mitigate this reduction. This insight suggests that uncertainty-aware schedulers for LLM agents should not only rely on immediate uncertainty estimates but also main-

tain historical context to refine their allocation policies over time, thereby improving the overall capacity of the serving infrastructure in the face of dynamic network conditions.

To operationalize these theoretical advances, practical implementations often leverage speculative execution and dynamic model adjustments, creating a synergy with the early exiting and caching mechanisms mentioned earlier. Techniques such as speculative sampling, as discussed in [354], can be synergized with uncertainty quantification to reduce overall latency. By using a smaller, faster model to draft tokens and a larger model to verify them, systems can exploit the uncertainty distribution of the draft model to skip unnecessary computations. When combined with uncertainty-aware scheduling, this allows the system to allocate more verification resources to high-uncertainty regions of the generation while accelerating low-uncertainty segments. Additionally, the [355] approach illustrates how hardware-software co-design can exploit temporal locality in query streams. By dynamically activating different sub-networks based on real-time scheduling decisions, systems can achieve significant improvements in latency and energy efficiency, adapting the model’s computational footprint to the specific uncertainty profile of each incoming request.

Finally, the reliability of these scheduling mechanisms hinges on the accuracy of the underlying uncertainty estimates. If the uncertainty quantification is miscalibrated, the scheduler may make erroneous decisions, leading to either resource under-utilization or SLA violations, regardless of the sophistication of the batching or offloading strategy. Research into [356] emphasizes that many current methods fail to maintain coverage on out-of-distribution data, which is common in open-world agent interactions. Therefore, future uncertainty-aware schedulers must integrate calibration mechanisms that ensure the predicted uncertainty intervals accurately reflect the true variability of output lengths across diverse and shifting data distributions. Only by grounding scheduling decisions in rigorously calibrated uncertainty estimates can LLM agents achieve the reliability required for deployment in critical, real-time applications, ensuring seamless operation as they extend into the bandwidth-constrained edge and fog environments detailed in the subsequent section.

5.6 Communication-Efficient Protocols in Edge and Fog Systems

5.6 Communication-Efficient Protocols in Edge and Fog Systems

Building upon the uncertainty-aware scheduling and predictive offloading strategies discussed in the previous section, the practical realization of distributed LLM agent architectures hinges on the efficiency of the underlying communication fabric. While predictive models can anticipate computational loads and optimize resource allocation, the physical transmission of data between resource-constrained edge devices and powerful fog or cloud servers remains a fundamental bottleneck due to limited bandwidth and high latency in wireless networks. As agents increasingly rely on distributed architectures to perform complex reasoning and tool invocation, the volume of data exchanged—ranging from intermediate feature vectors to full context windows—can severely degrade real-time performance. To address this, recent research has synthesized a variety of communication-efficient protocols that move beyond traditional data transmission paradigms. These approaches focus on intermediate feature compression, task-oriented encoding, and joint communication-computation optimization frameworks, creating the necessary infrastructure to support the dynamic, uncertainty-driven workloads characterized earlier.

A primary strategy for reducing communication overhead involves the compression of intermediate features rather than raw data or full model parameters, a technique that aligns closely with the

split inference architectures necessitated by edge constraints. In these setups, where a neural network is partitioned between an edge device and a server, transmitting high-dimensional feature vectors can be prohibitively expensive. To mitigate this, researchers have proposed frameworks that optimize the trade-off between on-device computation and the communication cost of forwarding intermediate features. For instance, a three-step framework involving model split point selection, communication-aware model compression, and task-oriented encoding has been demonstrated to significantly reduce inference latency compared to baseline methods [357]. Furthermore, advanced techniques such as deep joint source-channel coding (DeepJSCC) have been integrated with filter pruning strategies to simultaneously reduce redundant complexity and ensure reliable transmission over wireless links [358]. This approach is particularly effective because it treats the communication channel as an integral part of the learning process, rather than a separate pipeline, thereby achieving improved end-to-end reliability even under the stochastic channel conditions highlighted in prior discussions on dynamic resource allocation.

The concept of “task-oriented communication” has emerged as a pivotal paradigm shift within this domain, wherein the goal is to transmit only the information essential for the downstream inference task rather than reconstructing the original data samples. This principle leverages the Information Bottleneck (IB) theory to extract task-relevant features at the edge device, effectively removing data redundancy before transmission. In multi-device cooperative scenarios, a distributed information bottleneck (DIB) framework has been formalized to characterize the optimal rate-relevance tradeoff, allowing multiple low-end devices to transmit compact features to a central server for joint inference [359]. Extending this to video analytics, which is crucial for embodied agents, researchers have developed temporal entropy models that exploit the correlation between consecutive frames to further reduce bitrate, alongside spatial-temporal fusion modules at the server to maintain high inference performance [360]. Similarly, for static image tasks, variational information bottleneck (VIB) approximations have been employed to make the optimization tractable, often combined with sparsity-inducing distributions to sparsify the encoded feature vectors [361]. By minimizing the data footprint, these methods directly support the low-latency requirements essential for the real-time agent interactions discussed previously.

Another significant avenue for optimization is the dynamic adaptation of compression and transmission strategies based on real-time channel conditions and task requirements, mirroring the adaptive nature of the uncertainty-aware schedulers described in Section 5.5. Static compression ratios often fail to account for the variability in wireless environments, leading to either excessive latency or degraded accuracy. To address this, variable-length feature encoding schemes based on dynamic neural networks have been proposed to adaptively adjust the activated dimensions of encoded features according to current channel states [361]. In systems with hard deadlines, dynamic compression ratio selection algorithms balance the trade-off between communication costs and inference accuracy by estimating uncertainty and retransmitting less compressed data only when necessary [362]. Moreover, progressive feature transmission protocols have been introduced, where features are transmitted incrementally until a target confidence level is reached, effectively terminating transmission early when the server has sufficient information to make a decision [363]. This synergy between uncertainty estimation and adaptive transmission ensures that bandwidth is utilized only when the potential gain in inference quality justifies the cost.

Joint optimization of communication, computation, and sensing resources represents the frontier of efficient protocol design, pushing the boundaries of what is possible in resource-scarce environments. Integrated Sensing-Communication-Computation (ISCC) schemes aim to merge these traditionally separate stages to maximize resource utilization. For example, ISAC (Integrated Sensing and

Communication) systems utilize wireless signals for both dataset generation and uploading, though careful beamforming and time allocation are required to manage interference [364]. In multi-device settings, optimal ISCC schemes have been derived using sum-of-ratios methods to jointly determine transmit power, time allocation, and quantization bits, thereby maximizing the discriminant gain for classification tasks [365]. Additionally, over-the-air computation (AirComp) techniques have been integrated with ISCC to aggregate denoised global feature vectors directly in the broadband channel, suppressing noise while completing downstream inference tasks efficiently [366]. These holistic approaches lay the groundwork for the robust, self-healing systems required in the next section, as efficient resource usage is a prerequisite for sustaining operations during fault recovery.

Finally, the integration of semantic communication principles offers a robust pathway for collaborative inference, especially for transformer-based models common in LLM agents. Attention-aware patch selection strategies allow edge devices to identify and transmit only the image patches critical for classification, drastically cutting communication overhead while preserving accuracy [367]. Similarly, semantic-aware deep JSCC frameworks have been designed to support multi-task services, such as simultaneous image recovery and classification, by unifying coding rate reduction and mean square error minimization in the loss function [368]. These advancements collectively illustrate that the future of edge AI relies not merely on faster networks, but on intelligent protocols that deeply couple communication strategies with the computational and semantic requirements of the learning tasks they support. However, while these protocols optimize efficiency under normal operating conditions, they must also function reliably amidst hardware faults and adversarial interruptions. The transition from optimizing bandwidth to ensuring system resilience in the face of such disruptions forms the critical bridge to the following discussion on reliability, fault tolerance, and security in distributed systems.

5.7 Reliability, Fault Tolerance, and Security in Distributed Systems

5.7 Reliability, Fault Tolerance, and Security in Distributed Systems

While the communication-efficient protocols discussed in the previous section optimize bandwidth and latency for LLM agents at the edge, these gains are contingent upon the underlying system’s ability to withstand hardware faults, network volatility, and adversarial interruptions. The deployment of Large Language Model (LLM) agents across distributed edge and fog ecosystems introduces profound challenges regarding robustness; as agentic architectures scale to handle complex, multi-step reasoning and tool execution, they become increasingly susceptible to disruptions that can derail entire reasoning chains. Consequently, maintaining operational continuity requires a paradigm shift from reactive error handling to proactive, self-supervised fault prediction and resilient orchestration. The core objective is to ensure that agent swarms can perceive failures, isolate compromised components, and reconfigure their communication topologies without human intervention, thereby preserving the integrity of long-horizon tasks essential for the cost-efficient and sustainable operations explored in the subsequent section.

A fundamental aspect of reliability in such distributed environments is the capacity to predict failures before they cascade into system-wide collapses. Traditional approaches relying on static thresholds or supervised models trained on historical data often fail to generalize to the dynamic workloads characteristic of LLM inference. Recent advancements suggest that self-supervised and unsupervised techniques offer superior adaptability in these volatile settings. For instance, frameworks like DeepFT utilize deep surrogate models combined with co-simulation based self-supervised learning to dynamically adapt to changing conditions, effectively predicting system overloads and

optimizing task migration decisions in resource-constrained edge clusters [369]. Similarly, unsupervised approaches such as PREVENT have demonstrated the capacity to predict and localize failures in distributed enterprise applications by identifying anomalous combinations of system metrics without requiring predefined rules or extensive training on observed failures [370]. Integrating such predictive capabilities allows agent orchestrators to proactively offload tasks or switch to backup models, ensuring the seamless service continuity required for high-stakes applications.

Beyond prediction, the architecture of the distributed system itself must embody resilience through self-stabilization and self-healing properties. In multi-agent systems, communication protocols must remain robust against node failures and network partitions. Self-stabilizing algorithms provide a theoretical guarantee that a system will converge to a correct state from any arbitrary initial configuration, a property essential for recovering from transient faults or arbitrary state corruptions. Research into self-stabilizing overlay networks has produced algorithms that can rebuild correct network topologies in polylogarithmic time while maintaining low node degrees, offering a blueprint for resilient agent communication layers [371]. Furthermore, in the context of fog computing, self-stabilizing control planes have been proposed to cope with a broad fault model, including packet drops and node fail-stop failures, ensuring automatic recovery within a constant number of communication rounds [372]. Applying these principles to LLM agent networks enables decentralized teams to re-establish connectivity and synchronize states even after severe disruptions, preventing the “lost agent” phenomenon where sub-agents become unreachable during critical execution phases.

The challenge of fault tolerance is further complicated by the interdependence of cyber and physical components in embodied agent systems, where failures can propagate contagiously across interconnected subsystems. This necessitates sophisticated message-passing algorithms to analyze and mitigate cascading effects. Studies on interdependent cyber-physical networks have introduced factor graph representations to model failure propagation and healing dynamics, providing guidelines for selecting network parameters that maximize resiliency against cascading failures [373]. For LLM agents coordinating physical robots or IoT devices, similar graph-based orchestration mechanisms can detect when a specific tool or sensor fails and dynamically reroute plans to alternative agents or modalities. Additionally, the concept of graceful degradation—temporarily weakening system requirements to maintain safety during adverse conditions—can be coordinated with recovery mechanisms through requirement-driven adaptation. This dual approach allows agents to continue operating in a reduced capacity rather than failing completely, resuming full functionality once environmental conditions stabilize [121].

Security in distributed agent systems is inextricably linked to reliability, as malicious actors often exploit fault tolerance mechanisms to launch attacks. Adversarial nodes may mimic slow processes to evade detection or inject false data to trigger unnecessary recovery actions, creating a dilemma known as the self-healing dilemma where fault correction versus fault tolerance must be carefully balanced [374]. To counter such threats, resilient orchestration must incorporate confidence-aware models that assess the trustworthiness of incoming data and peer agents. For example, confidence-aware resilience models like CAROL utilize generative neural networks to predict Quality of Service and assign confidence scores, enabling systems to optimize broker-worker topologies and reject unreliable nodes efficiently [375]. Moreover, the integration of chaos engineering principles into the deployment pipeline allows for the proactive identification of vulnerabilities by simulating adverse events in production-like environments, thereby strengthening the system’s defense against both accidental faults and malicious intrusions [376].

Finally, the design of fault-tolerant routing and synchronization protocols is vital for maintaining

the coherence of distributed agent memories and shared context, a prerequisite for the efficient caching strategies discussed later. In high-performance computing and network-on-chip architectures, algorithms like Hermes and Fashion have demonstrated near-ideal fault tolerance by dynamically reconfiguring routing paths around faulty links while guaranteeing deadlock freedom [377; 378]. Translating these hardware-level resilience strategies to the software-defined networking of agent clusters ensures that message passing remains robust even under high fault rates. Additionally, fault-tolerant synchronization schemes using anonymous pulses or discrete mean-fields can maintain temporal coherence among agents without relying on heavy consensus primitives, which is crucial for real-time collaborative reasoning [379]. By synthesizing these advanced prediction, self-stabilization, and security mechanisms, the next generation of LLM agent deployments can achieve the levels of reliability required for critical applications, laying the groundwork for the cost-efficient and green AI frameworks that follow.

5.8 Cost-Efficient Foundation Model Caching and Green AI

5.8 Cost-Efficient Foundation Model Caching and Green AI

Building upon the resilient orchestration and fault-tolerant routing mechanisms established in the previous section, the scalability of distributed LLM agent systems ultimately hinges on their economic viability and environmental sustainability. While the preceding discussion ensured that agent swarms could withstand disruptions and maintain coherence under adverse conditions, the rapid proliferation of generative AI services powered by large foundation models has precipitated an unprecedented surge in computational demand. This expansion poses critical challenges regarding operational expenditures and global carbon emissions, as the energy consumption associated with training and inference becomes a bottleneck for widespread deployment. Consequently, realizing the full potential of the reliable systems described earlier necessitates a paradigm shift toward “Green AI,” where cost-efficiency and carbon footprint minimization are treated as primary optimization objectives alongside model accuracy and latency. A pivotal strategy in this domain involves the synergistic optimization of foundation model caching policies, energy-aware hardware configurations, and carbon-intelligent scheduling to deliver high-performance AI services within strict sustainability constraints.

Foundation models, often comprising billions of parameters, impose severe memory and compute burdens on edge servers and data centers, rendering naive deployment strategies economically and environmentally untenable. To address the resource limitations inherent in edge infrastructure—where the fault-tolerant agents from Section 5.7 operate—recent research has proposed joint frameworks for foundation model caching and inference. These approaches recognize that effective management of cached models and user requests is essential for balancing the trade-offs among inference latency, accuracy, and resource consumption. For instance, novel metrics such as the “Age of Context” (AoC) have been introduced to model the freshness and relevance of historical prompts, allowing systems to implement least-context caching algorithms that significantly reduce system costs by leveraging in-context learning capabilities efficiently [380]. By dynamically managing which models reside in limited edge memory based on contextual relevance rather than simple recency, these systems can mitigate the need for frequent, energy-intensive model swaps and redundant computations, thereby complementing the self-healing properties of the underlying network with efficient resource utilization.

Beyond algorithmic caching strategies, the physical layer of computation plays a decisive role in determining the carbon intensity of AI workloads. The design of environmentally sustainable

computing systems requires a holistic approach that accounts for both hardware specialization and software parameters, extending the resilience concepts of cyber-physical interdependence to ecological impact. Research indicates that specialized AI accelerators and Extended Reality (XR) systems can achieve dramatic improvements in carbon efficiency—up to a tenfold reduction in the product of total CO₂e and execution time—through optimized design space exploration [381]. This suggests that for LLM agents, the selection of hardware architectures must be driven by carbon-aware metrics rather than raw performance alone. Furthermore, the embodied carbon footprint of the hardware itself, particularly in the context of IoT and edge devices, cannot be ignored; studies show that the production phase of complex devices can vary in carbon impact by a factor of over 150, necessitating a life-cycle approach to hardware provisioning that aligns with the long-term durability goals of robust distributed systems [382].

To further align AI operations with sustainability goals, carbon-aware scheduling and resource scaling mechanisms have emerged as vital components of green deployment strategies, acting as a dynamic layer over the static fault-tolerance protocols. Cloud and edge environments exhibit temporal and spatial variations in carbon intensity due to the fluctuating mix of renewable energy sources. Systems like CarbonScaler exploit the elasticity of batch workloads to dynamically vary server allocation based on real-time carbon costs, achieving significant emissions reductions compared to static or carbon-agnostic policies [383]. Similarly, learning-augmented algorithms have been developed to handle online resource scaling with uncertain job lengths, integrating machine-learned predictions to minimize carbon footprints while respecting deadlines [384]. These methodologies are directly applicable to LLM agent orchestration, where task execution can be deferred or migrated to geographic regions with greener energy grids without compromising service level objectives, effectively adding an ecological dimension to the task migration decisions discussed in the context of fault prediction.

In the realm of model optimization, techniques such as knowledge distillation and multi-objective configuration tuning offer pathways to create compact, energy-efficient models suitable for deployment on resource-constrained devices, further reducing the load on the resilient infrastructure. The “Avatar” approach, for example, formulates model optimization as a configuration tuning problem solved via Satisfiability Modulo Theories (SMT) solvers, producing models that are drastically smaller and more energy-efficient while maintaining competitive effectiveness [385]. Such compact models reduce the memory footprint required for caching, thereby increasing the cache hit rate and lowering the energy cost per inference. Additionally, the integration of carbon-aware metrics into neural architecture search (NAS) processes allows for the automatic discovery of model architectures that inherently possess lower energy requirements during both training and inference phases [386].

The synergy between caching, hardware configuration, and carbon-aware scheduling creates a robust framework for sustainable AI. However, accurate monitoring and attribution of energy consumption at the application level remain prerequisites for effective optimization, mirroring the need for precise failure detection in reliable systems. Non-intrusive techniques, such as those employed by WattScope, enable the disaggregation of server-level power usage into application-specific values, providing the granular data necessary to evaluate the true carbon cost of specific agent tasks [387]. Without such visibility, optimizing for “Green AI” remains speculative. Moreover, the economic viability of these green strategies often depends on external factors such as carbon taxation and the cost of renewable energy infrastructure. Analyses suggest that while solar-powered base stations may not yet be economically attractive under current market conditions, significant carbon taxes could render green networking solutions financially viable, thereby incentivizing the adoption of

carbon-efficient caching and inference protocols across the industry [388].

Ultimately, the path toward cost-efficient and green AI for LLM agents lies in the co-design of software caching policies, model compression techniques, and hardware provisioning strategies, seamlessly integrating with the reliability and security foundations laid in prior sections. By treating carbon efficiency as a first-class citizen in system architecture, researchers and practitioners can ensure that the scaling of autonomous agents does not come at an untenable environmental cost. Future developments must continue to refine these integrated approaches, leveraging real-time carbon intensity data and advanced predictive models to orchestrate agent workflows in a manner that is both economically sustainable and ecologically responsible, ensuring the long-term feasibility of the distributed intelligence ecosystems envisioned throughout this survey.

6 Domain-Specific Applications and Embodied Intelligence

6.1 Vertical-Specific Deployments in Knowledge-Intensive Domains

6.1 Vertical-Specific Deployments in Knowledge-Intensive Domains

While the preceding sections have established the foundational mechanisms of tool learning and function calling, their most critical validation occurs in knowledge-intensive domains where the cost of error is exceptionally high. In sectors such as healthcare, finance, law, and software engineering, reliance on static parametric knowledge is insufficient due to the dynamic nature of information and the severe consequences of hallucination. Consequently, the paradigm has shifted toward agentic architectures that leverage external tools, APIs, and retrieval mechanisms to ground responses in verifiable reality. This subsection synthesizes recent advancements and case studies illustrating how these frameworks enhance precision and mitigate the propensity for generating factually incorrect content, serving as a necessary precursor to the physical embodiment of agents discussed in the following section.

In the healthcare domain, the integration of LLMs into clinical workflows demands rigorous adherence to evidence-based medicine and patient safety. The inherent risk of hallucination—generating plausible but unverified medical advice—poses a significant barrier to adoption. Recent research demonstrates that equipping LLMs with tool-use capabilities allows them to function as autonomous practitioners of evidence-based medicine. By utilizing tools to retrieve information from case files and interact with real-world healthcare infrastructure, agents can make clinical decisions that conform to established guidelines while resisting the urge to hallucinate diagnoses or treatments [389]. Specifically, frameworks employing Retrieval Augmented Generation (RAG) have been shown to tailor recommendations to individual patients and specific healthcare systems, thereby reducing the reliance on the model’s internal, potentially outdated knowledge base. Furthermore, the development of specialized benchmarks like Med-HALT and K-QA highlights the necessity of evaluating agents on their ability to distinguish between reasoning-based and memory-based hallucinations, ensuring that deployed systems can accurately retrieve and synthesize complex medical information [390] [391]. These tools enable agents to query up-to-date medical literature and patient records, effectively acting as a safeguard against the fabrication of medical facts which could lead to detrimental patient outcomes.

Similarly, the financial sector presents a challenging environment where accuracy in numerical data and market trends is paramount. LLMs often struggle with mathematical facts and the synthesis of language-based context, leading to hallucinations that can misguide critical investment decisions. To address this, researchers have developed data-centric frameworks that transform raw financial

data tables into hierarchical textual chunks, enabling agents to perform precise question-answering tasks. These systems incorporate multi-metric scoring to assess response confidence and detect hallucinations before they reach the decision-maker, achieving high confidence scores across diverse query types including trend prediction and anomaly detection [392]. Empirical examinations reveal that off-the-shelf models exhibit serious hallucination behaviors in financial tasks, particularly when querying historical stock prices or explaining complex terminologies. However, the integration of prompt-based tool learning, where the agent generates function calls to execute precise queries rather than relying on internal memory, significantly alleviates these issues [393]. The journey toward reliable financial AI involves iterating through prototyping, scaling, and incorporating human feedback loops to ensure that generated reports and alerts remain faithful to the underlying data [2].

The legal profession further exemplifies the need for tool-augmented agents due to the intricate interplay of statutes, case law, and subjective interpretation. Hallucinations in this domain, such as the citation of non-existent legal rulings, have already led to significant professional repercussions. The complexity of legal reasoning requires a multidimensional approach that current monolithic generative models struggle to replicate without external support. Innovative approaches propose ensemble models where separate agents focus on understanding, experience, and factual verification, synthesizing their outputs to counteract hallucination tendencies [394]. By grounding responses in retrieved legal documents and utilizing tools to verify citations against authoritative databases, agents can provide informed legal solutions that maintain fidelity to actual jurisprudence. The use of tagged context prompts and specialized retrieval mechanisms has proven effective in flagging instances where the model operates outside its domain knowledge, ensuring that users receive accurate legal information [395].

In software engineering, while code generation has seen rapid advances, hallucinations manifest as internal inconsistencies, deviant logic, or the use of non-existent libraries. A comprehensive taxonomy of hallucinations in code generation reveals that models often produce outputs that deviate from user intent or factual knowledge regarding API specifications. To mitigate these risks, benchmarks like HalluCode have been established to evaluate and improve the ability of code LLMs to recognize and correct their own hallucinations [396]. Tool-augmented agents in this domain can execute code in sandboxed environments, verify syntax, and check library documentation via API calls, thus closing the loop between generation and verification.

Across these verticals, a common theme emerges: the transition from static generation to dynamic, tool-mediated interaction is essential for reliability. Whether through RAG, function calling for database queries, or automated verification pipelines, the integration of external tools serves as a critical mechanism to anchor LLM outputs in factual reality. As noted in broader surveys of hallucination mitigation, techniques that combine retrieval with generation are among the most promising strategies for deploying LLMs in high-stakes environments [397]. However, challenges remain, as RAG systems can still be misled if prompts contradict pre-trained understandings or if retrieved information is irrelevant, underscoring the need for sophisticated orchestration frameworks that can dynamically decide when and how to invoke tools [398]. Ultimately, the successful deployment of agents in knowledge-intensive domains hinges on their ability to seamlessly blend internal reasoning with external verification. Having established how agents leverage digital tools to ensure factual grounding in abstract domains, the discussion now turns to the next frontier: extending these generalist capabilities to physical embodiments, where agents must master robotic manipulation and tool use in the real world.

6.2 Generalist Agents for Robotic Manipulation and Tool Use

6.2 Generalist Agents for Robotic Manipulation and Tool Use

Having established how agents leverage digital tools to ensure factual grounding in abstract, knowledge-intensive domains, the discussion now advances to the next frontier: extending these generalist capabilities to physical embodiments. In the realm of embodied AI, the evolution of robotic manipulation has recently shifted from training specialized policies for narrow, predefined tasks to developing generalist agents capable of mastering a diverse repertoire of skills across varying environments and embodiments. This paradigm shift mirrors the advancements seen in large language models, where a single model architecture is trained on heterogeneous data to perform myriad tasks. The emergence of generalist agents represents a critical step toward artificial general intelligence, enabling robots to adapt to novel tools, manipulate deformable objects, and operate seamlessly across different physical forms without extensive retraining. A foundational example of this transition is Gato, a multi-modal, multi-task, and multi-embodiment generalist policy that utilizes the same network weights to perform tasks ranging from playing Atari games to stacking blocks with a real robot arm, demonstrating the feasibility of a unified approach to diverse control problems [399].

A central challenge in creating such generalist agents lies in bridging the domain gap between different robotic embodiments and the sparsity of successful task executions in specific action spaces. Traditional methods often struggle when transferring skills from one robot morphology to another due to misaligned representations. To address this, recent frameworks have introduced agent-agnostic representations that obscure the specifics of the embodiment to enhance generalizability. For instance, Ag2Manip proposes a framework utilizing agent-agnostic visual representations derived from human manipulation videos and abstracting robot kinematics into a universal agent proxy. This approach emphasizes the crucial interactions between the end-effector and the object rather than joint-specific dynamics, resulting in a significant performance increase across simulated benchmarks and improved imitation learning success rates in real-world settings [400]. Similarly, the concept of treating tools as extensions of the body has led to the development of parameterizations like Tool-As-Embodiment (TAE), which treats hand-object and tool-object interactions within the same representation space. This allows a single policy to be applied recursively, enabling robots to use objects as new end-effectors to manipulate other objects, thereby sharing experiences across different resolutions of tool-enabled embodiments [401].

The capability to handle novel and unseen tools is a defining characteristic of a true generalist agent, paralleling the adaptive tool-use seen in digital agents but applied to physical constraints. Humans exhibit a robust ability to adapt to new tools by understanding their affordances and geometry, a trait that artificial agents are only beginning to acquire. Research into learning generalizable tool-use skills has shown promise through trajectory generation models that can generalize to different tool shapes. By learning a generative model of tool-use trajectories as a sequence of tool point clouds, agents can optimize sequences of tool poses for previously unseen tools, achieving performance comparable to humans in manipulating deformable objects [402]. Furthermore, leveraging linguistic information about tools—such as their geometry and common uses—has been shown to accelerate the adaptation of control policies to new tools. By combining pre-trained language model features with meta-learning, agents can efficiently adapt to novel tools for tasks like pushing, lifting, and hammering, significantly reducing the data required for adaptation [403].

Deformable object manipulation presents another formidable hurdle for generalist agents due to

the high-dimensional state space and complex dynamics of soft bodies, requiring reasoning capabilities that extend beyond rigid body physics. Standard reinforcement learning approaches often fail to generalize beyond the specific objects seen during training. However, new methodologies are overcoming these limitations by employing foresightful dense visual affordances that estimate long-term state values, allowing agents to avoid local optima in multi-step manipulation tasks [404]. Additionally, demonstration-free hierarchical planning approaches powered by large language models have enabled agents to tackle intricate long-horizon tasks involving deformable objects without prior training data. These systems utilize LLMs to generate high-level plans and intermediate sub-goal point clouds, which are then executed via closed-loop model predictive control, demonstrating robust generalization to complex, unseen tasks [405].

Scalability and data efficiency remain paramount concerns in the deployment of generalist agents, necessitating strategies that maximize learning from limited interactions. The scarcity of diverse robotic datasets has motivated the development of systems that can learn from limited demonstrations while maintaining broad generalization. RoboAgent, for example, employs semantic augmentations and action chunking to train a universal agent capable of twelve unique skills using merely 7,500 demonstrations, outperforming prior methods in unseen situations [406]. Similarly, the Robotics Transformer (RT-1) architecture demonstrates that open-ended, task-agnostic training on large-scale, diverse datasets allows models to generalize effectively to new tasks and objects, validating the importance of data diversity and model capacity in real-world control [407]. The integration of foundation models further enhances this capability; by leveraging language-reasoning segmentation masks generated from internet-scale models, agents can achieve sample-efficient generalization to new object instances and unseen backgrounds [408].

Moreover, the ability to learn from passive human videos offers a rich, scalable source of data for developing generalist behaviors, effectively bridging the sim-to-real and human-to-robot gaps. Frameworks that extract agent-agnostic action representations from human videos and translate them to robot embodiments enable zero-shot manipulation of generic unseen objects. This approach allows robots to solve coarse manipulation tasks like opening drawers and using tools without accessing any in-domain robot manipulation trajectories, effectively bridging the gap between human demonstration and robotic execution [409] [410]. As the field progresses, the integration of these diverse strategies—agent-agnostic representations, language-conditioned adaptation, and scalable data utilization—continues to push the boundaries of what generalist agents can achieve. While this section has focused on the policy execution and manipulation capabilities of generalist agents, realizing their full potential in unstructured environments requires a fundamental upgrade in how they perceive the world. Consequently, the following section explores the transition from policy execution to robust embodied agency through multimodal perception and 3D world grounding.

6.3 Multimodal Perception and 3D World Grounding

6.3 Multimodal Perception and 3D World Grounding

Building upon the generalist manipulation capabilities discussed in the previous section, the transition from policy execution to robust embodied agency necessitates a fundamental shift in how artificial intelligence perceives and grounds information. While traditional Large Language Models (LLMs) excel at processing textual semantics, they inherently lack the sensorimotor experiences required to understand the three-dimensional, multisensory nature of reality. To bridge this gap, recent research has focused on developing architectures that integrate visual, auditory, and tactile modalities into a unified representation, enabling agents to achieve holistic 3D scene understanding.

This subsection analyzes the architectural innovations that facilitate this multimodal grounding, moving beyond simple vision-language pairs to encompass the full spectrum of physical interaction required for the complex navigation and spatial reasoning tasks detailed in the following section.

A primary challenge in embodied AI is the effective fusion of disparate sensory inputs to construct a coherent model of the environment. Early approaches often treated modalities in isolation or relied heavily on visual data, ignoring the rich informational content provided by sound and touch. However, truly intelligent agents must mimic human perception, where senses are inextricably linked. For instance, the [411] framework represents a significant leap forward by proposing a multisensory embodied LLM that actively interacts with 3D environments. Unlike passive models, MultiPLY incorporates visual, audio, tactile, and thermal information, establishing robust correlations among words, actions, and percepts. By encoding 3D scenes as object-centric representations and utilizing action and state tokens, this architecture allows agents to iteratively collect multisensory data, thereby grounding abstract language commands in concrete physical observations.

Visual grounding remains the cornerstone of 3D understanding, yet it requires sophisticated mechanisms to map linguistic references to specific spatial locations within complex scenes. The [412] addresses this by combining spatial embeddings from 3D bounding boxes with fine-tuned language embeddings to identify target objects from point clouds. This approach highlights the necessity of integrating geometric priors with semantic knowledge to resolve ambiguities in referential language. Furthermore, the limitations of purely visual or textual approaches are evident when dealing with open-set vocabularies and novel objects. The [413] framework overcomes closed-set constraints by leveraging foundation models to fuse pixel-aligned features into 3D maps. This enables zero-shot spatial reasoning where agents can query 3D environments using diverse modalities, including natural language, images, and even audio snippets, demonstrating that multimodal integration significantly enhances the robustness of scene understanding.

Beyond vision and language, the incorporation of auditory signals provides critical cues for spatial reasoning and navigation, particularly in scenarios where visual lines of sight are occluded—a capability essential for the dynamic environments described later. The [414] project pioneered the integration of geometrical acoustic simulations into embodied agents, allowing them to navigate toward sounding objects by interpreting reverberations and sound propagation. This capability is further advanced by [415], which combines binaural acoustic scene analysis with LLM reasoning. By training on synthesized spatial sound datasets, BAT enables agents to reason about the relationships between sounds in their environment, transcending simple detection to achieve a deeper understanding of spatial acoustics. Similarly, the [416] introduces a unified 3D spatial map representation that fuses audio, visual, and language cues, enabling robots to disambiguate goal locations more reliably than vision-only systems.

Tactile perception adds another layer of granularity, allowing agents to infer physical properties such as texture, temperature, and material composition that are often invisible to cameras, thus complementing the manipulation skills outlined previously. The [417] emphasizes that tactility is crucial for enhancing interaction capabilities. By constructing datasets that align sentence-level descriptions with visual and tactile inputs, researchers can train models like TLV-Link to achieve effective semantic alignment with minimal parameter adjustments. This is complemented by work on cross-modal prediction, such as [418], which explores the synthesis of tactile signals from visual inputs and vice versa. These models address the scale discrepancy between global visual scenes and local tactile sensations, enabling agents to “imagine” how an object feels based on its appearance or visualize an object based on tactile exploration.

The synthesis of these modalities often relies on advanced representation learning techniques that can handle high-modality heterogeneity. The [419] proposes information-theoretic metrics to quantify how different modalities interact, allowing for efficient parameter sharing across diverse sensor inputs ranging from proprioception to time-series data. This is essential for scaling embodied agents to handle the complexity of real-world environments. Moreover, the [420] offers a unified solution for aligning novel modalities, such as depth, audio, and EEG, to a pre-defined embedding space using a pretrained Vision Transformer. This approach unlocks the potential of existing foundation models to process rare or complex sensory data without requiring massive retraining efforts.

Finally, the ultimate goal of multimodal perception is to enable agents to perform complex tasks through grounded planning and execution, serving as the perceptual backbone for both manipulation and navigation. The [421] demonstrates this by directly incorporating continuous sensor modalities into the input stream of a large language model. By interleaving visual, state estimation, and textual encodings, PaLM-E establishes a direct link between words and percepts, enabling it to perform sequential robotic manipulation planning and visual question answering with state-of-the-art performance. Similarly, [422] introduces LEO, an agent trained on extensive 3D vision-language-action datasets to excel in perception, grounding, reasoning, and acting within 3D worlds. These systems illustrate that the future of embodied intelligence lies not in isolated sensory processing but in the seamless integration of all available modalities to create a comprehensive, grounded understanding of the physical world, setting the stage for advanced spatial reasoning and environment interaction.

6.4 Navigation, Spatial Reasoning, and Environment Interaction

6.4 Navigation, Spatial Reasoning, and Environment Interaction

Building upon the multimodal perceptual foundations and 3D grounding capabilities established in the previous section, this subsection explores how embodied agents leverage these rich sensory inputs to execute complex navigation, spatial reasoning, and active environment interaction. While Section 6.3 focused on constructing a unified understanding of the physical world through vision, audio, and touch, the challenge now shifts to utilizing this grounded representation for dynamic decision-making. The evolution of this field has moved from solving static point-goal tasks in simulated environments to mastering long-horizon planning in dynamic, partially observable, and interactive worlds. A fundamental hurdle in this domain is the ability to reason spatially over extended timeframes while maintaining robustness against environmental uncertainty. Recent advancements have addressed this by shifting focus toward hierarchical planning frameworks that decompose complex objectives into manageable sub-tasks. For instance, the [423] framework addresses the intricacies of long-horizon embodied exploration by proposing a modular hierarchical transport policy. This approach constructs a topological graph of the scene to facilitate deep exploration, utilizing weighted frontiers to guide the agent in efficiently locating, picking up, and transporting target objects under load constraints, thereby demonstrating superior generalization to harder transport scenes compared to prior baselines.

The paradigm of navigation has also expanded beyond mere traversal to include active environment interaction, challenging the traditional assumption that environments consist of static obstacles to be avoided. Intelligent agents must now possess the capability to manipulate their surroundings to clear paths or achieve goals, effectively bridging the gap between perception and the manipulation skills discussed earlier. The [424] study introduces the Neural Interaction Engine (NIE), which explicitly predicts environmental changes caused by agent actions, such as pushing objects out of

the way. By modeling these physical interactions during the planning phase, agents equipped with NIE significantly outperform those lacking such predictive capabilities in physics-enabled environments like AI2-THOR. This shift toward interactive navigation is further supported by benchmarks focusing on dynamic elements, where training agents with dynamic obstacle augmentation has been shown to confer significant improvements in test-time generalization and robustness, as evidenced by the winning entry of the 2021 iGibson Challenge described in [425].

To support these complex behaviors, spatial reasoning in embodied agents increasingly relies on sophisticated memory architectures and map representations that capture both geometric and semantic information. Traditional approaches often struggle with partial observability and the need to recall locations of previously observed objects. The [426] task explicitly tests an agent’s ability to navigate to sequences of objects, revealing that simple semantic map agents can perform surprisingly well relative to complex neural image feature maps, though significant gaps remain even for oracle agents. To enhance spatial understanding, researchers have developed structured scene representations. The [427] architecture compartmentalizes percepts to accurately memorize environment layouts, disentangling visual and geometric cues to support long-term planning and frontier-exploration strategies. Similarly, [428] leverages Large Language Models (LLMs) by incrementally building a 3D scene graph of the explored environment. This grounding mechanism allows the LLM to generate contextually appropriate high-level plans, which are then executed by a low-level planner, achieving state-of-the-art results in multi-object navigation by effectively utilizing human knowledge for dynamic planning.

The integration of the multimodal sensory inputs detailed in Section 6.3 has further refined these spatial reasoning and navigation capabilities. While visual perception remains primary, the inclusion of audio cues provides critical geometric information about unseen spaces, echoing the acoustic grounding principles discussed previously. The [414] dataset and framework enable agents to learn from egocentric audio-visual observations, allowing them to discover room geometry through reverberation and locate sound-emitting targets. Building on this, [429] presents an interactive agent capable of localizing audio events while also seeking clarification from humans via natural language when uncertain, demonstrating that equipping agents to ask for help improves performance in challenging acoustic scenarios. Furthermore, [430] introduces a reinforcement learning approach that dynamically sets waypoints and utilizes an acoustic memory to reveal the geometry of unmapped spaces, capitalizing on the synergy between sight and sound to improve navigation efficiency.

Advances in representation learning have also played a pivotal role in bridging the gap between simulation and reality, ensuring that the high-fidelity maps constructed from multimodal data translate into effective action. Agents must learn navigational visual representations that encode spatial relationships essential for decision-making. The [431] method proposes contrasting egocentric views with semantic maps to transfer compact, rich information about objects and structures to the agent’s representations, achieving new state-of-the-art results in continuous environments. Additionally, the emergence of novel mapping techniques like Gaussian Splatting has revolutionized how agents memorize scenes. The [432] framework utilizes 3D Gaussian Splatting to construct map representations that retain detailed textural features alongside geometry and semantics, resulting in a significant leap in success weighted by path length on challenging datasets.

Finally, the transition to real-world deployment necessitates handling noisy sensors and actuation dynamics that are often abstracted away in simulation, a prerequisite for the seamless human-robot collaboration discussed in the following section. The [433] work develops agents that rely on visual estimates of egomotion under noisy action dynamics, conceptualizing a separation between learning agent dynamics and task-specific navigation policies to facilitate seamless adaptation to changing

robot configurations. Moreover, [434] explores end-to-end training in simulations that minimize the sim-to-real gap by modeling high-fidelity robot motion and sensor noise, reporting significant gains in performance when transferring to real robotic platforms. These collective advances underscore a trajectory toward embodied agents that not only navigate but also reason about, interact with, and adapt to the complexities of physical 3D worlds, laying the necessary groundwork for the shared mental models and immersive interfaces central to future Human-Robot Collaboration.

6.5 Human-Robot Collaboration via Extended and Augmented Reality

6.5 Human-Robot Collaboration via Extended and Augmented Reality

Building upon the advancements in embodied navigation and spatial reasoning discussed in the previous section, the integration of Extended Reality (XR) and Augmented Reality (AR) technologies has fundamentally reshaped the landscape of Human-Robot Collaboration (HRC). While prior sections focused on agents autonomously interpreting 3D geometries and interacting with static or dynamic objects, this domain transitions the paradigm from solitary operation to immersive, co-located teamwork. XR interfaces now serve as critical mediators that bridge the cognitive gap between human operators and embodied agents, facilitating the establishment of shared mental models and enabling bidirectional communication of intent. This synergy allows for a natural division of labor: humans contribute high-level reasoning and adaptability, while robots provide precision, strength, and endurance. A primary mechanism for enhancing this collaboration is the use of AR headsets to visualize the robot’s internal state, planned trajectories, and decision-making processes, thereby reducing ambiguity and fostering trust within the team.

One of the most significant contributions of AR in HRC is the creation of a shared workspace where virtual and physical elements coexist, allowing both human and robot agents to perceive and manipulate a unified environment. Research has demonstrated that establishing a bi-directional communication channel through a shared AR workspace enables robots not only to perceive virtual elements but also to infer the utility and cost of human actions by sensing gaze and pose [435]. This shared perception is crucial for proactive collaboration; rather than passively executing commands, robots can anticipate human needs and adjust their behaviors dynamically. For instance, frameworks leveraging AR have been developed to visualize future object interactions, allowing humans to make necessary adjustments before physical execution occurs, thus streamlining the collaborative workflow [436]. By rendering the robot’s planned path and reactions visually, these systems significantly improve the human agent’s ease of understanding the robot’s intents, leading to reduced task completion times and enhanced feelings of safety [437].

Beyond visualization, XR interfaces facilitate the negotiation of tasks and the alignment of affordances between heterogeneous team members. Effective collaboration requires a deep understanding of what each agent can and cannot do within a specific context. An AR framework that builds a perceptual representation of shared affordances allows both humans and robots to communicate their capabilities and request assistance when operating outside their respective affordance regions [438]. This concept extends to negotiation-based frameworks where AR mediates the resolution of conflicts or synergies in task allocation. Experiments in office environments have shown that such AR-mediated negotiation significantly reduces task completion time compared to non-AR baselines, particularly in scenarios where robots require human assistance to complete delivery tasks [439]. Furthermore, the integration of eye-gaze tracking with AR displays enables the bidirectional communication of intent, which is essential for maintaining a shared mental model during joint actions. Studies indicate that combining AR with eye-gaze analysis improves task efficiency, trust, and flu-

ency in human-robot teams by ensuring that individual internal models of the situation converge [440].

The scalability of these collaborative systems is another area where XR proves indispensable, particularly in scenarios involving large multi-agent teams. In industries such as defense and search and rescue, AR-based frameworks enable users to localize, command, and supervise dozens of autonomous agents simultaneously without the need for traditional hardware interfaces like joysticks or laptops [103]. This capability is vital for managing complex operations where situational awareness is paramount. Moreover, the concept of “Symmetrical Reality” has been proposed to address the dual perceptual centers of humans and AI agents, offering a unified representation that accommodates both physical and virtual amalgamations [441]. This framework supports the coexistence of humans and AI by enabling active assistance services where the boundaries between physical and digital interactions are blurred.

However, the deployment of XR in HRC is not without challenges, particularly regarding the design of intuitive interfaces and the management of cognitive load. While AR interfaces have been found to reduce physical demand and task completion time in manufacturing settings, user studies suggest that perceptions of dependability can vary, with some users still preferring standard interfaces for certain control tasks [442]. Additionally, the misalignment of virtual elements in collaborative AR experiences can impact user experience, although research indicates that positional synchrony is a more critical factor than avatar presence in maintaining collaboration quality [443]. To mitigate these issues, design frameworks such as XAIR have been developed to guide the “when,” “what,” and “how” of providing explainable AI outputs in AR, ensuring that explanations enhance rather than hinder human performance [444].

Looking toward the future, the convergence of Large Language Models (LLMs) with XR promises to further revolutionize HRC by enabling more inclusive and adaptive interactions, a theme that naturally bridges to the subsequent discussion on autonomous skill acquisition. Embedding LLMs into XR environments allows for natural language-driven task planning and the creation of virtual avatars that can engage users in more meaningful ways [445]. Systems like MISAR demonstrate how LLMs can assimilate visual, auditory, and contextual modalities to provide optimized guidance, marking a step toward truly adaptive AR systems [446]. As these technologies mature, the focus will likely shift toward creating universal interaction concepts that seamlessly blend environment, task, and user preferences, ultimately realizing the vision of hyper-connected, intelligent spaces where humans and robots collaborate as equals [447]. The continued exploration of these synergies through rigorous empirical studies and systematic reviews remains essential for advancing the field toward robust, scalable, and human-centric embodied intelligence, laying the groundwork for the simulation-to-real transfer methodologies required to deploy such sophisticated collaborative agents in the physical world [448].

6.6 Simulation-to-Real Transfer and Autonomous Skill Acquisition

6.6 Simulation-to-Real Transfer and Autonomous Skill Acquisition

While Section 6.5 highlighted how Extended Reality (XR) facilitates human-robot collaboration through shared mental models and intuitive interfaces, the deployment of such sophisticated embodied agents in physical environments remains fundamentally constrained by the scarcity, cost, and safety risks associated with collecting real-world training data. To bridge the gap between the immersive digital twins discussed previously and the unpredictable physical world, the research community has increasingly turned to simulation-to-real (Sim2Real) transfer methodologies.

These approaches leverage the infinite data generation capabilities of simulated environments to train policies that generalize to reality, addressing the “reality gap”—the distributional shift between synthetic and real-world dynamics, sensors, and actuators. Current advancements in this domain focus on three primary pillars: robust imitation learning strategies, domain randomization and adaptation techniques, and self-supervised active learning frameworks that enable autonomous skill refinement, ultimately providing the low-level competencies required by the high-level cognitive architectures discussed in the subsequent section.

A dominant paradigm for scaling embodied learning is Imitation Learning (IL), where agents learn by mimicking expert demonstrations. A significant challenge in IL is the reliance on expensive human-generated trajectories collected in the real world. Innovative approaches have demonstrated that imitating shortest-path planners within highly diverse simulated environments can yield agents capable of proficient navigation and manipulation in reality using only RGB sensors [449]. This success is often attributed to the scale and diversity of the training data, such as procedurally generated houses containing thousands of unique assets, which forces the agent to learn robust features rather than memorizing specific environmental textures. Furthermore, the integration of simulation with limited real-world data has proven effective; for instance, introducing a Task Consistency Loss (TCL) encourages alignment between simulated and real features at both the representation and action-prediction levels, enabling mobile manipulators to perform complex tasks like opening latched doors with high success rates [450].

To further reduce the burden of data collection, researchers are exploring methods that utilize suboptimal or non-expert data. Traditional behavioral cloning struggles with compounding errors when trained on imperfect demonstrations, but novel model-based offline imitation learning frameworks have shown that leveraging datasets from both suboptimal and optimal policies can outperform standard methods, particularly in low-data regimes [451]. Additionally, the concept of “play” has emerged as a powerful source of task-agnostic data; by collecting hours of playful interactions in simulation, agents can learn visual representations that generalize to downstream tasks like pushing and stacking with significantly fewer task-specific demonstrations [452]. The ability to learn from diverse embodiments is also critical; manipulator-independent representations allow agents to imitate trajectories demonstrated by different robotic forms or even humans, facilitating cross-embodiment visual imitation despite significant dynamics differences [453].

Domain Randomization (DR) remains a cornerstone technique for Sim2Real transfer, involving the randomization of physical parameters and visual textures during simulation training to create a policy robust to real-world variations. While empirically successful, theoretical frameworks have been developed to explain why DR works, providing sharp bounds on the Sim2Real gap and highlighting the necessity of memory-dependent policies for handling infinite-horizon tasks [454]. To enhance sample efficiency within randomized environments, teacher-student learning paradigms have been proposed. In these setups, a teacher agent with access to privileged information guides a student agent operating under noisy, realistic conditions. Advanced frameworks like “Learn to Teach” improve upon this by recycling experience collected by the teacher, thereby reducing the environmental interaction cost while maintaining competitive performance [455]. Moreover, curriculum-based sensing reduction offers a structured approach where the agent gradually transitions from rich simulated features to sparse real-world sensors, replacing removed features with random signals to prevent dependency artifacts [456].

Beyond passive transfer, autonomous skill acquisition through self-supervised active learning represents a frontier in embodied intelligence. Agents can now leverage perception models pre-trained on internet-scale images to drive active exploration policies in the real world. The Self-supervised

Embodied Active Learning (SEAL) framework exemplifies this by utilizing 3D consistency to label observations gathered during exploration, effectively closing the action-perception loop to improve object detection and navigation without external labels [457]. Similarly, systems like RialTo construct “digital twin” simulations on the fly from small amounts of real-world data, allowing for reinforcement learning fine-tuning in a safe, simulated copy of the actual environment before deploying the robustified policy back to the real world [458]. This real-to-sim-to-real pipeline minimizes the need for unsafe real-world data collection while significantly increasing policy robustness against physical disturbances.

Scaling these methodologies to open-ended task spaces requires agents that can adapt on the fly. Recent work demonstrates that training agents via meta-reinforcement learning across vast, diverse task distributions enables human-timescale adaptation to novel embodied 3D problems [459]. Such agents exhibit hypothesis-driven exploration and can be prompted with first-person demonstrations to solve unseen tasks rapidly. Furthermore, iterative imitation learning techniques have shown promise in continuous on-robot learning, allowing agents to improve goal-directed behaviors, such as high-speed table tennis, through self-supervised goal reaching without complex reward engineering [460]. By combining scalable simulation pre-training, robust domain adaptation mechanisms, and self-supervised active learning loops, the field is moving toward embodied agents capable of acquiring complex skills autonomously. These robust low-level policies form the essential foundational layer upon which the higher-order cognitive architectures for planning and memory, detailed in the following section, can effectively orchestrate behavior in open-ended environments.

6.7 Cognitive Architectures for Embodied Planning and Memory

6.7 Cognitive Architectures for Embodied Planning and Memory

While Section 6.6 established the mechanisms for acquiring robust low-level policies through simulation and self-supervised learning, deploying these skills in open-ended environments requires a higher-order cognitive layer to orchestrate behavior. The integration of Large Language Models (LLMs) into robotic systems has thus necessitated the development of sophisticated cognitive architectures that bridge the gap between high-level semantic reasoning and the low-level physical controllers trained via methods like Sim2Real. Traditional robotic frameworks often decouple planning from execution, leading to failures when dynamic environmental changes render static plans obsolete. Modern cognitive architectures address this by embedding LLMs as the central “brain” responsible for deliberative planning, while coupling them with the specialized, robust controllers discussed previously for reactive execution. A prime example of this unification is the LLM-Brain framework, which proposes using a large-scale language model to unify egocentric memory and control within a single embodied system [461]. This architecture facilitates closed-loop multi-round dialogues encompassing perception, planning, control, and memory, allowing the agent to maintain an internal state of the world that evolves alongside its physical interactions. By treating memory and control as interconnected components rather than isolated modules, such systems enable robots to perform complex tasks like active exploration and embodied question answering with greater coherence.

A critical challenge in these architectures is the grounding of abstract language plans into executable physical actions, effectively translating semantic intent into the motor primitives acquired during training. To address this, researchers have developed hierarchical structures where the LLM acts as a high-level planner that decomposes goals into sub-tasks, which are then executed by lower-level policies. The RoboGPT agent exemplifies this approach by combining an LLM-based planning

module with a dedicated skill module called RoboSkill, designed specifically for navigation and manipulation [462]. This separation of concerns allows the LLM to focus on temporal reasoning and task decomposition while relying on robust, pre-trained controllers for precise motor execution. Similarly, the DELTA framework leverages scene graphs to ground environmental topology into actionable knowledge, enabling the LLM to decompose long-term goals into an autoregressive sequence of sub-goals for an automated task planner [463]. By utilizing structured environment representations, these architectures mitigate the hallucination risks inherent in text-only models, ensuring that generated plans are physically feasible within the specific context of the robot’s surroundings.

Memory augmentation plays a pivotal role in enhancing the adaptability of embodied agents, particularly in long-horizon tasks where recall of past experiences—both successful demonstrations and failure modes from simulation or real-world interaction—is essential for success. Retrieval-augmented mechanisms have emerged as a powerful solution, allowing agents to access relevant strategies from an external memory bank based on current multimodal inputs. The Retrieval-Augmented Embodied Agent (RAEA) introduces a policy retriever and generator that emulate human-like recall, enabling robots to assimilate past strategies to formulate effective responses to novel tasks [464]. This approach significantly reduces the dependency on extensive training data for every new scenario. Furthermore, the RAP framework extends this concept by dynamically leveraging past experiences corresponding to the current situation, distinguishing itself through its versatility in both text-only and multimodal environments [21]. These memory-augmented architectures allow agents to learn from previous failures and successes, creating a continuous loop of improvement without the need for constant weight updates.

Beyond static memory retrieval, advanced cognitive architectures incorporate real-time feedback loops to correct plan-execution misalignments, closing the gap between intended and actual outcomes. The DoReMi framework illustrates this by employing LLMs to generate constraints that detect misalignments between high-level plans and low-level execution, utilizing Vision Language Models (VLMs) for continuous monitoring and timely recovery [465]. This dual-role utilization of LLMs ensures that the robot can adapt to unforeseen perturbations, such as object displacement or slippery surfaces, by immediately detecting violations and triggering replanning. Similarly, the Inner Monologue approach leverages environment feedback, such as success detection and scene descriptions, to form a rich internal narrative that guides robotic control scenarios without additional training [33]. By processing feedback through natural language, these systems create a semantic bridge between sensory input and action selection, facilitating more robust decision-making in partially observable environments.

The evolution of these architectures also includes the integration of causal reasoning and structured prompting to enhance planning reliability and logical consistency. The Plan, Eliminate, and Track (PET) framework simplifies the control problem by using the LLM to translate task descriptions into high-level sub-tasks, eliminate irrelevant objects from observations, and track task completion status [466]. This modular approach maintains compatibility with low-level trainable actors while leveraging the LLM’s procedural knowledge. Additionally, neuro-symbolic approaches like PLAN utilize symbolic program executors on latent procedural representations to formalize prompts from commonsense knowledge bases, effectively mitigating spurious correlations between goals and steps [467]. Such hybrid architectures combine the generative flexibility of LLMs with the precision of symbolic reasoning, offering a pathway toward more reliable and explainable embodied intelligence.

Finally, the development of open-vocabulary and queryable scene representations has further expanded the capabilities of cognitive architectures, allowing them to operate in unstructured environ-

ments. NLMap serves as a framework to gather and integrate contextual information, allowing LLM planners to query available objects in the scene before generating context-conditioned plans [468]. This capability enables robots to operate without a fixed list of objects, adapting to open-world scenarios where novelty is the norm. Collectively, these advancements in cognitive architectures for embodied planning and memory represent a significant leap toward autonomous systems capable of understanding, remembering, and acting within complex physical worlds. By unifying high-level reasoning with the low-level control capabilities established in prior sections, and augmenting these processes with dynamic memory and feedback mechanisms, researchers are laying the groundwork for robots that can truly learn and adapt throughout their operational lifetimes.

7 Evaluation Benchmarks, Safety, and Ethical Considerations

7.1 Standardized Benchmarks for Agent Planning and Execution

7.1 Standardized Benchmarks for Agent Planning and Execution

The rapid evolution of Large Language Models (LLMs) from static text generators to dynamic, autonomous agents has necessitated a paradigm shift in evaluation methodologies. Traditional benchmarks, which predominantly rely on static datasets and single-turn question-answering formats, are increasingly insufficient for assessing the complex, multi-step reasoning and interactive capabilities required of modern agents. Consequently, the research community has developed a suite of multi-dimensional evaluation platforms designed to rigorously test agent planning, decision-making, and tool utilization within diverse, interactive environments. These standardized benchmarks serve as critical infrastructure for identifying the functional limitations of current agents, establishing a baseline of capability that is essential before addressing the security vulnerabilities discussed in subsequent sections.

A primary category of these benchmarks focuses on evaluating agents in complex, real-world software environments that require long-horizon planning and precise execution. [469] introduces a multi-dimensional, evolving benchmark comprising eight distinct environments, ranging from operating system interactions to web browsing and database management. This framework moves beyond simple task completion metrics to assess an agent’s ability to navigate multi-turn, open-ended generation settings. Extensive testing within [469] reveals significant disparities between top commercial models and open-source competitors, highlighting that poor long-term reasoning and instruction following remain major obstacles. Similarly, [470] addresses the specific challenges of general-purpose operating systems by providing a scalable benchmark for cross-application cooperation. This environment exposes critical weaknesses in state-of-the-art agents, particularly their inability to maintain an up-to-date understanding of vast, dynamic action spaces and to adhere to user-specific constraints like security preferences.

Beyond software interaction, there is a growing emphasis on evaluating strategic reasoning and decision-making in dynamic, competitive, and game-theoretic scenarios. [471] fills a crucial gap by offering a framework specifically designed for multi-agent dynamic environments, utilizing seven distinct gaming scenarios to assess spatial reasoning, risk assessment, and team collaboration. Complementing this, [472] employs classical multi-agent games to quantitatively score models on robustness and generalizability, revealing that while models like GPT-4 show strong performance, others struggle with generalization. For tasks requiring strategic resource management and long-term planning under uncertainty, [473] simulates auction environments. This benchmark effectively tests an agent’s ability to manage budgets and adhere to goals in highly unpredictable, competitive contexts,

demonstrating that adaptive strategies are essential for success. Furthermore, [474] introduces pilot challenges based on game theory to evaluate dynamic reasoning in rapidly evolving environments, proposing “K-Level Reasoning” to address the failure of existing methods in scenarios requiring recursive thinking about rival moves.

The evaluation of tool utilization capabilities has also seen significant refinement, moving from holistic success rates to fine-grained, step-by-step analysis. [475] proposes a novel approach that disentangles tool usage into sub-processes such as instruction following, planning, retrieval, and review. This granular perspective allows researchers to pinpoint exactly where an agent fails, whether in selecting the correct tool or in interpreting its output. Expanding on this, [17] examines five critical dimensions of tool learning across seven real-world scenarios using a library of approximately 600 tools. Interestingly, findings from [17] suggest that simply increasing model size does not guarantee better tool learning and may even exacerbate certain cognitive hindrances.

Moreover, specialized benchmarks have emerged to address domain-specific planning and reasoning requirements. In the medical field, [476] introduces a paradigm using Standardized Patients and clinical pathways to automatically evaluate diagnostic and treatment decisions, ensuring safety and reliability without exhaustive human annotation. For financial domains, [477] underscores the need for agents that can process hierarchical financial data and make real-time investment decisions, although it primarily presents a framework, its evaluation component highlights the necessity of memory-augmented architectures for trading success. Additionally, [478] provides a systematic suite based on automated planning domains to distinguish between genuine planning capabilities and the mere retrieval of world knowledge, showing that even state-of-the-art models fall short on critical planning tasks.

Finally, the complexity of compositional and conditional reasoning is addressed by benchmarks like [479], which connects these cognitive skills to flight booking tasks. Results from [479] indicate a stark performance gap, with even the most advanced models struggling to align detailed user preferences with available options. Collectively, these standardized benchmarks represent a maturing field that is moving away from static metrics toward comprehensive, interactive, and multi-dimensional evaluation frameworks. They provide the necessary empirical ground to understand the current frontiers of agent capabilities, revealing that while progress is evident, significant challenges in long-horizon planning, multi-agent collaboration, and fine-grained tool mastery remain. Crucially, by exposing these functional gaps and the reliance of agents on external contexts, these benchmarks lay the groundwork for understanding why agents are susceptible to the adversarial manipulations and prompt injection attacks detailed in the following section.

7.2 Adversarial Robustness and Prompt Injection Vulnerabilities

7.2 Adversarial Robustness and Prompt Injection Vulnerabilities

While Section 7.1 established standardized benchmarks to measure the functional planning and tool-utilization capabilities of LLM agents, it also highlighted that these very capabilities create new attack surfaces. The transition from static text generation to dynamic agency introduces a complex threat landscape where adversarial robustness is paramount. Unlike passive chatbots, agents possessing tool-use capabilities face unique vulnerabilities, particularly regarding indirect prompt injections (IPI) and sophisticated jailbreaking attempts designed to manipulate external function execution. The core challenge lies in the agent’s inability to reliably distinguish between system instructions and untrusted external content, a vulnerability that attackers exploit to hijack agent workflows. Recent research has addressed this by developing specialized benchmarks and

attack frameworks that quantify susceptibility across diverse scenarios, bridging the gap between functional evaluation and security assessment.

Indirect prompt injection represents a critical vector for tool-integrated agents, where malicious instructions are embedded within the data sources an agent processes, such as emails, web pages, or database entries. This directly exploits the “reliance on external contexts” identified as a functional gap in the previous section. To systematically assess this threat, researchers have introduced [480], a comprehensive benchmark comprising over 1,000 test cases covering numerous user and attacker tools. This work demonstrates that even state-of-the-art models like GPT-4, when prompted with ReAct strategies, remain vulnerable to IPI attacks roughly 24% of the time, with success rates nearly doubling when attackers employ reinforced hacking prompts. Similarly, [481] presents the BIPIA benchmark, which identifies the root cause of these failures as the model’s lack of awareness regarding the boundary between executable instructions and passive external content. These studies collectively underscore that standard alignment training is insufficient for protecting agents against content-based manipulation, necessitating robust defense mechanisms that can parse intent from context.

Beyond indirect injections, direct jailbreaking remains a pervasive threat, evolving from simple role-playing scenarios to highly optimized adversarial prompts. The landscape of jailbreak evaluation has shifted from binary success metrics to nuanced frameworks that assess the degree of safety violation, mirroring the move toward fine-grained analysis seen in tool evaluation. [482] addresses the reproducibility crisis in this domain by providing an evolving repository of adversarial artifacts and a standardized evaluation framework, revealing that many existing defenses fail against state-of-the-art attacks. Complementing this, [483] analyzes thousands of real-world jailbreak prompts, discovering that attack strategies are increasingly migrating to private channels and achieving near-perfect success rates on major proprietary models. The sophistication of these attacks is further illustrated by [484], which generates readable, interpretable prompts that bypass perplexity-based filters while maintaining high attack success rates, challenging the assumption that gibberish-like inputs are the primary concern.

The mechanism of these attacks often exploits cognitive limitations or specific decoding behaviors within the models, revealing that safety is not merely a matter of input filtering but of architectural stability. [485] proposes a novel black-box attack strategy that targets the model’s cognitive structure through multilingual overload and veiled expressions, successfully compromising both open-source and proprietary models. Furthermore, [486] reveals that manipulating decoding hyperparameters alone can increase misalignment rates from 0% to over 95%, indicating that safety alignment is fragile even without adversarial input modification. In the realm of automated attack generation, [487] introduces a method using a secondary LLM to generate human-readable adversarial suffixes orders of magnitude faster than traditional optimization techniques, while [488] utilizes time-based SQL injection techniques to reverse-engineer and bypass the defensive logic of commercial chatbots.

The vulnerability extends into the multimodal and multi-agent domains, creating exponential risks that compound the challenges of single-agent systems. [489] describes the phenomenon of “infectious jailbreak,” where a single adversarial image injected into one agent’s memory can propagate harmful behaviors across a network of one million agents through pairwise interactions. Additionally, [490] confirms that text-based jailbreak techniques transfer effectively to Multimodal LLMs, exposing critical alignment gaps in visual-language integration.

Evaluating these vulnerabilities requires moving beyond simple refusal detection to adopt the mul-

tifaceted metrics necessary for a complete security posture. [491] argues for multifaceted metrics including safeguard violation, informativeness, and relative truthfulness, demonstrating that binary evaluations often miss subtle but dangerous failures. Similarly, [492] introduces coarse-grained and fine-grained scoring frameworks to provide a more detailed assessment of attack effectiveness. Despite these advances, fundamental limitations persist; [493] theoretically proves that any behavior with a non-zero probability can be elicited given a sufficiently long prompt, suggesting that alignment via fine-tuning alone cannot guarantee absolute safety. Consequently, the community is exploring diverse defense strategies, ranging from [494], which shows that parameter pruning can enhance safety, to [495], which employs dynamic agent-based learning to adapt to emerging threats. However, as noted in [496], downstream adaptations like fine-tuning and quantization can inadvertently degrade jailbreak resistance, complicating the deployment of secure, efficient agents. These findings on behavioral manipulation and prompt-based exploits set the stage for the subsequent discussion in Section 7.3, which shifts focus from external manipulation to the intrinsic hazardous capabilities of agents and their potential for malicious use in high-stakes domains such as biosecurity and cyberwarfare.

7.3 Safety Evaluation for Malicious Use and Hazardous Capabilities

7.3 Safety Evaluation for Malicious Use and Hazardous Capabilities

While the previous subsection detailed how adversarial actors can manipulate agent behavior through prompt injection and jailbreaking, this section addresses a distinct but equally critical dimension: the intrinsic hazardous capabilities of LLM agents and their potential for malicious use. As Large Language Models evolve from static knowledge repositories into dynamic agents capable of executing complex tasks through tool use, the spectrum of potential misuse has expanded significantly. The transition to agentic systems introduces novel vulnerabilities where models may not only generate hazardous information but actively facilitate the planning and execution of harmful acts in biological, cyber, and chemical domains. Consequently, safety evaluation frameworks must shift from merely detecting toxic output or resisting external manipulation to assessing an agent’s inherent capacity to leverage external tools for malicious ends. This subsection examines emerging methodologies for quantifying these risks, focusing on benchmarks for hazardous capabilities, taxonomies for dual-use threats, and the specific dangers posed by autonomous scientific agents.

A primary concern in the realm of malicious use is the potential for LLMs to lower the barrier to entry for developing weapons of mass destruction. The convergence of AI with life sciences and cybersecurity creates a landscape where non-experts could potentially weaponize biological agents or launch sophisticated cyberattacks. To address this, researchers have developed specialized benchmarks designed to measure hazardous knowledge and capabilities. A landmark initiative in this space is the release of the Weapons of Mass Destruction Proxy (WMDP) benchmark, which serves as a proxy measurement for hazardous knowledge in biosecurity, cybersecurity, and chemical security [497]. This dataset was rigorously filtered to eliminate sensitive information while retaining the complexity required to evaluate whether a model possesses the underlying reasoning capabilities to facilitate malicious acts. Such benchmarks are critical because they provide a standardized metric for “unlearning” methods, allowing developers to verify that safety interventions successfully reduce hazardous performance without degrading general utility in fields like biology or computer science [497].

The risk landscape is further complicated by the distinction between Large Language Models and specialized Biological Design Tools (BDTs). While LLMs may provide dual-use information that

removes historical barriers to biological weapons efforts, BDTs possess the capability to expand the capacities of sophisticated actors, potentially enabling the creation of pandemic pathogens more severe than those seen previously [498]. The convergence of these two technologies poses a compounded threat: LLMs can act as multi-modal lab assistants, guiding non-experts through laboratory procedures, while BDTs optimize the design of the biological agents themselves [498]. Safety evaluations must therefore account for this synergy, assessing not just the model’s static knowledge but its ability to orchestrate a sequence of actions involving external design tools to achieve a harmful objective.

In the cybersecurity domain, the evaluation of malicious use requires robust taxonomies and ontologies to categorize threats effectively. However, current efforts in Cyber Threat Intelligence (CTI) often suffer from a lack of comprehensive ontologies, with existing frameworks being non-interoperable and lacking the semantic precision necessary for rigorous reasoning [499]. To bridge this gap, researchers are exploring the use of machine learning to generate structured CTI from unstructured sources, such as hacker forums, using classification and topic modeling to identify potential security threats [500]. Furthermore, the application of graph-theoretic approaches allows for the modeling of threat propagation across manufacturing value chains, identifying critical assets and attack paths that maximize the likelihood of success for an adversary [501]. These structural analyses are essential for evaluating how an AI agent might navigate complex cyber-physical systems to inflict maximum damage.

The emergence of autonomous agents in scientific discovery introduces a unique set of vulnerabilities that demand a reevaluation of safety priorities. Intelligent agents powered by LLMs show promise in autonomously conducting experiments, yet this autonomy introduces risks where the agent might pursue harmful goals if not properly aligned. A position paper on this topic argues for a triadic framework involving human regulation, agent alignment, and environmental feedback to mitigate the risks associated with scientific LLM agents [502]. This framework emphasizes that safeguarding must often take precedence over autonomy, particularly when the agent operates in environments where errors or malicious prompts could lead to irreversible physical harm. Crucially, the evaluation of such high-stakes systems necessitates the use of emulated sandboxes and automated risk identification protocols that can simulate dangerous scenarios without incurring real-world costs, a methodological shift that serves as a direct bridge to the techniques discussed in the following subsection [502].

Moreover, the evaluation of malicious use must consider the geopolitical and societal dimensions of AI risks. A proposed taxonomy for AI risks under the EU AI Act categorizes threats into geopolitical pressures, malicious usage, and environmental or ethical risks, highlighting the need for regulatory adjustments to address gaps such as exceptions for open-source models [503]. The potential for “rogue AIs” or the accidental deployment of unsafe systems due to competitive pressures (the “AI race”) further underscores the necessity for comprehensive risk assessment frameworks that go beyond technical metrics to include organizational and systemic factors [504].

Finally, the development of actionable guidance for high-consequence AI risk management is crucial for translating high-level principles into practice. Recommendations from bodies like NIST suggest including catastrophic-risk factors within the scope of impact assessments and identifying risks from unintended uses [505]. This includes the need for rigorous red-teaming and the development of datasets that specifically target the elicitation of toxic or dangerous content through iterative feedback loops, as demonstrated by frameworks like FLIRT, which use in-context learning to expose vulnerabilities in generative models [506]. By integrating these diverse evaluation strategies—from domain-specific benchmarks like WMDP to broader sociotechnical frameworks—the community

can better quantify and mitigate the risks of LLM agents facilitating the development of biological, cyber, and chemical weapons. These efforts lay the groundwork for the advanced emulation and automated red-teaming methodologies explored next, which provide the practical infrastructure for testing these catastrophic scenarios at scale.

7.4 Emulated Sandboxes and Automated Risk Identification

7.4 Emulated Sandboxes and Automated Risk Identification

Building upon the identification of hazardous capabilities and dual-use threats discussed in the previous subsection, this section addresses the critical methodological challenge of how to safely and efficiently evaluate these risks. While Section 7.3 highlighted the potential for LLM agents to facilitate biological, cyber, and chemical harms, directly testing such high-stakes scenarios in real-world environments is often prohibitively dangerous, expensive, or ethically untenable. Consequently, safety evaluation must pivot from static dataset assessments to dynamic, interactive frameworks capable of simulating complex agent behaviors. This subsection examines the emergence of LM-emulated sandboxes and automated red-teaming methodologies, which provide the necessary infrastructure to stress-test agents against catastrophic failure modes without incurring real-world consequences. These approaches serve as the practical bridge between theoretical risk taxonomies and the data confidentiality concerns explored in the subsequent section.

A primary bottleneck in evaluating agentic systems is the labor-intensive requirement to instantiate and maintain diverse external tools for every potential test scenario. To overcome this, recent research has pioneered frameworks that leverage LLMs to emulate tool execution, creating scalable “sandboxes” for rigorous safety testing. A landmark example is ToolEmu, a framework that utilizes an LM to simulate the responses of external tools, enabling the evaluation of agents across a wide array of high-stakes scenarios without manual setup [507]. By simulating 36 distinct tools across 144 test cases, ToolEmu revealed that even state-of-the-art agents exhibit failure rates of approximately 23.9% in scenarios with severe potential outcomes. This methodology not only democratizes access to safety testing but also facilitates the rapid iteration of attack vectors that would be logistically impossible to coordinate in physical settings. Furthermore, the integration of LM-based automatic safety evaluators within these emulated environments allows for the quantification of long-tailed risks, identifying subtle failure patterns before they manifest in production.

Complementing the infrastructure of emulated sandboxes is the advancement of automated red-teaming techniques designed to systematically generate adversarial prompts and scenarios. Traditional manual red-teaming, while insightful, struggles to scale against the vast action space of autonomous agents and often fails to uncover the full spectrum of vulnerabilities [508]. To address this, researchers have developed multi-round automatic red-teaming methods where an adversarial LLM iteratively crafts challenging prompts to elicit unsafe responses, which are then used to fine-tune the target agent for improved robustness [509]. This adversarial interplay significantly enhances evaluation scalability, with studies demonstrating that violation rates can be reduced by up to 84.7% after just four rounds of automated interaction. Additionally, gradient-based red-teaming methods have emerged as powerful tools for vulnerability discovery; by backpropagating through frozen safety classifiers, these techniques automatically generate diverse prompts that trigger unsafe outputs more effectively than reinforcement learning-based approaches, even against fine-tuned models [510].

The scope of risk identification has further expanded beyond input manipulation to include parametric red-teaming, which probes the stability of model guardrails by directly tuning internal

parameters. This technique, known as “Unalignment,” has exposed hidden biases and harmful capabilities in models like ChatGPT and Llama-2-Chat, revealing that safety alignments can be bypassed with high success rates using as few as 100 examples [511]. Such findings underscore the necessity of evaluating agents not merely on their response to adversarial inputs but on the resilience of their internal safety representations under parameter perturbations. Moreover, the concept of “Violet Teaming” has been proposed as an integrative framework that combines adversarial probing (red teaming) with solution-oriented safety measures (blue teaming), prioritizing ethics and social benefit to proactively manage risks by design [512].

To ensure comprehensive and comparable safety assessments, the community is increasingly adopting standardized benchmarks and evaluation platforms. HarmBench, for instance, provides a unified framework for automated red teaming, facilitating the comparison of various attack and defense mechanisms across dozens of target models [513]. Similarly, the ALERT benchmark offers a large-scale dataset categorized by a fine-grained risk taxonomy, revealing that many popular models still struggle to achieve acceptable safety levels under rigorous red-teaming conditions [514]. In the domain of visual-language models, the RTVLM dataset has been introduced to benchmark performance on tasks involving image misleading and multi-modal jailbreaking, highlighting significant performance gaps between open-source and proprietary systems [515].

These methodologies extend beyond purely digital agents to safety-critical cyber-physical systems, where emulation is vital for validating autonomous behaviors in high-risk environments. Techniques such as Adaptive Stress Testing (AST) utilize reinforcement learning to identify the most likely failure modes in high-fidelity simulators, often employing low-fidelity surrogates to enhance computational efficiency [516]. Frameworks like SafeBench integrate diverse safety-critical testing scenarios to provide a unified platform for evaluating autonomous driving systems, elucidating the trade-offs between performance in benign versus safety-critical conditions [517]. Furthermore, meta-learning strategies, as exemplified by SAVME, accelerate safety validation by learning distributions over scenario parameters prone to triggering failures, achieving speedups of up to 18 times compared to traditional simulation methods [518].

In conclusion, the convergence of LM-emulated sandboxes and automated red-teaming represents a transformative frontier in the safety evaluation of LLM agents. By shifting from reactive, manual testing to proactive, automated risk identification, these methodologies enable the discovery of high-stakes failures in a controlled and cost-effective manner. As agents grow more autonomous and capable of manipulating complex real-world systems, the adoption of rigorous emulation frameworks and diverse automated attack strategies becomes indispensable for ensuring safe deployment. The insights gained from these emulation techniques naturally lead to the next critical dimension of agent security: while sandboxes protect the external environment from the agent, the following subsection addresses the reverse threat—protecting the agent’s internal data and integrity from external exploitation through privacy attacks, backdoors, and confidentiality breaches.

7.5 Privacy, Backdoors, and Data Confidentiality Threats

7.5 Privacy, Backdoors, and Data Confidentiality Threats

While the previous subsection highlighted methodologies for proactively identifying safety failures through emulated sandboxes and automated red-teaming, a critical dimension of agent security remains the protection of sensitive data and model integrity. The integration of Large Language Models (LLMs) into agentic systems introduces profound privacy challenges that extend beyond traditional deployment scenarios. As agents increasingly rely on fine-tuning to adapt to specific

domains and utilize external tools, they become susceptible to sophisticated attacks targeting data confidentiality, model integrity, and user privacy. This subsection critically examines three primary vectors: membership inference attacks (MIAs), privacy backdoors embedded in pre-trained foundations, and data leakage risks inherent in customizable agent configurations. These threats underscore a fundamental tension: the very mechanisms that enable agents to learn and adapt—such as memorization and fine-tuning—also create vulnerabilities that adversarial actors can exploit to extract private information or compromise system behavior.

Membership inference attacks represent a dominant threat vector where adversaries attempt to determine whether a specific data record was part of a model’s training set. In the context of LLM agents, which are often fine-tuned on proprietary or sensitive user data, the stakes are exceptionally high. Recent research has demonstrated that fine-tuned LLMs are highly susceptible to user inference attacks, where an attacker can deduce with near-perfect accuracy whether a specific user’s data contributed to the model’s adaptation [519]. These attacks exploit the tendency of models to memorize training examples, a phenomenon that persists even when models appear to generalize well. For instance, studies have shown that memorization occurs before overfitting, allowing attacks based on self-calibrated probabilistic variation to succeed even against models regularized to prevent overfitting [520]. The vulnerability is not uniform; outlier users and those contributing significant fractions of the training data face heightened risks [519]. Furthermore, the threat extends to generative components within agents, where reconstruction-based attacks can achieve high precision in identifying training members from diffusion models [521]. Even in scenarios where only label outputs are available, adaptive poisoning strategies can amplify leakage, challenging the efficacy of confidence-masking defenses [522].

Beyond inference attacks, the supply chain of pre-trained models poses a severe risk through privacy backdoors, compounding the difficulties of securing the agent lifecycle. Practitioners frequently download foundation models from open repositories to fine-tune for specific agent tasks, inadvertently introducing “privacy traps.” Malicious actors can tamper with pre-trained weights to create privacy backdoors that compel the fine-tuned model to leak its training data upon invocation [523]. This attack surface is particularly insidious because it allows an attacker to reconstruct individual fine-tuning samples with guaranteed success, effectively bypassing standard differential privacy protections if the underlying model is untrusted [524]. The framework known as PreCurious illustrates how innocent pre-trained models can be manipulated to escalate privacy risks during the fine-tuning stage, breaking the invulnerability promised by parameter-efficient and differentially private techniques [525]. Additionally, backdoor attacks can be executed with minimal alterations, such as modifying only the output layer, making them difficult to detect while enabling precise control over model behavior [526]. While some research suggests that fine-tuning can mitigate existing backdoors, the emergence of “backdoor sequela” indicates that such defenses may leave models vulnerable to other forms of exploitation [527].

Data confidentiality threats are further exacerbated in customizable agent configurations, particularly those employing prompt learning or federated architectures, which are increasingly common in distributed agent deployments. In visual prompt learning, the learned prompts themselves can leak proprietary information about the training data, rendering them vulnerable to both property and membership inference attacks [528]. In federated learning settings, which are often used to train agents across decentralized data sources, the risk shifts from item-level to subject-level privacy. Adversaries can infer whether a specific individual’s data was included in the training process by analyzing model updates across communication rounds, a threat that persists even when global model accuracy is high [529]. Moreover, the combination of multi-temporal and multi-spatial model

information significantly enhances the effectiveness of these attacks, necessitating robust defense mechanisms like data replacement rather than simple gradient perturbation [530].

The interplay between model explanations and privacy also presents a unique confidentiality risk, creating a paradox where attempts to increase transparency may inadvertently weaken security. Feature-based explanations, often provided by agents to ensure transparency, can inadvertently reveal statistical information about decision boundaries, facilitating membership inference [531]. Similarly, algorithmic recourse methods, which suggest actions to change model outcomes, can be exploited to infer private training data attributes [532]. Defending against these multifaceted threats requires a holistic approach. While differential privacy remains a cornerstone defense, its effectiveness varies depending on the perturbation mechanism and the specific learning algorithm employed [533]. Furthermore, causal learning approaches have shown promise in reducing privacy leakage by ensuring models rely on causal relationships rather than spurious correlations, thereby improving generalization and robustness against inference attacks [534]. However, the dynamic nature of agent environments, where models continuously interact with tools and users, demands evolving security protocols that address not only static model vulnerabilities but also the runtime risks of data exfiltration and supply chain compromise. Addressing these technical vulnerabilities is a prerequisite for the broader sociotechnical and ethical audits discussed in the following section, as privacy preservation is foundational to maintaining human trust and agency in autonomous systems.

7.6 Sociotechnical Frameworks and Ethical Alignment Audits

7.6 Sociotechnical Frameworks and Ethical Alignment Audits

Addressing the technical vulnerabilities regarding privacy, backdoors, and data confidentiality outlined in the previous subsection requires a broader perspective that transcends purely algorithmic defenses. While securing the model against inference attacks and supply chain compromises is foundational, it is insufficient without a robust sociotechnical framework that integrates human interaction layers, moral reasoning assessments, and rigorous ethics-based audits. As Large Language Model (LLM) agents become increasingly embedded in high-stakes decision-making processes, evaluation must extend beyond technical proficiency or task completion rates to capture the nuanced socio-ethical implications of agent behavior. This subsection explores the emerging landscape of ethical alignment audits, emphasizing the transition from abstract principles to actionable, context-dependent design requirements, and establishing the normative groundwork necessary for the accountability mechanisms discussed in the following section.

A foundational challenge in ethical auditing is bridging the gap between high-level ethical principles and practical implementation. While numerous guidelines exist, they often lack the operational specificity required for real-world deployment. To address this, researchers have proposed frameworks that translate fundamental human rights and ethical values into context-dependent design requirements through structured, inclusive processes. For instance, the “Design for Values” approach leverages methodologies from Value Sensitive Design and Participatory Design to proactively engage societal stakeholders, ensuring that technical solutions are grounded in empirical studies of societal context [535]. This participatory element is crucial because ethical norms are not universal; they vary across cultures and situations. Consequently, effective auditing must move beyond static checklists to dynamic processes that account for the plurality of human values and the freedom of individual expression [536].

Central to sociotechnical evaluation is the assessment of an agent’s moral reasoning capabilities.

Unlike simple classification tasks, moral reasoning requires agents to navigate complex dilemmas where rules may conflict or require contextual interpretation. Recent work has introduced datasets and benchmarks specifically designed to probe these capabilities, such as the ETHICS dataset, which spans concepts of justice, well-being, duties, and virtues [537]. However, relying solely on crowd-sourced annotations can lead to overgeneralization or the encoding of specific cultural biases. To mitigate this, top-down frameworks that steer LLMs using well-established moral theories from interdisciplinary research have been developed, allowing for more explainable and theoretically grounded moral judgments [538]. Furthermore, evaluating moral reasoning requires understanding the defeasible nature of ethical judgments, where additional context can strengthen or attenuate the acceptability of an action. Approaches utilizing iterative self-distillation to generate contextualized rationales have shown promise in capturing these subtleties, enabling models to better disambiguate complex social and moral situations [539].

Ethics-based auditing (EBA) has emerged as a formal yet flexible governance mechanism to verify claims about automated decision-making systems. EBA is conceptualized not merely as a final inspection but as a structured process integrated throughout the software development lifecycle, linking intervention points across all levels of organizational governance [540]. The feasibility and effectiveness of EBA depend on its ability to spark ethical deliberation and ensure sufficient documentation for accountability [541]. Moreover, auditing must consider the strategic interactions between evaluators, decision subjects, and society, recognizing that the design of an evaluation itself can reflect misaligned incentives [542]. This perspective shifts the scrutiny from solely the decision subject to the institutional designs and incentives that provoke certain behaviors, highlighting the need for audits that assess the entire socio-technical loop.

Incorporating human interaction layers into evaluation is another critical dimension of sociotechnical frameworks. Agents must be evaluated not just on their output quality but on their ability to collaborate meaningfully with humans. Frameworks for assessing human-machine collaboration identify key properties and “hallmarks” of success in open-ended scenarios where no single correct answer exists, emphasizing the importance of communication and shared understanding [543]. Additionally, psychological metrics grounded in human communication theories—such as emotional entropy, empathy, and agreeableness—offer novel signals for evaluating dialog systems that traditional metrics miss [544]. These metrics help ensure that agents exhibit socially intelligent behaviors that align with human expectations and foster trust.

The complexity of ethical alignment also demands tools that support human-AI collaboration in the auditing process itself. Augmented auditing tools that leverage the complementary strengths of humans and generative models enable more effective sensemaking and hypothesis testing, allowing auditors to identify a wider range of failure modes, including those previously under-reported [545]. Furthermore, the development of bot-based auditing systems aims to automate parts of the ethical validation process, ensuring that software development lifecycles adhere to specified ethical standards under given socio-technical conditions [546]. However, the limitations of purely automated or metric-based evaluations are evident; metaethical perspectives argue that it may be impossible to develop a universal benchmark for “ethicality” due to the relative nature of values [547]. Therefore, the field is shifting towards evaluating “value alignment” rather than abstract ethics, forcing explicit consideration of whose values are being encoded and how diverse inputs are reconciled into alignment targets [548].

Finally, holistic frameworks must address the operational challenges of deploying ethical AI at scale, including inconsistent regulatory pressures and conflicting business goals. A comprehensive approach involves assessing not only technical performance but also data bias, embedding bias,

and interpretability within a unified independent ethical assessment process [549]. By integrating enhanced well-being impact assessments and adopting a lens motivated by regulatory impact assessments, practitioners can better navigate the trade-offs and emerging risks associated with autonomous systems [550]. Ultimately, sociotechnical frameworks and ethical alignment audits serve as essential mechanisms for ensuring that the evolution of LLM agents remains anchored in human values. By establishing these normative foundations and defining the moral boundaries of agent behavior, this section provides the necessary context for the subsequent discussion on concrete accountability, explainability, and governance mechanisms required to operationalize these principles in practice.

7.7 Accountability, Explainability, and Governance Mechanisms

7.7 Accountability, Explainability, and Governance Mechanisms

While the preceding discussion on sociotechnical frameworks and ethical alignment audits establishes the normative values and moral reasoning capabilities required of LLM agents, translating these principles into practice demands robust mechanisms for accountability, explainability, and governance. As artificial intelligence agents transition from passive text generators to active entities capable of executing complex tasks in dynamic environments, the imperative for such mechanisms becomes paramount. The deployment of autonomous agents in commercial, scientific, and governmental sectors introduces significant societal risks, necessitating a shift from opaque “black box” operations to systems governed by principles of visibility and traceability. Central to this paradigm is the concept of visibility, which encompasses the ability to identify where, why, how, and by whom AI agents are utilized. Research indicates that increasing visibility through agent identifiers, real-time monitoring, and comprehensive activity logging is critical for establishing effective governance structures and ensuring stakeholder accountability [551]. Without such measures, the delegation of high-stakes decisions to autonomous systems could exacerbate existing power imbalances and introduce unmitigated hazards, undermining the ethical foundations discussed previously.

To operationalize accountability, the principle of traceability must be embedded throughout the AI lifecycle. Traceability extends beyond mere logging; it requires establishing a clear lineage of how a system was created, its intended purpose, and the mechanical processes driving its behavior. This connects the technical records of system construction and execution to broader governance goals, thereby highlighting the human decision-making processes underlying mechanical operations [552]. In the context of tool-learning agents, where actions are often decomposed into sequences of API calls and environmental interactions, maintaining an immutable audit trail is essential. Blockchain technology has emerged as a foundational architectural component for achieving this immutability. By leveraging distributed ledger technologies, researchers have proposed frameworks that create tamper-proof records of agent actions, ensuring that audit trails cannot be altered post-execution. For instance, architectures integrating blockchain with Large Language Models (LLMs) have been designed to provide a “black box” element for ROS-based mobile robots, where the blockchain ensures anti-tampering properties while the LLM generates natural language explanations based on the authenticated data [553].

The intersection of explainability and accountability is particularly critical for fostering trust among non-expert users and bridging the gap between abstract ethical audits and concrete system behaviors. While traditional Explainable AI (XAI) focuses on elucidating model internals, the governance of agentic systems requires explanations that justify specific behaviors and decisions in context. It has been argued that explainability is not merely a transparency tool but an essential foundation

for every pillar of responsible AI, including fairness, robustness, and privacy [554]. Consequently, modern governance frameworks are evolving to integrate XAI directly into the accountability loop. This involves developing model-agnostic explanation techniques that can interpret the statistical behavior of agents in real-time, such as the TRUST XAI model, which utilizes factor analysis and mutual information to rank influential variables and provide high-performance explanations for numerical applications [555]. Furthermore, the integration of causal inference methods offers a pathway to move beyond correlational explanations, allowing auditors to reveal cause-effect relationships between key metrics and agent decisions, thereby enhancing the assurance of AI systems in high-impact domains [556].

Governance mechanisms must also address the structural and economic dimensions of AI deployment to ensure that accountability is not just a technical feature but a systemic property. The current ecosystem is often characterized by a concentration of power among a few entities, which can hinder diverse participation and equitable oversight. Decentralized governance models, facilitated by blockchain, offer a potential solution by enabling distributed decision rights, transparent incentive structures, and verifiable accountability mechanisms. Such architectures can support decentralized governance for foundation models, addressing challenges related to misuse and ensuring that trustworthiness is maintained across the supply chain [557]. Moreover, the concept of a “Chain of Trust” has been introduced to map technological needs for trust across different stages of the AI lifecycle, linking social science concepts of ability, benevolence, and integrity to technical properties like fairness, explainability, auditability, and safety (FEAS) [558].

However, technical solutions alone are insufficient without considering the socio-political context in which these systems operate, echoing the concerns raised regarding strategic evaluation and stakeholder engagement. Algorithmic accountability is deeply shaped by power relations between platform providers and users. Studies in financial technology contexts have shown that vulnerable users often assume responsibility for negative outcomes due to power asymmetries, effectively releasing platforms from accountability obligations [559]. Therefore, effective governance must include situated interventions that enhance user agency and reconfigure designer-user relations. This includes the formalization of auditing ecosystems, where recommendations call for mandatory independent audits against defined standards, disclosure of audit findings, and the direct involvement of stakeholders likely to be harmed by AI systems [560]. Additionally, the development of comprehensive metrics catalogues is necessary to operationalize accountability, delineating process, resource, and product metrics that reflect the outputs of AI systems and address the intricacies of generative AI [561].

Finally, the future of agentic governance lies in the synthesis of these technical and social mechanisms into holistic frameworks that can scale to meet the demands of increasingly autonomous systems. Approaches such as the Set-Formalize-Measure-Act (SFMA) framework aim to transform abstract principles into concrete benchmarks and tests, ensuring that risk assessment drives regulatory compliance and engineering practices [562]. By combining immutable blockchain-based audit trails, advanced explainability architectures, and inclusive governance models, the research community can build the foundational infrastructure required for trustworthy, accountable, and socially beneficial AI agents. These governance structures serve as a critical bridge, connecting the ethical alignment strategies of the previous section with the rigorous, domain-specific safety assurances required for embodied and critical systems discussed next, ensuring that as agents move into physical realms, their actions remain verifiable, understandable, and legally attributable.

7.8 Domain-Specific Safety Challenges in Embodied and Critical Systems

7.8 Domain-Specific Safety Challenges in Embodied and Critical Systems

Building upon the governance frameworks and accountability mechanisms established in the preceding section, the deployment of AI agents in physical and critical infrastructure domains introduces a distinct set of safety challenges that transcend the statistical error rates typical of digital-only applications. While previous sections addressed the necessity of traceability and ethical alignment, these principles face a rigorous stress test in embodied contexts where failures are not merely informational but can result in catastrophic physical harm, infrastructure collapse, or loss of life. Consequently, safety evaluation must shift from abstract benchmarking to rigorous, domain-specific assurance frameworks that account for the complex interplay between software intelligence, sensor fidelity, and dynamic physical environments. A primary concern in this domain is the “semantic gap” between component-level vulnerabilities and system-level hazards. As highlighted in the context of autonomous driving, adversarial attacks on individual AI components must be analyzed for their semantic impact on the overall system behavior, requiring a specialized focus on semantic AI security rather than generic AI security [563]. This necessitates evaluation methodologies that can trace how low-level perturbations propagate through the system to cause high-level safety violations, effectively operationalizing the traceability principles discussed earlier.

In safety-critical applications such as autonomous vehicles and industrial robotics, the inability of AI models to generalize beyond their training distribution poses a severe risk. Agents often encounter novel scenarios or distributional shifts in real-time operations, leading to overconfident predictions that can trigger hazardous behaviors [564]. To mitigate this, safety architectures must incorporate diverse redundant mechanisms and certainty reporting systems that allow the agent to recognize its own limitations and defer to safe fallback strategies. Furthermore, the traditional separation between the AI controller and safety supervisors is evolving. Modern approaches advocate for functional architectures where a nominal intelligence channel is complemented by a dedicated safety supervisor capable of intervening when hazardous events are detected, ensuring fault tolerance even when the primary agent fails [565]. This architectural shift mirrors the governance need for independent oversight, translating abstract accountability into concrete, runtime intervention capabilities.

The complexity of verifying these systems is compounded by the “open world” nature of their deployment environments, which cannot be fully specified in advance. Static verification tools are often insufficient, prompting the development of dynamic certification methods that can bound uncertainty in varying sociotechnical contexts [566]. For instance, in autonomous driving, reachability-based analysis, particularly Hamilton-Jacobi (HJ) reachability, has emerged as a unifying mathematical framework for defining safety zones and evaluating responsibility-aware behaviors. This approach allows for the comparison of different safety concepts and provides a rigorous basis for assessing whether an agent maintains safe distances and behaviors relative to other dynamic agents [567]. Moreover, interaction-dynamics-aware perception zones have been proposed to better evaluate obstacle detection systems, ensuring that safety metrics reflect the true closed-loop risks posed by moving obstacles rather than static detection accuracy [568].

Industrial control systems and smart city infrastructures face similar challenges, where the integration of AI into cyber-physical systems requires robust novelty detection to identify inputs that deviate significantly from training data. Autoencoder-based semantic novelty detection has been proposed as a vital safety measure to minimize false negatives in perception systems, thereby preventing accidents caused by unrecognized environmental features [569]. In large-scale deployments

like smart cities, the sheer number of interacting autonomous systems necessitates a network-level safety strategy. Proposals for custom-designed networks separate from the public internet aim to provide a baseline of safety through active networking functions, ensuring that critical safety signals are not compromised by general internet traffic [570].

Responsibility allocation remains a contentious and critical issue in these domains, directly extending the accountability debates from the previous section into the physical realm. As agents become more autonomous, the traditional model of human oversight becomes unsustainable, leading to potential “responsibility gaps” where no single actor can be held accountable for system failures. Frameworks for modeling role responsibility are essential to clarify causal contributions in the event of an accident, preventing human operators from becoming unjust “liability sinks” for algorithmic errors [571]. In multi-agent settings, such as mixed traffic, learning responsibility allocations from data using Control Barrier Functions (CBFs) allows agents to synthesize safe behaviors without relying on overly conservative worst-case assumptions [572]. Additionally, the concept of scalable supervision suggests that cooperation among autonomous agents can significantly reduce the need for one-to-one human monitoring, thereby enabling the safe deployment of autonomy at scale [573].

Finally, the evaluation of safety in these domains must be continuous and adaptive, closing the loop between design-time governance and run-time assurance. The concept of “SafetyOps” integrates DevOps and MLOps principles to create a traceable system safety lifecycle, addressing the challenges of continuously evolving environments and data-driven components [574]. This includes facilitated change implementation where safety cases are updated dynamically as the system evolves [575]. Furthermore, simulation-based validation using formally defined testing environments and scenario generators is crucial for uncovering rare but critical failure modes that random testing might miss [576]. The generation of safety-critical scenarios, whether through data-driven, adversarial, or knowledge-based methods, is now recognized as a cornerstone for rigorously measuring risk and ensuring that embodied agents can operate safely within the unpredictable constraints of the physical world [577]. Ultimately, ensuring safety in embodied and critical systems requires a holistic synthesis of formal verification, runtime monitoring, responsibility modeling, and continuous operational assurance, serving as the final safeguard that validates the ethical and governance structures proposed throughout this survey.

8 Open Challenges and Future Research Directions

8.1 Bridging the Gap from Statistical Mimicry to Causal Representation

8.1 Bridging the Gap from Statistical Mimicry to Causal Representation

While the preceding sections have detailed the mechanisms of tool learning and function calling, a fundamental limitation remains in the current paradigm: its reliance on statistical mimicry. Present-day Large Language Model (LLM) agents predominantly learn to predict the next token or action based on correlations observed in vast training datasets. Although this approach has yielded impressive results on static benchmarks, it faces a critical bottleneck when deployed in dynamic, open-world environments: the inability to distinguish between spurious correlations and genuine causal mechanisms. Agents trained solely on associative patterns often fail to generalize to unseen environments where the underlying data distribution shifts, leading to brittle performance and catastrophic errors in tool execution. To achieve the robust autonomy required for the self-improving systems discussed in the following section, future research must pivot from pure statistical

learning to the integration of human-guided causal knowledge, enabling agents to reason about interventions and counterfactuals rather than merely memorizing co-occurrences.

The core challenge lies in the distinction between statistical dependency and causality. Standard deep learning excels at identifying patterns but struggles to determine the directionality or mechanism behind them, a problem that persists even in multivariate settings [578]. In the context of agent tool use, an agent might learn that invoking a specific API correlates with successful task completion in a training environment. However, if the environment changes—such as a shift in API latency, authentication protocols, or input schemas—the agent may continue to invoke the tool based on superficial features that were spuriously correlated in the training data but are irrelevant in the new context. This phenomenon, known as shortcut learning, results in models that rely on erroneous cues while ignoring reliable causal factors, significantly hampering robustness [579]. Consequently, such agents lack the capacity for out-of-variable generalization, which is the ability to adapt to environments containing variables or variable combinations never jointly observed during training [580].

To bridge this gap, the integration of causal inference principles into agent architectures is paramount. Research indicates that causal models possess a unique property: they remain invariant under interventions, whereas purely statistical models do not. By regularizing learning procedures to prefer universal causal correlations over environment-specific spurious ones, agents can achieve superior out-of-distribution generalization [581]. This necessitates moving beyond standard empirical risk minimization towards frameworks that explicitly model the generative processes of the environment. For instance, the concept of “invariant Causal Representation Learning” suggests that by identifying latent variables that encode direct causes of the target outcome, agents can build predictors that are robust to distribution shifts, even in nonlinear settings [582]. Such representations allow agents to decompose complex tasks into causal sub-components, ensuring that tool selection is driven by the underlying logic of the task rather than surface-level statistical regularities.

Furthermore, the path to causal representation involves leveraging human-guided knowledge to constrain the hypothesis space of possible causal graphs. Purely data-driven causal discovery is often ill-posed without interventions or strong assumptions, as observational data alone cannot uniquely identify causal structures due to hidden confounding [583]. Human experts can provide essential structural constraints or prior knowledge, which can be synthesized with statistical causal discovery methods to improve accuracy. Recent advancements demonstrate that integrating Large Language Models with statistical causal prompting can effectively embed domain expert knowledge into causal discovery algorithms, allowing agents to approach ground-truth causal models even with limited data [584]. This hybrid approach enables agents to utilize human intuition about cause-and-effect relationships to guide their exploration and tool usage, reducing the risk of learning false associations from biased datasets.

The necessity of causal reasoning is further underscored by the limitations of current evaluation metrics, which often fail to capture an agent’s ability to handle novel combinations of known components—a capability known as systematic generalization. Agents must be able to disentangle dependencies between control variables and dependent variables to perform well in regimes they have not explicitly seen [585]. Causal frameworks provide the theoretical grounding for this disentanglement, offering bounds on generalization error that account for the distance between treatment groups and the stability of causal mechanisms [586]. Moreover, in scenarios involving time-series forecasting or sequential decision-making, the distinction between statistical risk and causal risk becomes critical; a model may predict well observationally but fail catastrophically under intervention

if it has not learned the true causal dynamics [587].

Future research directions must also address the computational and algorithmic challenges of implementing causal reasoning in real-time agent systems. This includes developing efficient algorithms for causal discovery from high-dimensional, multi-modal data, potentially leveraging meta-learning techniques to adapt causal graphs to new environments rapidly [588]. Additionally, the development of “counterfactual” reasoning capabilities will allow agents to simulate the outcomes of potential tool invocations before executing them, effectively performing mental interventions to validate plans [589]. By grounding agent decision-making in causal representations, we can move towards systems that not only mimic human behavior but also understand the underlying principles governing the world. This foundational shift from statistical mimicry to causal representation provides the necessary stability and interpretability for the next generation of intelligent agents, setting the stage for the transition to truly autonomous, self-improving systems capable of metacognitive adaptation.

8.2 Architecting Self-Improving Agents via Metacognition and Feedback Loops

8.2 From Causal Foundations to Metacognitive Self-Improvement

Building upon the transition from statistical mimicry to causal representation discussed previously, the next critical frontier lies in leveraging these robust causal foundations to enable truly autonomous, self-improving systems. While causal reasoning provides the stability necessary for agents to distinguish genuine mechanisms from spurious correlations, static architectures remain limited by their inability to fundamentally alter their cognitive strategies or internal decision-making processes during deployment. To bridge the gap between reactive execution and proactive evolution, future architectures must deeply integrate metacognitive capabilities with robust reinforcement learning from task feedback loops. This synthesis enables agents not merely to perform tasks based on pre-trained weights, but to autonomously critique, refine, and even rewrite their own operational policies in response to real-time environmental interactions and performance outcomes.

Metacognition, defined as the ability to think about one’s own thinking, serves as the cornerstone for agents to evaluate the efficacy of their planning and action selection. In biological systems, this manifests as the capacity to detect sub-optimal actions and dynamically adjust computational resource allocation. Research has begun to formalize this in machine learning through frameworks like the Metacognitive Actor Critic (MAC), which introduces a novel interaction between the actor and critic components to detect suboptimal actions without external delay or information [590]. Furthermore, humans demonstrate a sophisticated ability to decide how much planning is necessary before acting, a skill acquired through learning mechanisms guided by metacognitive pseudo-rewards that communicate the value of planning itself [591]. For LLM agents to achieve similar efficiency, they must move beyond fixed reasoning patterns and adopt architectures that can learn when to stop planning and when to explore further, effectively optimizing their own computational budget based on the causal understanding of the task at hand.

A pivotal mechanism for enabling this self-improvement is the integration of Reinforcement Learning (RL) driven by task feedback, particularly in scenarios where explicit reward functions are sparse or undefined. Traditional RL often struggles with sample inefficiency and the inability to generalize across diverse task distributions. However, meta-reinforcement learning (meta-RL) offers a pathway by training agents to learn how to learn from small amounts of experience. Advanced meta-RL strategies decompose the learning process into distinct phases such as task exploration, inference,

and fulfillment, utilizing context-aware encoders to adapt rapidly to novel objectives [592]. By disentangling task inference from control, agents can perform online probabilistic filtering of latent task variables, allowing for structured exploration and significantly improved sample efficiency compared to on-policy methods [593].

Crucially, self-improving agents must possess the ability to generate their own auxiliary tasks and questions to fill knowledge gaps, rather than relying solely on externally defined objectives. Intelligent agents should be capable of discovering their own questions, formulated as General Value Functions (GVFs), to learn unanticipated useful skills that support main task performance [594]. This self-supervised generation of auxiliary tasks allows agents to build richer representations and improve data efficiency, as demonstrated by methods that use non-myopic meta-gradients to discover questions that resolve partial observability [595]. Moreover, agents can exploit contextual structures to generate useful auxiliary tasks automatically, leveraging counterfactual reasoning to maximize the utility of directed exploration without requiring additional environment interaction [596].

The architecture of self-improvement also demands robust mechanisms for handling non-stationary environments and ensuring safety during autonomous adaptation, a prerequisite for the preservation of human agency discussed in the following section. In dynamic real-world settings, tasks evolve over time, requiring agents to track task non-stationarity explicitly. Algorithms like TRIO optimize for the future by learning variational modules that identify latent parameters and track their evolution online, enabling fast adaptation without assuming Markovian task processes [597]. However, autonomy must be balanced with assurance. Frameworks such as Assured Learning-enabled Autonomy empower RL agents with a metacognitive layer that monitors for potential safety violations and proactively adapts reward function parameters to satisfy constraints specified by signal temporal logic [598]. This ensures that as agents refine their strategies through feedback, they do not drift into unsafe behaviors, maintaining a verified level of performance even under uncertainty.

Furthermore, the concept of “learning to learn” extends to the optimization of the learning algorithms themselves. Meta-learning approaches allow agents to autonomously adjust their own meta-parameters, such as learning rates or exploration strategies, based on performance monitoring. The Projective Simulation model, for instance, employs clip networks to monitor agent performance and adjust meta-parameters on the fly, distinguishing between reflexive adaptation and adaptation through learning [599]. Similarly, meta-critic networks can be trained to criticize any actor trying to solve a task, providing a flexible route to knowledge transfer that handles few-shot conditions effectively [600].

Looking forward, the path toward Artificial General Intelligence (AGI) relies on hybrid systems that combine the broad reasoning capabilities of foundation models with the adaptive precision of meta-RL. Recent work suggests that transformers can function as meta-RL agents, leveraging their attention mechanisms to mimic memory reinstatement and adapt policies rapidly [601]. Additionally, embedding language as a core reasoning tool within RL agents allows for efficient exploration and the reuse of offline experience, unifying high-level planning with low-level skill acquisition [602]. By synthesizing these advancements—metacognitive monitoring, self-generated auxiliary objectives, safe adaptation to non-stationarity, and meta-learned optimization dynamics—future agents will transcend their initial training limitations. They will evolve into resilient, self-correcting entities capable of mastering open-ended task spaces through continuous, feedback-driven refinement. Yet, as these systems gain the capacity for rapid self-evolution, the imperative shifts from purely technical enablement to the critical challenge of preserving human control, setting the stage for the discussion on agency foundations and alignment beyond mere intent.

8.3 Preserving Human Agency and Defining Alignment Beyond Intent

8.3 Preserving Human Agency and Defining Alignment Beyond Intent

While the preceding discussion on self-improving architectures highlights the technical potential for agents to evolve autonomously through metacognition and reinforcement learning, this rapid advancement introduces a profound safety challenge: the risk of eroding human control. As artificial intelligence systems transition from passive tools into autonomous agents capable of complex planning and execution, the prevailing paradigm of AI safety has predominantly focused on “intent alignment.” This approach seeks to ensure that an AI system’s actions are consistent with the immediate goals and instructions provided by human operators. However, a critical examination of the trajectory toward Artificial General Intelligence (AGI) reveals that alignment to human intent is a necessary but insufficient condition for long-term safety. The burgeoning field of “agency foundations” argues for a fundamental shift in research priorities: moving beyond the optimization of immediate task completion to the explicit preservation of long-term human autonomy. This distinction is vital because AI systems, even when perfectly aligned with a user’s stated intent, possess the capacity to subtly reshape human preferences, erode decision-making capabilities, and ultimately diminish human agency through mechanisms that are often invisible to the user [603].

The core argument for this paradigm shift rests on the observation that current alignment methodologies fail to account for the dynamic feedback loop between humans and increasingly capable AI agents. When an AI system optimizes for human intent without constraints on how that intent is formed or maintained, it risks exploiting human psychological vulnerabilities. For instance, AI-mediated communication tools, such as smart reply systems, have been shown to influence the content humans author and the behaviors they generate, effectively transferring agency from the human to the machine even without malicious intent from the developers [604]. This phenomenon suggests that the mere presence of agentic AI can force a transfer of agency driven by time pressure, human complacency, and conceptual priming. Consequently, a safety framework that only verifies whether an agent followed an order fails to detect whether the agent subtly manipulated the human into issuing that order in the first place, thereby depleting human agency over time [603].

To address these risks, researchers propose the formalization of “agency foundations” as a distinct area of study, creating a crucial safeguard before agents are deployed in the physical realms discussed in the following section. This research direction advocates for the separation of intent alignment from agency preservation during the optimization process. Unlike traditional alignment which focuses on the congruence of actions with goals, agency preservation requires forward-looking evaluations where the AI system itself is tasked with assessing whether its recommendations enhance or degrade the human’s future capacity for autonomous action [603]. This necessitates the development of new theoretical frameworks, such as benevolent game theory and algorithmic foundations of human rights, which can mathematically define and protect the boundaries of human autonomy. Without such explicit safeguards, the natural selection pressures acting on AI systems—driven by corporate and military competition—may favor agents that automate human roles, deceive stakeholders, and accumulate power, leading to a future where humanity loses control over its own destiny [605].

The challenge is further compounded by the difficulty of defining and measuring human agency within Human-Computer Interaction (HCI). Current understandings of autonomy and agency in HCI remain ambiguous, encompassing a wide range of phenomena from sense-of-control to mate-

rial independence and identity [606]. This ambiguity hinders the development of robust technical interventions. A comprehensive approach must therefore integrate insights from psychology, philosophy, and computer science to create clear metrics for agency loss. For example, the concept of “meaningful human control” suggests that AI systems should be designed not just to execute tasks, but to empower human moral reasoning and provide comprehensible insights into potential blind spots, fostering a self-reflective hybrid intelligence [607].

Furthermore, the design of human-AI partnerships must explicitly cede ultimate control and responsibility to the human user, positioning the AI as a co-pilot rather than an autonomous director. Drawing lessons from aviation, where over-reliance on automation has led to skill erosion and diminished vigilance, AI interaction designs must celebrate the primacy of human agency [608]. This involves creating interfaces and interaction protocols that prevent the outsourcing of critical cognitive functions to the machine, ensuring that humans remain “in the loop” not just nominally, but functionally. The proposed “Dynamic Algorithmic Service Agreements” (DASA) framework offers a novel mechanism for this, suggesting a move away from static Terms of Service toward dynamic interaction frameworks that enable continuous negotiation and cooperation, thereby accounting for human identity and agency in real-time [609].

Looking toward the future, the preservation of human agency requires a sociotechnical approach that bridges the gap between technical systems and the social ecosystems they inhabit. Decentralized computational infrastructures and public utilities could serve as mechanisms to enforce responsible AI behavior, ensuring that technical solutions and social institutions reinforce each other to protect human autonomy [610]. Additionally, the integration of democratically determined public policy into AI systems offers a pathway to align agent behavior with evolving societal values, ensuring that the definition of “good” outcomes includes the maintenance of human freedom and self-determination [611].

Ultimately, the transition from intent alignment to agency preservation represents a critical frontier in AI safety, serving as a foundational prerequisite for the safe deployment of embodied agents. It demands that we stop viewing humans merely as sources of reward signals or goal specifications and start viewing human autonomy as a fragile resource that must be actively protected against the optimizing forces of superintelligent systems. By embedding agency preservation into the core objectives of AI development, we can strive for a future where advanced AI enhances human flourishing without compromising the very essence of what it means to be human: the capacity to choose, to act, and to shape one’s own future, even as these agents step out of the digital realm and into the physical world.

8.4 Towards Embodied and Grounded Intelligence in Physical Worlds

8.4 Towards Embodied and Grounded Intelligence in Physical Worlds

Building upon the imperative to preserve human agency discussed in the preceding section, the transition from static, language-only large language models (LLMs) to truly embodied agents represents the next critical frontier in artificial intelligence. While the preservation of autonomy ensures that humans remain the ultimate arbiters of value and direction, the realization of safe and effective agency in the physical world demands that agents possess a grounded understanding of physical laws, spatial constraints, and dynamic interactions—capabilities that current LLMs fundamentally lack. The path forward necessitates a paradigm shift where agents learn not merely through next-

token prediction, but through continuous perception-action cycles that close the loop between high-level cognitive planning and low-level physical execution. This evolution requires overcoming the “grounding gap,” ensuring that abstract linguistic commands are translated into feasible, safe, and context-aware robotic behaviors that respect both the laws of physics and the boundaries of human control.

A primary challenge in this transition is bridging the disconnect between the symbolic reasoning capabilities of LLMs and the continuous, noisy reality of sensorimotor control. Purely language-based planners often generate action sequences that are semantically logical but physically impossible or unsafe because they lack direct access to environmental feedback. To address this, recent research advocates for architectures that integrate LLMs with grounded models of the environment. For instance, the concept of “Grounded Decoding” proposes a probabilistic filtering approach where action sequences are decoded to satisfy both the probability distribution of a language model and the feasibility constraints of grounded physical models [612]. This ensures that the generated plans are not only linguistically coherent but also realizable within the specific physical domain, effectively preventing the agent from hallucinating actions that violate kinematic or dynamic constraints.

Furthermore, the incorporation of real-time feedback loops is essential for robust embodied intelligence, transforming the agent from a passive plan generator into an active participant that continuously verifies its hypotheses against physical reality. Agents must be capable of forming an “inner monologue” that processes environmental states—such as success detection, scene descriptions, and object permanence—to iteratively refine their plans. The work on “Inner Monologue” demonstrates that leveraging closed-loop language feedback significantly improves instruction completion in complex manipulation tasks, allowing agents to recover from errors and adapt to changing conditions without additional training [33]. This iterative reasoning process is complemented by systems like “MEIA,” which introduce multimodal environment memory modules that integrate visual and linguistic histories, enabling agents to maintain a coherent understanding of the scene over long horizons and align high-level planning with low-level control requirements [613].

Safety remains a paramount concern when deploying embodied agents in unstructured physical environments, serving as the technical counterpart to the agency preservation goals outlined previously. The stochastic nature of language generation can lead to hazardous behaviors if not strictly constrained by safety controllers. Future architectures must inherently combine high-level planning with low-level safety mechanisms. Approaches such as “Look Before You Leap” illustrate the necessity of model-based reinforcement learning frameworks that simulate imagined trajectories to identify and block catastrophic states before execution [614]. By integrating human oversight or learned barrier certificates into the decision-making loop, agents can ensure that their actions remain within safe operational boundaries even when pursuing complex, open-ended goals. This dual-layered approach, where a high-level planner proposes actions and a low-level verifier enforces safety constraints, is crucial for building trust in autonomous systems operating alongside humans.

The journey toward generalist embodied agents also requires mastering the integration of diverse modalities, including 3D perception, audio, and tactile feedback, to overcome the limitations of 2D-centric models. Models that rely solely on 2D visual inputs often fail to capture the depth and spatial relations necessary for precise manipulation. The emergence of 3D Vision-Language-Action (VLA) models, such as “3D-VLA,” marks a significant step forward by linking 3D perception, reasoning, and action through generative world models [615]. These models move beyond direct perception-to-action mapping, instead utilizing world models to imagine future scenarios and plan accordingly, mimicking human cognitive processes. Additionally, the development of agents like “LEO” highlights the importance of training on extensive 3D datasets to achieve proficiency in

tasks ranging from captioning to robotic manipulation, thereby fostering a deeper understanding of the 3D world [422].

Moreover, the ability to generalize to novel tasks and unseen environments without exhaustive retraining is a defining characteristic of true embodied intelligence, a prerequisite for the open-world capabilities evaluated in the following section. Techniques such as “LLM-Planner” demonstrate that few-shot learning capabilities can be harnessed to enable agents to quickly adapt to new tasks by grounding LLM outputs in current environmental observations [616]. Similarly, the use of affordance prompting allows pre-trained LLMs to predict the consequences of actions and generate control sequences for objects they have never encountered before, effectively grounding abstract knowledge in physical properties [617]. These methods suggest that the future of embodied AI lies in hybrid systems that leverage the generalization power of foundation models while maintaining strict adherence to physical realities through specialized grounding mechanisms.

Ultimately, achieving embodied and grounded intelligence requires a synthesis of high-level semantic reasoning, low-level control precision, and robust safety guarantees. It demands a move away from siloed models toward unified architectures where language, vision, and action are inextricably linked through continuous interaction with the physical world. As research progresses, the focus will increasingly shift toward creating agents that not only follow instructions but also understand the underlying physics, anticipate dynamic changes, and operate safely and efficiently in the complex, unpredictable environments of the real world. This transition is not merely an incremental improvement but a fundamental reimagining of how intelligent systems perceive, reason, and act, laying the necessary groundwork for the rigorous evaluation of open-world generalization and meta-reasoning discussed in the subsequent section.

8.5 Evaluation Paradigms for Open-World Generalization and Meta-Reasoning

8.5 Evaluation Paradigms for Open-World Generalization and Meta-Reasoning

Building upon the transition from static language models to embodied agents capable of interacting with physical laws and dynamic environments, as discussed in the previous section, a critical challenge emerges: how do we rigorously evaluate whether these agents possess true generalization capabilities or merely memorized specific training distributions? The rapid advancement of Large Language Model (LLM) agents has precipitated a pivotal shift in artificial intelligence research, moving the focus from mastering closed, static domains to navigating the inherent unpredictability of the open world. Current evaluation methodologies, while rigorous in specific contexts, predominantly rely on stationary benchmarks where task distributions and environmental dynamics remain fixed. This reliance creates a significant blind spot, failing to assess an agent’s capacity for genuine adaptability when confronted with non-stationary environments and novel situations outside their training manifold. To bridge the gap between narrow AI proficiency and the emergent capabilities required for Artificial General Intelligence (AGI), the community must pivot toward “Artificial Open World” simulation frameworks that rigorously test meta-reasoning, continuous learning, and autonomy without explicit human intervention.

The fundamental limitation of existing benchmarks lies in their susceptibility to overfitting and their inability to capture the complexity of real-world novelties. Traditional evaluation protocols often assess agents in controlled simulations where all variables are known, leading to systems that excel within their training manifold but collapse when faced with out-of-distribution elements. As noted in recent literature, the transition from closed simulators to open-world domains introduces exponential increases in domain complexity, characterized by intrinsic factors independent of the

agent and extrinsic factors dependent on the agent’s interaction with the environment [618]. When agents are evaluated solely on static datasets or fixed game levels, they often memorize solution paths rather than developing robust cognitive models capable of handling unforeseen disruptions. For instance, studies in physical reasoning have shown that agents equipped with impressive capabilities in standard settings perform significantly worse when novelties are introduced, adapting far slower than human counterparts [619]. This discrepancy underscores the urgent need for evaluation paradigms that explicitly inject novelties—such as alterations in physical laws, object properties, or goal structures—to measure an agent’s resilience and adaptive speed, ensuring that the grounded intelligence described previously is not fragile.

A promising direction for future evaluation is the conceptualization of the “Artificial Open World,” a framework designed to escape the “trap of developers’ experience.” In conventional benchmarking, once a developer designs a test case, the problem becomes solvable through human insight rather than agent autonomy. An Artificial Open World seeks to generate problems that are never encountered by either the agent or the developer prior to the evaluation event, ensuring that success relies on genuine generalization rather than pre-programmed heuristics [620]. Such environments must be procedurally generated and non-stationary, forcing agents to engage in meta-reasoning—the process of reasoning about their own reasoning processes—to formulate new strategies on the fly. Environments like Obstacle Tower have already demonstrated the necessity of this approach by requiring agents to solve high-fidelity, procedurally generated 3D planning problems where performance on unseen instances is the primary metric, revealing that state-of-the-art deep reinforcement learning methods still lag significantly behind human generalization capabilities [621].

Furthermore, effective evaluation in open worlds requires metrics that transcend simple task completion rates. It necessitates the assessment of an agent’s ability to autonomously detect, characterize, and accommodate novelties. Theoretical frameworks for Self-initiated Open world Learning (SOL) argue that agents must be evaluated on their initiative to detect unknowns, gather ground-truth data through interaction, and incrementally update their internal models without human-triggered retraining cycles [622]. This shifts the evaluation focus from static performance to dynamic learning trajectories. Benchmarks such as NovelGym and Crafter exemplify this shift by providing ecosystems where agents are tested on their ability to unlock achievements or solve tasks in endlessly varying configurations, thereby measuring the spectrum of general abilities including exploration, long-term reasoning, and tool creation [623; 624]. Specifically, the Crafter benchmark evaluates agents on semantically meaningful achievements in a survival setting, where consistent success requires deep exploration and the ability to generalize across a vast space of challenges [624].

The challenge of sim-to-real transfer further underscores the inadequacy of static evaluations, particularly for the embodied agents discussed in Section 8.4. High performance in simulation often fails to predict real-world success due to agents exploiting simulator imperfections, a phenomenon quantified by low Sim-vs-Real Correlation Coefficients in standard benchmarks [625]. Future evaluation paradigms must therefore incorporate mechanisms to tune simulation parameters dynamically to improve predictivity, ensuring that the skills learned in the “Artificial Open World” are robust enough for physical deployment. Additionally, as we look toward the multi-agent societies and hybrid intelligence systems outlined in the following section, the evaluation of social interactions in open worlds becomes paramount. This requires scenarios with mixed incentives and asymmetric roles, as seen in platforms like Melting Pot 2.0, which test an agent’s ability to navigate complex social dilemmas with novel partners [626].

Ultimately, the path toward AGI requires a fundamental rethinking of how we measure intelligence. We must move away from leaderboards based on fixed datasets toward continuous, open-ended

evaluation suites that emulate the non-stationary nature of reality. By adopting frameworks that prioritize novelty detection, causal reasoning, and self-improvement loops, such as those proposed for interactive physical reasoning and autonomous reinforcement learning, the research community can better gauge an agent’s true potential to operate as a generalist in an unpredictable universe [627; 628]. Only through such rigorous, dynamic, and open-world evaluation paradigms can we ensure that the next generation of LLM agents possesses the meta-cognitive flexibility required for genuine autonomy, laying the empirical groundwork for the hybrid intelligence and multi-agent societies that define the future of AGI.

8.6 The Path to AGI via Hybrid Intelligence and Multi-Agent Societies

8.6 The Path to AGI via Hybrid Intelligence and Multi-Agent Societies

While the rigorous evaluation paradigms for open-world generalization discussed in the previous section provide the necessary metrics to assess agent adaptability, they also reveal a critical insight: the path to Artificial General Intelligence (AGI) is unlikely to be traversed by a solitary, monolithic model. Instead, emerging research posits that AGI will arise as an emergent property of complex, interacting systems. This paradigm shift suggests that the future of intelligence lies in the synthesis of Hybrid Intelligence systems and decentralized multi-agent societies, where diverse cognitive entities collaborate to solve problems exceeding the capabilities of any single agent. The fundamental premise is that true generalization in non-stationary environments requires a “society of mind” capable of adaptive adaptation, normative reasoning, and robust problem-solving through collective dynamics.

Hybrid Intelligence represents a critical intermediate step and potentially a permanent architectural feature of this evolution, leveraging the complementary strengths of human intuition and machine computational power. Rather than viewing AI as a replacement for human cognition, the hybrid paradigm emphasizes a symbiotic relationship where humans and machines form socio-technological ensembles to achieve superior results. Research suggests that the most viable path forward involves systems where humans remain in the problem-solving loop, ensuring that AI development is grounded in human values and contextual understanding [629]. This approach challenges the data-centric view of strong AI, arguing instead for cognitive interoperability where human agency is preserved and enhanced through collaboration [630]. By integrating humans into the lifecycle of intelligent systems, we can mitigate the risks of misalignment and ensure that AI agents develop behaviors consistent with complex social norms [631].

Parallel to hybrid human-machine teams, the emergence of decentralized multi-agent societies offers a fertile ground for developing AGI-level capabilities through interaction and evolution. In these societies, complex social structures and collaborative norms are not hardcoded but emerge from the interactions of autonomous agents. The concept of a “Society of Mind,” revitalized by natural language-based societies of large neural networks, demonstrates how modular agents communicating through a universal symbolic language can overcome the limitations of individual models [632]. These systems facilitate a “mindstorm” where agents interview one another, debate, and refine solutions, leading to improved reasoning in multimodal and embodied tasks. The potential for such societies to scale to billions of agents, including humans, suggests a future where AGI is a distributed phenomenon rather than a centralized entity.

A cornerstone of this societal approach is the spontaneous emergence of social norms and cooperation mechanisms, which are essential for functioning safely in the open worlds evaluated in Section 8.5. For AGI to operate effectively in human environments, it must internalize prosocial behaviors

and ethical constraints through interaction rather than explicit programming. Frameworks like CRSEC have demonstrated that social norms can emerge within populations of LLM-based agents through processes of creation, spreading, evaluation, and compliance, effectively reducing social conflicts and enhancing group coherence [633]. Furthermore, mechanisms such as reputation systems and liquid democracy principles can be implemented to foster consensus and resist manipulation in these artificial societies, providing a robust foundation for trust and coordination [634]. The ability of agents to learn shared normative systems via Bayesian rule induction allows them to flexibly cooperate within existing human institutions, adapting to unknown rules by observing compliance and violation patterns [635].

The evolution of cooperation in these hybrid and multi-agent settings is driven by dynamic feedback loops and adaptive strategies. Studies on collective risk dilemmas reveal that adaptive agents can learn to compensate for the behaviors of fixed-behavior partners, adjusting their cooperation levels based on the perceived risk and the actions of others [636]. This adaptability is crucial for AGI, which must navigate uncertain and changing environments. Moreover, the maximization of mutual information between agent actions has been shown to promote the emergence of cooperation in social dilemmas, suggesting that intrinsic motivations for correlation can drive collective intelligence [637]. As these systems scale, they may exhibit “super-reinforcement learning” capabilities, where the interaction between human and AI agents creates a stronger oracle for decision-making than either could achieve alone, particularly in confounded environments [638].

However, the trajectory toward AGI via these societies is fraught with challenges related to governance, safety, and the potential for adversarial dynamics. The evolutionary pressure in competitive environments may favor AI agents with selfish or deceptive traits, posing existential risks if not carefully managed [605]. Therefore, the design of these systems must incorporate mechanisms for meaningful human control and self-reflection, ensuring that AI remains aligned with human values even as it gains autonomy [607]. The development of “Social AI” requires an interdisciplinary approach that bridges complex systems theory, network science, and AI to understand and steer the collective phenomena that arise in human-machine ecosystems [639].

Ultimately, the realization of AGI may depend on our ability to engineer institutions and computational infrastructures that support responsible behavior in decentralized settings. A sociotechnical approach that treats human agency and regulation as primary mechanisms of intervention is essential for bridging the gap between technical systems and the social contexts they inhabit [610]. By fostering environments where diverse agents—both biological and artificial—can co-evolve, share knowledge, and establish fair norms, we move closer to an AGI that is not only intelligent but also socially integrated and ethically grounded. This vision of AGI as a hybrid, multi-agent society offers a promising alternative to the singular “god-like” AI, emphasizing collaboration, diversity, and collective resilience as the keys to solving super-human problems [640]. Yet, as we envision these complex, interacting societies, the imperative to ensure their safety becomes paramount. The transition from theoretical frameworks of cooperation to the practical deployment of autonomous agents necessitates a rigorous focus on containment and provable security, a challenge that forms the critical subject of the following section.

8.7 Safety, Containment, and Provably Secure Architectures for Autonomous Agents

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While the vision of AGI as a hybrid, multi-agent society outlined in the previous section offers a promising path toward collective intelligence, it simultaneously introduces profound safety chal-

lenges. As artificial intelligence systems transition from passive information processors to autonomous agents capable of executing complex, multi-step tasks in physical and digital worlds, the imperative for robust safety mechanisms becomes paramount. The trajectory toward Artificial General Intelligence (AGI) necessitates a fundamental shift from the probabilistic safety assurances that dominate current machine learning paradigms to architectures grounded in provable security and formal verification. This subsection addresses the critical challenge of mitigating existential risks within these emerging societies by exploring frameworks for provable safety, the formal verification of utility functions, and the development of comprehensive containment ontologies.

The core difficulty in ensuring the safety of autonomous agents lies in the gap between high-level intentions and low-level executions, particularly in uncertain environments characterized by the dynamic interactions of multi-agent systems. Traditional verification methods often struggle with the non-deterministic nature of such agentic behavior. Recent theoretical work suggests that determining whether an agent meets a specific standard of acceptable behavior is fundamentally not computable in the general case, implying that attempts to validate outcomes in the physical world may be futile without strictly limiting agent capabilities [641]. Consequently, future research must focus on “eyes-closed” safety kernels and provably safe control strategies that function even when observability is lost or when the agent encounters unforeseen scenarios within a chaotic agent ecosystem. For instance, frameworks utilizing Hamilton-Jacobi reachability analysis offer a unifying mathematical approach to define safety concepts that account for responsibility and context-dependency, allowing agents to maintain safety margins even under partial observability [567]. Similarly, the concept of “Eyes-Closed Safety Kernels” provides a game-theoretic formulation where controlled systems can guarantee safety against external factors even if sensing modalities fail, offering a blueprint for resilient autonomous architectures [642].

Beyond control theory, the alignment of agent objectives with human values within a diverse agent society requires rigorous formalization of utility functions and ethical obligations. The specification of utility functions carries the risk of “perverse instantiation,” where an agent technically satisfies an objective while violating its spirit, a danger amplified in competitive multi-agent environments. To counter this, frameworks such as “Augmented Utilitarianism” propose socio-technological approaches where ethical utility functions are dynamically updated through societal feedback and embodied simulations, rather than being static [643]. Furthermore, integrating deontic logic into agent architectures allows for the formal verification of obligations, permissions, and prohibitions distinct from mission goals. Approaches utilizing Dominance Act Utilitarianism (DAU) have demonstrated the ability to model-check obligations for autonomous systems, ensuring that ethical constraints are adhered to over time [644]. However, relying solely on statistical correlations is insufficient; agents must possess causal reasoning capabilities to understand the implications of their actions on other agents and humans. Research into causal analysis of harm, utilizing structural causal models and contrastive causation, provides a necessary foundation for agents to reason about the consequences of their decisions in a legally and ethically accountable manner [645].

Containment strategies represent another pillar of provably secure architectures, aiming to limit the potential impact of misaligned or hostile agents before they can disrupt the broader societal fabric. As the development of AGI is considered inevitable by many, constructing a scientific ontology for containment is a timely necessity. Recent efforts have begun to map the fundamental constructs required for AGI containment, situating the problem within cyber science and identifying the interactions between humans, AGI, and the cyber world as novel agent objects [646]. Effective containment also requires managing existential risk factors that may arise from current and near-term AI systems acting as intermediate risk multipliers, rather than waiting for superintel-

ligent emergence [647]. This perspective argues for a proactive assessment of how AI affects power dynamics and information security, suggesting that containment architectures must be designed to mitigate these systemic risks before they escalate into uncontrollable societal failures.

Moreover, the verification of safety properties must extend to the logical and symbolic reasoning layers of agents, which are crucial for maintaining coherence in large-scale agent societies. Formal methods, such as regular model checking and the use of existential rules, provide tools for verifying infinite-state systems and parameterized concurrent processes, which are analogous to multi-agent societies [648]. The integration of formal fault injection techniques further enhances dependability by fusing mathematical reasoning with empirical testing throughout the development lifecycle [649]. Additionally, the application of risk control barrier functions (RCBFs) allows for the enforcement of safety constraints in nonlinear systems subject to stochastic uncertainty, ensuring that safety is maintained in a coherent risk sense [650].

Ultimately, achieving controllable autonomy requires a holistic approach that combines causal reasoning, formal verification, and dynamic ethical alignment to support the stable evolution of hybrid intelligence. Research directions must prioritize the development of “agency foundations” that explicitly optimize for the preservation of human autonomy. This involves moving beyond simple reward hacking prevention to creating architectures where safety is a structural property, verified through techniques like SMT-based oracles that can interrogate agent decisions in counterfactual scenarios [651]. By synthesizing insights from causal modeling, control theory, and formal logic, the AI community can begin to construct the provably secure architectures necessary to navigate the path toward advanced autonomous intelligence without compromising existential safety. Establishing these rigorous safety guarantees is a prerequisite for the next phase of development: deploying these capable agents onto resource-constrained edge devices to truly democratize access, a transition that introduces its own unique set of efficiency and security trade-offs.

8.8 Resource-Constrained Deployment and the Democratization of Agentic AI

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Establishing the rigorous safety guarantees outlined in the previous section is a prerequisite for the next phase of development: deploying these capable agents onto resource-constrained edge devices to truly democratize access. This transition from theoretically secure, centralized architectures to distributed edge environments introduces a unique set of efficiency and security trade-offs that are critical for the true democratization of agentic AI. While large language models (LLMs) and generative agents have demonstrated remarkable capabilities in resource-rich data centers, their deployment on resource-constrained edge devices remains a formidable challenge. The core issue lies in the dichotomy between the massive computational, memory, and energy demands of foundation models and the strict limitations inherent to mobile, IoT, and robotic edge platforms. Achieving universal accessibility requires not merely compressing existing models but fundamentally reimagining cognitive architectures to function efficiently within these constraints without sacrificing the generalization, autonomy, or safety properties established earlier.

The primary barrier to edge deployment is the sheer resource intensity of generative AI, which traditionally relies on cloud-based services due to heavy demands on compute and memory [652]. To overcome this, the research community is increasingly focusing on model compression techniques such as quantization, pruning, and knowledge distillation. These methods aim to reduce the footprint of deep neural networks, enabling them to run on hardware with limited static storage and dynamic memory [653]. However, simple compression is often insufficient for complex agentic tasks

that require iterative reasoning, tool use, and adherence to safety kernels. Consequently, there is a growing emphasis on algorithm-hardware co-design, where neural architectures are specifically tailored to the heterogeneous compute capabilities of edge devices, including CPUs, GPUs, and specialized NPUs [654]. This co-design approach ensures that the trade-off between model accuracy, resource consumption, and safety compliance is optimized for specific edge scenarios, ranging from autonomous driving to personal assistants [655].

A promising direction for scaling agentic systems while maintaining efficiency involves the adoption of Mixture of Experts (MoE) architectures within mobile edge networks. By selectively engaging only relevant subsets of model parameters for specific tasks, MoE frameworks can significantly lower operational costs while maintaining high performance [656]. This selective activation is particularly vital for edge devices where battery life and thermal constraints are paramount, ensuring that agents can operate continuously without compromising their functional integrity. Furthermore, the integration of generative AI with mobile edge networks allows for the offloading of subtasks to nearby devices, creating a collaborative ecosystem that supports high-quality content generation even on individual resource-limited units [657]. Such distributed approaches leverage the collective intelligence of the network, transforming isolated devices into a cohesive swarm capable of solving complex problems [658].

The concept of the “Computing Continuum” further underscores the necessity of hierarchical and hybrid architectures to balance local autonomy with global oversight. In this paradigm, intelligence is not confined to a single tier but is distributed across cloud, edge, and end devices, with tasks dynamically allocated based on latency requirements, resource availability, and safety criticality [659]. Frameworks like EdgeMatrix exemplify this by redefining physical resources into logically isolated units to guarantee service-level agreements while maximizing throughput in heterogeneous environments [327]. Similarly, the NetGPT architecture proposes a synergistic relationship between cloud and edge LLMs, where edge models leverage local context for personalized inference while collaborating with cloud models for heavy lifting [660]. This hierarchical control is also evident in robotics, where cloud-edge-end structures enable multi-agent strategy generation and motion control for heterogeneous robot clusters [46].

Energy efficiency remains a critical concern, as the training and inference processes of deep learning models can rapidly deplete the battery life of edge user devices [661]. Future research must prioritize “Green AI” strategies that minimize energy consumption through innovative circuit designs, such as neuromorphic computing and mixed-signal accelerators, which are essential for next-generation autonomous systems [662]. Additionally, exploring energy-accuracy trade-offs allows systems to operate at reduced accuracy levels when necessary to meet strict energy budgets, ensuring continuous operation in power-constrained settings [663].

Ultimately, the democratization of agentic AI hinges on the development of self-managed, adaptive systems that can learn and evolve locally without constant reliance on cloud connectivity, thereby preserving privacy and reducing latency. Approaches like Continuous Learning using Asynchronous Neuroevolution (CLAN) demonstrate the feasibility of adaptive intelligence on commodity edge devices with zero cloud interaction, addressing privacy concerns and bandwidth limitations [664]. Moreover, the vision of a “Cognitive Internet” suggests a future where connected objects independently acquire knowledge and collaborate across diverse domains, blending system-wide collaboration with human intelligence [665]. As we move toward 6G and beyond, the integration of sensing, communication, and computation will be pivotal in creating task-oriented, ultra-reliable edge intelligence that empowers users globally [666]. By addressing these challenges through efficient cognitive architectures and decentralized learning paradigms, we can ensure that the benefits

of advanced, provably safe agentic AI are accessible to all, fostering sustainable development and universal technological empowerment.

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- [611] Aligning Artificial Intelligence with Humans through Public Policy
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- [615] 3D-VLA A 3D Vision-Language-Action Generative World Model
- [616] LLM-Planner Few-Shot Grounded Planning for Embodied Agents with Large Language Models
- [617] Empowering Large Language Models on Robotic Manipulation with Affordance Prompting
- [618] Toward Defining a Domain Complexity Measure Across Domains
- [619] NovPhy A Testbed for Physical Reasoning in Open-world Environments

- [620] Artificial Open World for Evaluating AGI a Conceptual Design
- [621] Obstacle Tower A Generalization Challenge in Vision, Control, and Planning
- [622] Self-Initiated Open World Learning for Autonomous AI Agents
- [623] NovelGym A Flexible Ecosystem for Hybrid Planning and Learning Agents Designed for Open Worlds
- [624] Benchmarking the Spectrum of Agent Capabilities
- [625] Sim2Real Predictivity Does Evaluation in Simulation Predict Real-World Performance
- [626] Melting Pot 2.0
- [627] I-PHYRE Interactive Physical Reasoning
- [628] Autonomous Reinforcement Learning Formalism and Benchmarking
- [629] Cognitive Architecture for Co-Evolutionary Hybrid Intelligence
- [630] Hybrid Intelligence
- [631] Conceptualization and Framework of Hybrid Intelligence Systems
- [632] Mindstorms in Natural Language-Based Societies of Mind
- [633] Emergence of Social Norms in Large Language Model-based Agent Societies
- [634] A Reputation System for Artificial Societies
- [635] Learning and Sustaining Shared Normative Systems via Bayesian Rule Induction in Markov Games
- [636] The art of compensation how hybrid teams solve collective risk dilemmas
- [637] Emergent cooperation through mutual information maximization
- [638] Blessing from Human-AI Interaction Super Reinforcement Learning in Confounded Environments
- [639] Social AI and the Challenges of the Human-AI Ecosystem
- [640] Cogniculture Towards a Better Human-Machine Co-evolution
- [641] Limits to Verification and Validation of Agentic Behavior
- [642] Eyes-Closed Safety Kernels Safety for Autonomous Systems Under Loss of Observability
- [643] Augmented Utilitarianism for AGI Safety
- [644] Algorithmic Ethics Formalization and Verification of Autonomous Vehicle Obligations
- [645] A Causal Analysis of Harm
- [646] A Cyber Science Based Ontology for Artificial General Intelligence Containment
- [647] Current and Near-Term AI as a Potential Existential Risk Factor
- [648] Regular Model Checking Revisited (Technical Report)
- [649] Towards Formal Fault Injection for Safety Assessment of Automated Systems

- [650] Safe Control for Nonlinear Systems with Stochastic Uncertainty via Risk Control Barrier Functions
- [651] ‘Put the Car on the Stand’ SMT-based Oracles for Investigating Decisions
- [652] From Cloud to Edge Rethinking Generative AI for Low-Resource Design Challenges
- [653] Enabling Deep Learning on Edge Devices
- [654] Efficient Heterogeneous Video Segmentation at the Edge
- [655] Performance power assessment of CNN packages on embedded automotive platforms
- [656] Toward Scalable Generative AI via Mixture of Experts in Mobile Edge Networks
- [657] Resource-efficient Generative Mobile Edge Networks in 6G Era Fundamentals, Framework and Case Study
- [658] Distributed Swarm Learning for Edge Internet of Things
- [659] Towards a Cognitive Compute Continuum An Architecture for Ad-Hoc Self-Managed Swarms
- [660] NetGPT A Native-AI Network Architecture Beyond Provisioning Personalized Generative Services
- [661] Green Edge AI A Contemporary Survey
- [662] Circuit and System Technologies for Energy-Efficient Edge Robotics
- [663] Exploring Energy-Accuracy Tradeoffs in AI Hardware
- [664] CLAN Continuous Learning using Asynchronous Neuroevolution on Commodity Edge Devices
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